

Original arithmetic current and effective cutoff control Complete source increments, observed evolution, and hidden return

Split-Zero research programme

22 September 2026
SZ-20260922-026

The original kernel-volume profile and the first-order arithmetic-current profile are connected by the exact identity

$$J'(t) = 2\psi(t) - 2(1+t)\psi'(t).$$

This edition proves a finite reconstruction from the actual source increments, sharper complete-kernel localization and local source-window bounds, and the original observed current's two signed directions. It calculates both extreme singular values of the compressed arithmetic evolution and its exact relation to the full evolution through the hidden kernel.

Both incoming manuscripts are printed in full, followed or preceded by the complete independent derivations that strengthen them. All original cutoffs, source minima, invariant rows, physical coordinates and mixed blocks are retained. The earlier complete mathematical source collection accompanies this reader as an unchanged byte prefix plus new sources.

The fixed-data simple-quartet domain is retained throughout. RC35–44 applies the earlier conductor theorem on its proved period domain: the terminal eigenclass has subexponential normalized current and exponentially balanced weights in the two observed sign sectors. Its residual individual sign is still governed by the exact complex pairing RC32. These results do not establish RH.

Contents

The two profiles and the actual hidden return	4
I Connecting the original cutoff and arithmetic current	5
The original kernel increments, observed arithmetic current, and hidden evolution	5
1. Original objects and the completed finite estimate	5
2. Exact map between the two retained profiles	5
3. Reconstruction from the actual source increments	6
4. Stronger shrinking-window and curvature bounds	7
5. Stronger original source cost and the actual current	8
6. Full first-order evolution and its hidden source	9
7. The prescribed eigenclass has an exact receiving map	9
8. Earlier heat receivers and verification scope	10
9. The earlier terminal-class bound completes another receiver	10
II Finite Gamma minimum and complete coefficient covariance	13
Finite Gamma evaluation and the complete power-band metric	13
1. Original measures and explicit finite constants	13
2. Exact evaluation minimum and the equilibrium profile	14
3. Harmonic measure gives the finite lower minimum	15
4. A trial polynomial controls the complete upper minimum	15
5. Uniform constants down to the first nonzero degree	16
6. Complete coefficient jets and both low cutoffs	17
7. Every row of the high covariance and one simultaneous right inverse	18
8. Finite every-vector rate and all reference cutoffs	18
Sources, scope, and reproducibility	19

III Effective original localization and every fixed subquotient	20
Effective localization in the full original kernel and word image	20
1. Original data, source constants, and the finite scalar input	20
2. Complete coefficient-to-quotient metric comparison	21
3. An explicit circle and the entire residue operator	21
4. The original invariant kernel, all constraints retained	22
5. The whole polynomial fibre and both real integration regions	23
6. Full power-band constants and the native estimate	24
7. A sharper exceptional-direction penalty	25
8. One growing radius and one growing band	26
9. Every positive word direction	26
10. The same finite improvement on every fixed subquotient	27
11. Provenance and scope	28
IV Original observed current: complete incoming proof	29
Original objects, source status, and the claim being proved	29
Two exact polynomial minima and the original current	29
The original conductor and its two lower test classes	30
A finite area bound, retaining the complex cross term	31
The first source direction and the kernel current	33
Evaluate all growing-degree errors	33
Two observed signs and the complete kernel-current bound	34
Prescribed signed observed classes	35
Evaluated return and scalar allowance	36
A new short-time law for the actual arithmetic propagator	36
Verification and the precise remaining boundary	38
V Signed-time evolution and full hidden-source defect	40
Signed-time evolution, extreme singular values, and the original hidden-source defect	40
1. The original spaces and their exact isometry	40
2. A finite noncommuting propagator estimate at both signed times	41
3. Exact finite shear, and the forward/backward singular receiver	41
4. Application to every original cutoff, with the required input displayed	42
5. The complete hidden-source defect and its connecting equations	43
6. Provenance and exact scope	44
VI Effective cutoff: complete incoming proof	45
Status, original objects, and principal results	45
1. The exact scalar profile	46
1.1 Equilibrium dictionary, with every scale retained	46
1.2 Hypergeometric coordinate and strict concavity	47
1.3 Certified scalar values and a new integrated value	48
2. Effective evaluation of the full Gamma kernel	48
2.1 Explicit Gamma envelopes and mass	48
2.2 Exact change of variable for evaluation, not a monic minimum	49
2.3 Harmonic-measure lower bound, including its entropy constant	49
2.4 Complete quantile trial upper bound	50
2.5 Uniform finite remainder, including the small-aspect boundary	50

3. Every vector in a growing power band	52
3.1 A finite low metric constant	52
3.2 Full high covariance, not its diagonal only	52
4. Quantitative localization of the original conductor kernel	53
4.1 A zero-free circle with a certified modulus bound	53
4.2 The exact finite-rank subtraction and the retained action defect	54
4.3 Finite relative spectral estimate and an effective choice	56
5. Original subquotients, observation layers, and the loss distribution	57
5.1 A simultaneous subquotient theorem	57
5.2 Every actual arithmetic observation layer	57
5.3 A measure of actual cutoff loss	58
6. The full original word spectrum through the cutoff interval	58
6.1 Uniform localization of the whole image	58
6.2 A direct finite low-end word band	59
6.3 Uniform positive energy band	60
7. The entire original cutoff-frequency surface	60
7.1 The kernel-energy logarithms cancel their common cutoff scale	60
7.2 All positive regularizers, without a hidden endpoint restriction	60
7.3 Every inverse exterior rank	61
7.4 The measured phase becomes an explicit moving interval	61
7.5 Averaging the original heat over the cutoff interval	62
8. Integration with edition 025 and new shrinking-window laws	62
8.1 Quantitative comparison between arbitrary cutoffs	62
8.2 A resolved local mean of the actual rank-one source responses	63
8.3 Negative mesoscopic curvature of the original determinant loss	63
8.4 An effective native quantile location	64
9. Verification and scope	65
References and exact source use	65

The two profiles and the actual hidden return

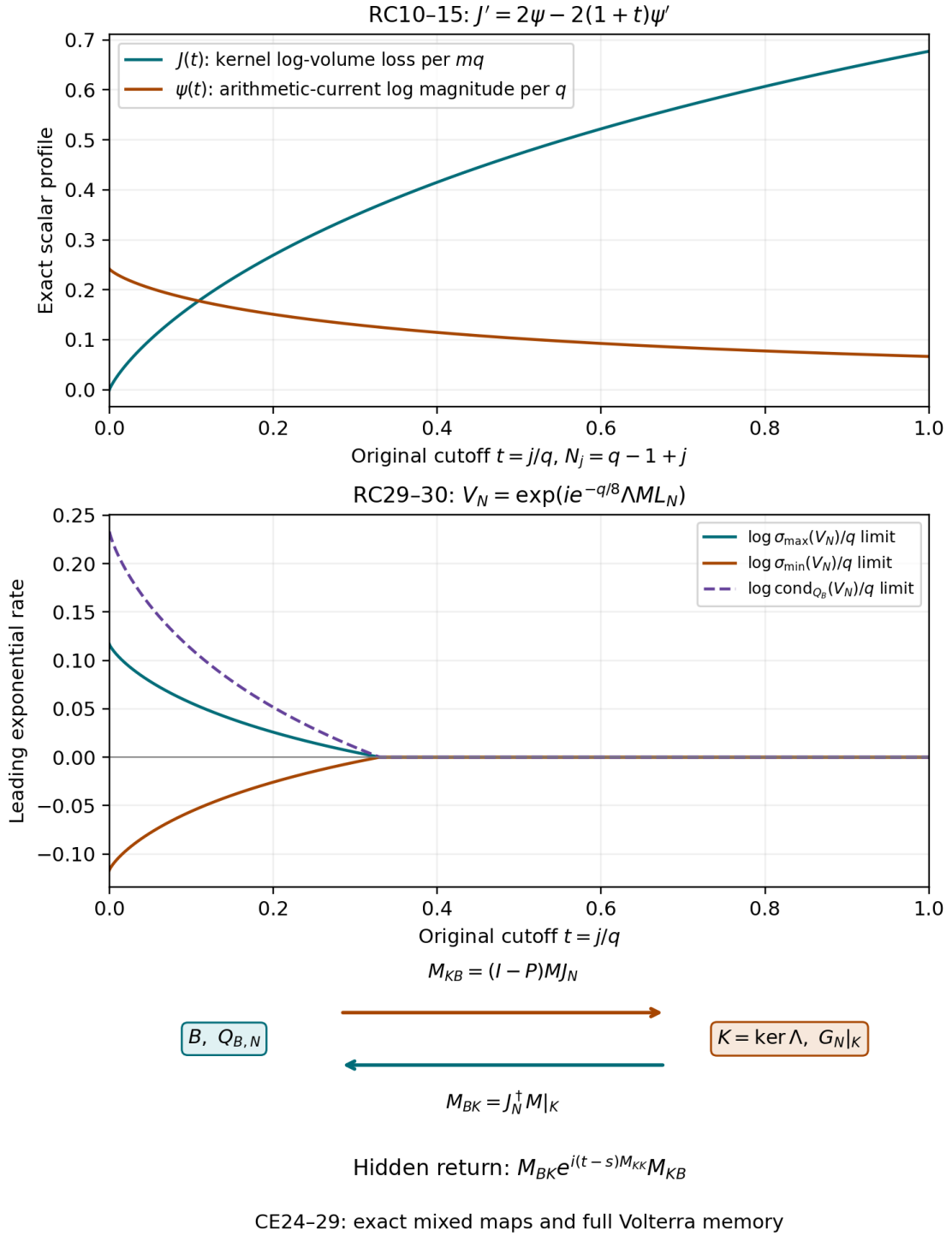


Figure 1: Top: the two evaluated profiles in their original cutoff coordinate. RC10-15 proves their connecting map and finite increment reconstruction. Middle: the limits of both extreme singular values and condition number for the original intrinsic evolution at time $e^{-q/8}$, CE19-23 and RC29-30. The displayed curves sample the universal elliptic formulas; they are not computed native-period matrices. Bottom: exact maps of the observed and hidden source spaces. The complete two-step return, with all mixed terms, occurs in the Volterra identity CE27. Elliptic conventions and formulas are B. C. Carlson's DLMF19.5 and 19.8, with original equation sources and complete derivations retained.

Part I

Connecting the original cutoff and arithmetic current

The original kernel increments, observed arithmetic current, and hidden evolution

22 September 2026. Result SZ-20260922-026. This continuation incorporates both the effective-cutoff manuscript and the subsequent observed-current manuscript. Their complete proofs and all earlier accepted sources accompany this edition. The calculations below connect their two scalar profiles exactly, improve the finite source-window estimates, and carry the stronger original source comparison into the current and evolution. Throughout, a stipulated simple quartet and its admitted original period, branch and arithmetic unit are fixed; these calculations neither construct such a zero nor establish RH.

1. Original objects and the completed finite estimate

Retain the entire original polynomial and physical coordinate

$$q = (k+1)^2, \quad q_0 = (k-7)^2, \quad \Delta = 16k - 48, \quad m = 8k - 16, \quad k \geq 17, \quad k \equiv 1 \pmod{4},$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad s = k/2 + iy. \quad (\text{RC1})$$

Let G_N be the complete minimum of the original source norm on $E = \mathbb{C}[y]/Q_k$, $\Lambda : E \rightarrow B$ the complete observation, and I_K the unchanged frame of $K = \ker \Lambda$. Put $H_{K,N} = I_K^* G_N I_K$. All indices

$$N_j = q - 1 + j, \quad t_j = j/q, \quad 0 \leq j \leq q+1, \quad \mathcal{R}f = f_{N_0} + f_{N_1} - f_{N_q} - f_{N_{q+1}} \quad (\text{RC2})$$

are retained. In particular, the two members of either endpoint pair are distinct.

The independent full Gamma proof EG23 improves the incoming finite scalar error to $83 + 3 \log(q+2)$ on $0 \leq M \leq \lfloor 1.1q \rfloor$. Its complete coefficient covariance and simultaneous right inverse are EG28-35. Inserting that bound into the proved full-kernel localization LX27-33 gives the explicit scalar

$$e_k = \frac{mE_{\text{loc}} + c_{\text{loc}}(24q + \log \kappa_k)}{mq}, \quad c_{\text{loc}} = \min(m, v_0 + N_R), \quad \kappa_k = u_k/\ell_k, \quad (\text{RC3})$$

where the exact finite circle, band, source constants and guards are LX10, LX17, LX19, LX23-24, LX27-28. Every term is specified there. The contour includes all symbol poles, the source comparison includes both real regions, and the codimension loss is a compression of the actual invariant kernel. It follows that

$$F_k(t_j) := \frac{1}{mq} \log \frac{\det H_{K,N_0}}{\det H_{K,N_j}}, \quad |F_k(t_j) - J(t_j)| \leq e_k. \quad (\text{RC4})$$

The choice LX34 satisfies all the finite guards eventually and gives $e_k \rightarrow 0$. No numerical value of a native period is required or asserted.

The same estimate reaches every fixed complete restriction or attained quotient: the original isometric compression LX30 has at most c_{loc} upper and c_{loc} lower exceptions; their total logarithmic deviation is at most $c_{\text{loc}}(24q + \log \kappa_k)$. This proves RC4 and its subquotient versions directly, with their actual ranks, rather than assigning a coefficient from the rank of the full space. Earlier IR6-10 therefore receive a specified finite error.

2. Exact map between the two retained profiles

Use B. C. Carlson's complete elliptic integral conventions, with modulus as argument. Their original equation sources DLMF19.5.1-2 and 19.8.12a-b were read and retained. Their derivation

and exact branch are also given in the preceding EP1-11. Put

$$F(z) = \frac{2}{\pi} K(\sqrt{z}) = \sum_{n \geq 0} \frac{\binom{2n}{n}^2}{16^n} z^n, \quad t = 4zF'(z)/F(z), \quad z = z_t \in (0, 1). \quad (\text{RC5})$$

The map is strictly increasing and onto: its logarithmic derivative is four times the positive variance of the displayed series distribution; at one every finite initial part of that distribution has vanishing mass. The exact cutoff profile is

$$J(t) = 2 \log F(z_t) - \frac{t}{2} \log z_t, \quad J(0) = 0, \quad J'(t) = -\frac{1}{2} \log z_t. \quad (\text{RC6})$$

The original arithmetic-current profile uses

$$r = \frac{1 - \sqrt{z}}{1 + \sqrt{z}}, \quad \kappa = \frac{2z^{1/4}}{1 + \sqrt{z}}, \quad 1 + t = \frac{E(\kappa)}{rK(\kappa)}, \quad \psi(t) = t \log \frac{\kappa}{E(\kappa)} + \log \frac{1 + r}{E(\kappa)}. \quad (\text{RC7})$$

Here r and z are scalar elliptic coordinates only; the physical coordinate and source matrices in RC1-4 are unchanged. The Landen formulas give

$$E(\kappa) = \frac{\pi}{2} (1 + t)(1 - \sqrt{z})F(z), \quad \psi(t) = (1 + t)L(t) + \frac{t}{4} \log z_t, \\ L(t) = \log \frac{4}{\pi(1 + t)(1 - z_t)F(z_t)}. \quad (\text{RC8})$$

These equalities follow by direct substitution in both terms of RC7: $\kappa/E = 4z^{1/4}/[\pi(1 + t)(1 - z)F]$ and $(1 + r)/E = 4/[\pi(1 + t)(1 - z)F]$.

The coefficient recurrence in RC5 gives

$$4z(1 - z)t'(z) = 4z(t + 1) - (1 - z)t^2. \quad (\text{RC9})$$

For clarity, differentiating RC8 yields

$$L'(t) = -\frac{1}{1 + t} + \frac{z'_t}{1 - z_t} - \frac{tz'_t}{4z_t}, \quad (1 + t)L'(t) + \frac{tz'_t}{4z_t} = 0;$$

the second equality is exactly RC9 after multiplication by $4z_t/z'_t$. Hence $\psi' = L + \frac{1}{4} \log z$. Subtracting gives the exact differential and integral maps

$$\boxed{J' = 2\psi - 2(1 + t)\psi', \quad \psi'' = -\frac{J''}{2(1 + t)}}, \quad (\text{RC10})$$

$$\boxed{\psi(t) = (1 + t) \left[a_0 - \frac{1}{2} \int_0^t \frac{J'(s)}{(1 + s)^2} ds \right], \quad a_0 = \log(4/\pi)}. \quad (\text{RC11})$$

Indeed differentiate $\psi/(1 + t)$ and integrate; RC8 gives its value a_0 at zero. Since $J'(s) = -\frac{1}{2} \log s + O(s)$, the integral converges at zero. Conversely integration of RC10, with $J(0) = 0$, proves

$$J(t) = 4 \int_0^t \psi(s) ds - 2(1 + t)\psi(t) + 2a_0. \quad (\text{RC12})$$

Thus these profiles determine each other together with their specified endpoint value. This supplies an exact connection between the already evaluated kernel-volume profile and the arithmetic-current growth profile.

3. Reconstruction from the actual source increments

Let b_N and ρ_N denote the two original nonnegative rank-one source and kernel leverages in the determinant update, with precisely the conventions of incoming ME2 and preceding IR14-16. The complete covariance update and the matrix determinant lemma give

$$A_{i,k} := \log \frac{\det H_{K,N_i}}{\det H_{K,N_{i+1}}} = \log \frac{1 + \rho_{N_i}}{1 + b_{N_i}} \geq 0, \quad F_k(t_j) = \frac{1}{mq} \sum_{i < j} A_{i,k}. \quad (\text{RC13})$$

For an explicit verification of the determinant step, a positive covariance C updated to $C + vv^*$ has inverse $C^{-1} - C^{-1}vv^*C^{-1}/(1 + v^*C^{-1}v)$. Restrict this inverse to the unchanged kernel frame and apply $\det(H - ww^*/d) = \det H(1 - w^*H^{-1}w/d)$. The resulting ratio is exactly the source quotient in RC13, with the complete original minimizing sections. This is the update used by the cited receiver, not a replacement set of columns. The nonnegative sign also follows directly from nesting of the full source fibres.

Define the finite observable

$$\hat{\psi}_k(t_j) = (1 + t_j) \left[a_0 - \frac{1}{2mq} \sum_{i=0}^{j-1} \frac{A_{i,k}}{(1 + t_{i+1})^2} \right]. \quad (\text{RC14})$$

It uses every actual increment up to the indicated cutoff. For every $0 \leq j \leq q + 1$,

$$\left| \hat{\psi}_k(t_j) - \psi(t_j) \right| \leq \frac{1 + t_j}{2} \left[e_k + \frac{2J(t_j)}{q} \right]. \quad (\text{RC15})$$

To prove it put $w_i = (1 + t_i)^{-2}$, $D_i = F_k(t_i) - J(t_i)$. Discrete integration by parts gives

$$\sum_{i < j} w_{i+1}(D_{i+1} - D_i) = w_j D_j + \sum_{i=1}^{j-1} (w_i - w_{i+1}) D_i,$$

since $D_0 = 0$. Its absolute value is at most $e_k w_1 \leq e_k$. On each mesh interval the variation of $(1 + t)^{-2}$ is at most $2/q$. Because dJ is positive, replacing its weight by the right-endpoint value costs at most $2J(t_j)/q$. Apply RC11 to prove RC15. No bound for an individual microscopic leverage is inferred from a cumulative error.

4. Stronger shrinking-window and curvature bounds

Fix $0 < a < 1$. On $[a, 1]$ let

$$v_a = \frac{9a}{80(1+a)}, \quad M_3 = 1280 \left(\frac{1+a}{a} \right)^3, \quad M_4 = 150 \left(\frac{80(1+a)}{9a} \right)^5. \quad (\text{RC16})$$

We prove $|J''| \leq M_3$, $|J''| \leq M_4$. In the probability $p_n = a_n z^n / F(z)$, put $V = \text{Var } n$ and let κ_3, κ_4 be its third and fourth cumulants. Termwise differentiation on compact disks gives

$$J'' = -\frac{1}{8V}, \quad J''' = \frac{\kappa_3}{32V^3}, \quad J'''' = \frac{\kappa_4 V - 3\kappa_3^2}{128V^5}. \quad (\text{RC17})$$

The rational series certificate EG19 implies $z < 11/20$ for $t \leq 1.1$. Also $z \geq t/(1+t)$, $F \leq (1-z)^{-1/2}$, so $V \geq p_0 p_1 = z/(4F^2) \geq v_a$. Bounding $a_n/F \leq 1$ in the raw moments and summing the differentiated geometric series at $z = 11/20$ gives

$$\mathbb{E}n^2 < 10, \quad \mathbb{E}n^3 < 47, \quad \mathbb{E}n^4 < 315, \quad \mathbb{E}n = t/4 \leq 11/40.$$

Thus $V < 10$, $|\kappa_3| < 47 + 3(11/40)10 + 2(11/40)^3 < 56$. The slightly tighter original values $\mathbb{E}n^2 \leq 6820/729$, $\mathbb{E}n^3 \leq 102740/2187$ give $|\kappa_3| < 55$. Moreover the fourth central moment is at most $315 + 6(11/40)^2 10 < 320$, since its other nonconstant terms are negative; hence $|\kappa_4| < 620 < 1000$. These inequalities in RC17 prove RC16: $55/(32v_a^3) < M_3$ and $(1000 \cdot 10 + 3 \cdot 55^2)/(128v_a^5) < M_4$.

Choose an integer $L \geq 1$, $h = L/q$, with $a \leq t_j - h < t_j + h \leq 1$, and take the complete actual product

$$G_{j,L}^c = \left[\prod_{i=j-L}^{j+L-1} \frac{1 + b_{N_i}}{1 + \rho_{N_i}} \right]^{1/(2mL)}.$$

RC4 and RC13, followed by the centered Taylor formula for J , prove

$$\left| \log \frac{G_{j,L}^c}{\sqrt{z_{t_j}}} \right| \leq \frac{e_k}{h} + \frac{M_3 h^2}{6}. \quad (\text{RC18})$$

Indeed $\log G^c = -[F_k(t_j + h) - F_k(t_j - h)]/(2h)$, and the two endpoint errors total at most e_k/h ; the centered average of J' differs from $J'(t_j)$ by at most $M_3 h^2/6$. The opposite sign agrees exactly with $\log \sqrt{z} = -J'$.

At $h_0 = (3e_k/M_3)^{1/3}$ and $L = \lceil qh_0 \rceil$, whenever the displayed interval guard holds, the bound is

$$\frac{3^{2/3}}{2} M_3^{1/3} e_k^{2/3} + \frac{M_3 h_0}{3q} + \frac{M_3}{6q^2}. \quad (\text{RC19})$$

This proves a smaller error than the one-sided shrinking window, using its exact original integer cutoffs.

For the second centered difference, the triangular integral of J'' cancels the odd term of its Taylor expansion. Since $\int_{-1}^1 u^2(1-|u|)du = 1/6$,

$$\left| \frac{F_k(t_j + h) - 2F_k(t_j) + F_k(t_j - h)}{h^2} - J''(t_j) \right| \leq \frac{4e_k}{h^2} + \frac{M_4 h^2}{12}. \quad (\text{RC20})$$

At $h_0 = (48e_k/M_4)^{1/4}$, $L = \lceil qh_0 \rceil$, its bound becomes

$$2\sqrt{M_4 e_k/3} + \frac{M_4(2h_0/q + q^{-2})}{12}. \quad (\text{RC21})$$

In particular the actual centered difference is negative as soon as this bound is below $1/80$, because $J'' = -1/(8V) < -1/80$. This is a finite guard with an explicit eventual realization from LX34; it is not an assumption of the desired sign.

5. Stronger original source cost and the actual current

The retained FW7 full-source comparison states for every finite polynomial degree n

$$\ell = \frac{a_k}{[1 + 4(n + M)^2]^M}, \quad u = A_k, \quad M = 21 + 4m_0, \quad \log(A_k/a_k) = O_h(k + \log k). \quad (\text{RC22})$$

Here M is a fixed source exponent, not the multiplication operator. Use exactly its stated $n = 2q + 2$; then $\log(u/\ell) = O_h(k + \log q)$, retaining both absolute constants and every source degree needed by the final raising column. The underlying polynomial proof uses the full weighted source inequality and successive Gamma multiplication bounds through degree $n + M$, followed by Cauchy-Schwarz between the reciprocal weights. This is the same source, not an assumed metric equivalence.

The retained LRS11 monic theorem, or its stated receiver FW47-49, gives on the fixed-data domain

$$\log(\nu_j^{(b),\sigma}/h_{Q_b+j}) = 2Q_b\psi(j/Q_b) + O_h(\log q), \quad 0 \leq j \leq 2Q_b, \quad b = k, k-8. \quad (\text{RC23})$$

All lower indices in OCP20 are at most $q_0 + O(k)$, and therefore lie in this proved interval eventually. For the remaining finite-index formulation OCP30 gives an explicit recurrence bound; no original cutoff is dropped. Integrating the logarithmic endpoint derivative bound from RC8-10 shows uniformly for $0 \leq r \leq q + 1$

$$q_0\psi((r + \Delta - v_0)/q_0) - q\psi(r/q) = O_h(k \log q). \quad (\text{RC24})$$

To verify this estimate, the argument displacement is $O(k/q)$. The integral of $1 + |\log t|$ on an interval of that length is $O((k/q) \log q)$; multiplying by q_0 and including $q - q_0 = O(k)$ proves it.

Retain the original current notation of the supplied complete TeX OCP1-28: e_N, f_N are its orthonormal consecutive source classes, $\epsilon_N = \sqrt{E_N F_N}/\omega_N$, C_N is their full observed two-column Gram, Θ_N its exact area certificate OCP22, and $\alpha_N = \|P_{K,N} e_N\|^2$. The conductor proof there keeps the exact finite determinant $|\det Y| \geq |\kappa_N \kappa_{N+1}| A_D B_D [(1 - x_N)(1 - y_N) - z_N]$, including its cross term. Its two source comparisons and RC23-24 now prove

$$\boxed{\begin{aligned} \log \epsilon_N &= q\psi(t_j) + O_h(k \log(q + 2)), & 0 \leq -\log \Theta_N &= O_{h,\varpi}(k \log(q + 2)), \\ \alpha_N &\leq e^{-2q\psi(t_j) + O_{h,\varpi}(k \log(q + 2))}, & \log \|C_N\|^+ &= O_h(k + \log q). \end{aligned}} \quad (\text{RC25})$$

Here C_N is the selfadjoint part of original multiplication; the last bound means $\log \max(1, \|C_N\|)$. Its finite bound is $2(N+1)\sqrt{u/\ell}$. To check each other error, the finite conductor norm $\mathcal{F}_N$ is polynomial in N with fixed exponents, and L_N in OCP20 has bounded positive logarithm. The logarithms of d_N, e_N° are $-2q\psi(t_j) + O(k \log q)$ by RC23-24. Thus OCP19 and OCP25 hold eventually; their margin $\zeta_N \geq 1/2$ has bounded logarithm. The ratio ξ_N/e_N° in the area certificate is compared before taking its bound: its logarithm is the difference in RC24 with error $O(\log q)$. Substitution

into OCP22 and OCP26 proves the area and leakage estimates in RC25. The exact monic difference formulas OCP5-7, with exponentially small subtraction ratios, prove its first estimate. This accounts for every possible growing error, without discarding either source column.

Write $\mathcal{C}_N = \begin{pmatrix} a & r \\ r & d \end{pmatrix}$. The observed centered Hermitian current has exact nonzero eigenvalues

$$\lambda_{\pm} = \epsilon_N [-\Im r \pm \sqrt{ad - (\Re r)^2}], \quad \lambda_+ \lambda_- = -\epsilon_N^2 \det \mathcal{C}_N. \quad (\text{RC26})$$

Indeed it is $i\epsilon_N(vu^* - uv^*)$, with $[u, v]^*[u, v] = \mathcal{C}_N$; multiplication of its two-by-two characteristic polynomial proves RC26. The full isometry of the actual quotient is CE2-3. Its finite area bound and $\mathcal{C}_N \preceq I$ give $\epsilon_N \Theta_N \leq \lambda_+, -\lambda_- \leq \epsilon_N$. Thus RC25 improves all OCP38-48 logarithmic errors to $O_{h,\varpi}(k \log q)$. In particular

$$\log \lambda_+ = \log(-\lambda_-) = q\psi(t_j) + O_{h,\varpi}(k \log q), \quad \|\mathcal{W}_{K,N}\| + |\text{tr } \mathcal{W}_{B,N}| \leq e^{O_{h,\varpi}(k \log q)}. \quad (\text{RC27})$$

The operator has inertia $(1, 1, \dim B - 2)$; the two specified original source superpositions in OCP41-45 have opposite signed currents. No sign for a different vector follows from this signature.

6. Full first-order evolution and its hidden source

Let $Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}$, $L_N = G_N^{-1} \Lambda^* Q_B$. The independent CE9-19 proof retains the full noncommuting Duhamel integral and both signed times. Its two operators are

$$U_N(T) = \Lambda e^{iT M} L_N, \quad V_N(T) = e^{iT \Lambda M L_N}, \quad T = e^{-bq}, \quad b > a_0/2. \quad (\text{RC28})$$

Their norms are taken in the original Q_B metric. CE20a lists the complete finite cost. Equation RC25 bounds it by $O(k \log q)$, so CE21-23 give

$$\begin{aligned} \log \|U_N(T)\| &= q(\psi(t_j) - b)_+ + O(k \log q), \\ \log \sigma_{\max} V_N(T) &= -\log \sigma_{\min} V_N(T) = q(\psi(t_j) - b)_+ + O(k \log q). \end{aligned} \quad (\text{RC29})$$

The equality here denotes the shared leading term with the stated errors. The smallest singular value follows from the exact intrinsic inverse $V_N(T)^{-1} = V_N(-T)$; the full observed operator has the different inverse defect CE26. At $b = 1/8$, retaining all four signs,

$$\boxed{\mathcal{R} \log \text{cond}_{Q_{B,N}} V_N(e^{-q/8}) = [4 \log(4/\pi) - 1/2]q + O_{h,\varpi}(k \log q).} \quad (\text{RC30})$$

The scalar $e^{kT/2}$ for the original physical action cancels exactly in this condition number and in the four-cutoff norm return. The coefficient in RC30 is approximately 0.4662579010819617787641475663; its certificate is twice the independently replayed outgoing-norm coefficient. Combining RC15 with the 1-Lipschitz function $x \mapsto (x-b)_+$ supplies actual-increment approximations to each leading exponent in RC29, with additional error at most $(1+t_j)(qe_k + 2J(t_j))/2$, and twice that for the condition number.

In the actual quotient isometry $J_N = L_N Q_B^{-1/2}$, put $P = J_N J_N^\dagger$. The full connecting defect is

$$\mathcal{D}_N = J_N^\dagger M(I - P)M J_N. \quad (\text{RC31})$$

CE24-29 prove its complete Volterra equation and the exact rank-one cancellation $J_N^\dagger R(I - P)R J_N = -z_N R_B$, because $R^2 = 0$ before observation and $R_B^2 = z_N R_B$ after observation. This retains all mixed and hidden source terms. RC25 strengthens its full bound to $\|\mathcal{D}_N\| \leq \exp[q\psi(t_j) + O(k \log q)]$, and the entire hidden current remains bounded as in RC27. These are quantitative controls on the actual defect and its maps.

7. The prescribed eigenclass has an exact receiving map

Let x be any original eigenclass $Mx = \omega x$, let $y = Px$, and keep its hidden part $x_K = (I - P)x$. The full action is $M = C + \epsilon f e^\dagger$, with $\|C\| \leq c$, $e \perp f$, and $\alpha = \|(I - P)e\|^2$. The eigenvalue equation implies

$$\epsilon |\langle e, x \rangle| \leq (|\omega| + c) \|x\|.$$

Since $\langle e, y \rangle = \langle e, x \rangle - \langle (I - P)e, x_K \rangle$, the exact current pairing and its bound are

$$\Phi_N(\Lambda x) = i[\langle y, My \rangle - \langle My, y \rangle] = -2\Im \omega \|y\|^2 + 2\Im \langle y, Mx_K \rangle, \quad (\text{RC32})$$

$$|\Phi_N(\Lambda x)| \leq 2[(|\omega| + c)\|x\| + \epsilon\sqrt{\alpha}\|x_K\|], \|y\|. \quad (\text{RC33})$$

For the bound, the selfadjoint part contributes zero and $|\Phi_N| \leq 2\epsilon|\langle e, y \rangle| |\langle f, y \rangle|$, with the last factor at most $\|y\|$. This proves RC33 and retains the complex interference term in RC32. By RC25 and $|\omega| \leq k\sqrt{\delta^2 + \gamma^2}$, its coefficient is $e^{O(k \log q)}$. If $\Lambda x = 0$, both sides vanish. Otherwise division by $\|y\|^2$ still retains the actual ratio $\|x\|/\|Px\|$. No individual projected sign or bound on that ratio is supplied by the operator signature. RC32–33 provide the exact return to that outstanding original vector calculation, strengthening the earlier complete residual identity NE13 without changing its observation.

8. Earlier heat receivers and verification scope

The new complete positive-word band LX42 has explicit error e_{word} for every positive energy. Insert this error into the preceding PH34–35. If $L_* = 1/J'(1)$, its complete measured heat error is bounded by

$$\alpha_k^{\text{angle}} + L_* \frac{e_{\text{word}}}{q} + \frac{1}{q} + \frac{L_*(1 + e^{-1})}{q}. \quad (\text{RC34})$$

Here α_k^{angle} is the same complete projection-angle defect as PH35. To verify the scalar part, a Gumbel variable G represents the heat smoothing, and $\mathbb{E}|G| \leq 1 + e^{-1}$. The limiting cutoff distribution is L_* -Lipschitz, so a center displacement e_{word}/q and the random displacement G/q cost at most the two displayed terms; the discrete cutoff measure costs $1/q$. The original angle comparison is added before this scalar comparison. Thus the effective arrival improves the word error while preserving the earlier sharper $O(1/q)$ smoothing term.

The auxiliary exact and interval checks accompanying this proof test the rational differential maps, source-window constants, complete finite maps and incoming scalar enclosures. They supplement the analytic proofs. They do not report evaluations of actual zeta zeros or period-dependent native matrices. Human provenance for the Gamma and orthogonal-polynomial inputs remains R. A. Askey and R. Roy, and T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw with W. P. Reinhardt, respectively: DLMF5.7.6, 18.22.8, 18.23.7. Complete author equation TeX and reading locators are preserved. The elliptic source is Carlson as cited above; the programme-specific maps and error estimates are proved here and in the accompanying EG, LX, OCP and CE proofs.

9. The earlier terminal-class bound completes another receiver

The earlier OB1–15 conductor and reverse-lift proof already controls the norm ratio occurring in RC33. We now apply it at its exact proved period domain. This is an integration of an established estimate, not a newly assumed bound.

Keep its prescribed terminal cardinal

$$v_+(S) = \frac{\chi_k(S)}{(S - k\rho)\chi'_k(k\rho)}, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad \omega = k\gamma - ik\delta. \quad (\text{RC35})$$

Then $Mv_+ = \omega v_+$ in the unchanged physical coordinate $S = k/2 + iy$. Let $j_- = k - 8$, $Q_0 = Q_{j_-}$, v_- its corresponding terminal cardinal, and v_A the same first nonzero conductor order denoted v_0 above. Evaluation at every lower-grid root proves

$$C_A v_+ = a_{88} v_-, \quad K \subset \ker C_A. \quad (\text{RC36})$$

Only the original (8, 8) shift reaches the upper terminal root. Every other evaluation is zero. Thus this is the original coefficient, with its phase, and not a freely chosen scalar.

Here is the explicit period domain proved in OC50–74. Retain $\gamma \geq 1000\delta$, the original five-orbit requirement and the admitted radius R_{adm} . Write

$$\tau = (\gamma^2 - \delta^2)/3, \quad d_h = 2\delta\gamma, \quad x = \delta + i\gamma, \quad h_c(w) = w^4 + 6\tau w^2 + (\delta^2 + \gamma^2)^2, \\ m_{\text{corner}} = \frac{23\tau^2 d_h}{500|h'_c(x)|^2}, \quad R_{\text{corner}} = \max\{R_{\text{adm}}, 1, 2\mathcal{M}_2/m_{\text{corner}}\}. \quad (\text{RC37})$$

The finite constant $\mathcal{M}_2 = 18432\kappa_V^2 M_*^8$ retains OC67-69:

$$\begin{aligned} \rho_0 &= |x|, \quad \kappa_V = \rho_0^2(1 + \rho_0 + \rho_0^2 + \rho_0^3)/d_h, \\ A_2 &= (|\beta|/3)5^{3/5}2^{2/5}, \quad B_2 = |\eta|5^{1/5}2^{4/5}, \\ K_2 &= \frac{2}{5}(12/5)^{3/2}A_2^{5/2} + \frac{4}{5}(4/5)^{1/4}B_2^{5/4}, \\ M_* &= \max_{\substack{1 \leq b \leq 4 \\ 0 \leq r \leq 3}} 5^{(r+1-b)/5} 2^{b/5} \frac{\Gamma((r+1)/5)}{\Gamma(b/5)} e^{K_2}, \quad \beta = 6\tau, \quad \eta = \rho_0^4. \end{aligned} \tag{RC38}$$

These are fixed functions of the original quartet. The complete source proof bounds the exact original period conjugation entries by $\mathcal{M}_2$ on $|z| = 2$, and their values at zero below by m_{corner} . Cauchy's coefficients give variation at most $\mathcal{M}_2/|\varpi|$ at the actual $z = \varpi^{-1}$. Extremal coefficients of the four-factor original biform are products of the four corresponding entries. Therefore, on $|\varpi| \geq R_{\text{corner}}$,

$$|a_{88}| \geq (m_{\text{corner}}/2)^4 > 0. \tag{RC39}$$

The algebraic estimate at zero keeps every original unit phase: OC59-60 bounds the squared-modulus difference of its two quadratic brackets, before taking their difference. The complete proof and coefficients are in the unchanged source bank at the precise locator above. No extension of RC39 to an exceptional period is made.

For completeness, the full finite terminal norm comparison used below is

$$\frac{\|P_N v_+\|_{G_N}^2}{\|v_+\|_{G_N}^2} \geq \mathfrak{l}_{k,N} := \frac{\ell_k}{u_k} \frac{|a_{88}|^2 |D_k(k\rho)|^2 g_{k,N}^{\Delta-v_A}}{\mathfrak{C}_{A,N}^2 \mathcal{T}_{N-\Delta}(z_0)^2 [2(N+1) + R]^{2\Delta}} > 0, \tag{RC40}$$

where every original quantity is

$$\begin{aligned} D_k(S) &= \chi_k(S)/\chi_{k-8}(S-8\rho), \quad R = k\sqrt{\delta^2 + \gamma^2}, \quad z_0 = -8\gamma + 8i\delta, \\ B_- &= 2q_0/\delta + 2 + \pi/2, \quad R_- = (k-8)\sqrt{\delta^2 + \gamma^2}, \\ g_{k,N} &= \{1 + B_-^2[R_-^2 + 4(N - v_A + 3)^2]\}^{-1}, \\ \mathcal{T}_n(z) &= e(n+1)(2n+1)^{(|z|+1)/2}, \\ \mathfrak{C}_{A,N} &= \sum_{u,w=0}^8 |a_{uw}| \mathcal{T}_N((2w-8)\gamma - i(2u-8)\delta). \end{aligned} \tag{RC41}$$

The values ℓ_k, u_k are the original complete source constants through degree $2q+2$.

Here is the entire receiving argument for RC40. The lower terminal cardinal minimum equals

$$\mathcal{E}_-^\sigma(q_0 - 1 + r) = \frac{H_r}{|Q'_0((k-8)\gamma - i(k-8)\delta)|^2}, \quad H_r = \min_{\deg p < r} \int \left| \frac{1}{z_- - y} - p(y) \right|^2 Q_0(y)^2 d\sigma.$$

The complete Jacobi resolvent gives $H_{r+1}/H_r = a_{r+1}^2 L / (1 + a_{r+1}^2 L)$, where $L = \|(z_- - J^{(r+1)})^{-1} e_0\|^2$. Its spectral measure has second moment a_{r+2}^2 , so Jensen gives $L \geq (R_-^2 + a_{r+2}^2)^{-1}$. Integration by parts using the digamma bound in OCP30 gives $a_{r+1} \geq (r+1)/B_-$, and the full Gamma multiplication bound gives $a_{r+2} \leq 2(q_0 + r + 2)$. Thus each of the $\Delta - v_A$ required original steps has ratio at least $g_{k,N}$.

The exact reverse lift is

$$v_+(S) = D_k(S) v_-(S-8\rho) / D_k(k\rho).$$

It maps every lower representative of degree $N - \Delta$ to an upper representative of degree N , with all free relations intact. Translation has norm at most $\mathcal{T}_{N-\Delta}(z_0)$: the Gamma generating function multiplies by $e^{z_0 \arctan t}$; Cauchy's formula at $t = n/(n+1)$ and summation of all its weighted coefficient diagonals give exactly $\mathcal{T}_n$. Successive multiplication by all Δ factors of D_k costs $[2(N+1) + R]^\Delta$. Conversely RC36 forces every lift of the observed terminal class to have lower conductor class $a_{88}v_-$, whose full minimum is bounded below by its lower cardinal norm divided by $\mathfrak{C}_{A,N}^2$. Apply both source comparisons and the consecutive-minimum bound. Their quotient is RC40, with every determinant factor retained.

Every factor of $D_k(k\rho)$ has modulus at least $2(k-7)\delta$; there are $\Delta = O(k)$ such factors. The finite formulas give

$$0 \leq -\log \mathfrak{l}_{k,N} = O_{h,\varpi}(k \log q) \tag{RC42}$$

uniformly on the original window. Its lower sign follows because RC40 compares with an actual norm ratio at most one. Its upper order follows from $-\log g = O(\log q)$, $\log(u/\ell) = O(k + \log q)$,

and fixed exponents in the translation constants. Consequently RC33 has the new normalized receiver

$$\boxed{\frac{|\Phi_N(\Lambda v_+)|}{\|\Lambda v_+\|_{Q_{B,N}}^2} \leq 2\mathfrak{l}_{k,N}^{-1/2} [|\omega| + c_N + \epsilon_N \sqrt{\alpha_N}] = \exp[O_{h,\varpi}(k \log q)]}. \quad (\text{RC43})$$

This closes the norm-ratio step on its proved period domain. The same explicit obstruction at other admitted periods remains governed by their actual conductor columns and RC32; no nonvanishing is assigned there.

There is a further exact consequence in the observed current's sign eigenspaces. Let $w_\pm$ be unit eigenvectors for RC26 in the actual observation metric, put

$$b = \Lambda v_+ / \|\Lambda v_+\|_{Q_B}, \quad p_\pm = |\langle w_\pm, b \rangle_{Q_B}|^2, \quad j_N = \Phi_N(\Lambda v_+) / \|\Lambda v_+\|_{Q_B}^2.$$

The spectral theorem, with all zero directions retained, gives $j_N = \lambda_+ p_+ + \lambda_- p_-$. Therefore

$$\boxed{|p_+ - p_-| \leq \frac{|j_N|}{-\lambda_-} + \left| \frac{\lambda_+}{-\lambda_-} - 1 \right| p_+ \leq \exp[-q\psi(t_j) + O_{h,\varpi}(k \log q)]}. \quad (\text{RC44})$$

The last inequality is RC25-27, RC43 and the exact trace bound $|\lambda_+ + \lambda_-| \leq 2\epsilon\sqrt{\alpha}$. It proves exponentially balanced weights of the original terminal class in the two sign sectors. It does not infer its residual sign: that still belongs to the complex pairing in RC32. The result now uses the earlier conductor comparison rather than sending that completed calculation back out as missing work.

Part II

Finite Gamma minimum and complete coefficient covariance

Finite Gamma evaluation and the complete power-band metric

This independent derivation checks HK1–HK11 and JB1–JB8 of the supplied *Effective native cutoff geometry*. It proves their finite scalar bounds and gives the stronger constants 83 and 111 below. It also proves the simultaneous bound for every vector of the prescribed coefficient band. The weight, both real half-lines, its complete mass, every coefficient row, and every affine minimum remain in the formulas.

The preceding programme proofs used here are PJ1–PJ44, original Gamma and jet maps, EL1–EL10, exact equilibrium constants, and EIQ1–EIQ19, the density and the Euler inequality on the complete positive half-line. Their complete source bodies are retained alongside this derivation. The new estimates below do not depend on a limiting zero distribution.

1. Original measures and explicit finite constants

For integer $n \geq 36$, retain

$$w(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad d\sigma(y) = w(y)dy, \quad d\nu_n = |y|^{2n}d\sigma, \quad d_n = \nu_n(\mathbb{R}). \quad (\text{EG1})$$

The evaluation kernel $K_{n,M}(0,0)$ is the squared norm of evaluation at zero on all complex polynomials of degree at most M , with norm from ν_n . Thus

$$K_{n,M}(0,0)^{-1} = \min\{\|P\|_{\nu_n}^2 : \deg P \leq M, P(0) = 1\}. \quad (\text{EG2})$$

Positive definiteness of every polynomial Gram matrix gives existence and uniqueness of the minimizer. The even and odd subspaces are orthogonal, so the minimizer is even and the kernels at degrees $2h$ and $2h+1$ agree exactly. At degree zero, $d_n K_{n,0}(0,0) = 1$.

The complete mass is $d_0 = \sqrt{2\pi}$. Indeed the Fourier transform of $g_a(u) = e^{au-e^u}$ is $\Gamma(a+it)$. Plancherel and $x = e^u$ give $\int |\Gamma(a+it)|^2 dt / (2\pi) = 2^{-2a}\Gamma(2a)$. The original coordinate is $y = 2t$; at $a = 1/4$ its factor two gives the asserted mass.

R. A. Askey and R. Roy's DLMF 5.11.1 and its remainder statement in 5.11(ii) imply, with $z = 1/4 + iy/2$, $y \geq 1$,

$$\log \Gamma z = (z - 1/2) \log z - z + \frac{1}{2} \log(2\pi) + R(z), \quad |R(z)| \leq \frac{\sec^2(\arg z/2)}{12|z|} \leq \frac{1}{3y}.$$

The original equation TeX is retained; the section's error statement, which is prose on the primary site, is recorded as such in the use ledger. Taking real parts, without changing the density, gives

$$\log(\sqrt{y}e^{\pi y/2}w(y)) = \frac{1}{2} \log 2 - \frac{1}{4} \log(1 + 1/(4y^2)) + y \arctan(1/(2y)) - \frac{1}{2} + 2\Re R(z).$$

The inequalities $x - x^3/3 \leq \arctan x \leq x$ and $\log(1+x) \leq x$ show that this expression lies between $\frac{1}{2} \log 2 - 37/48$ and $\frac{1}{2} \log 2 + 2/3$. These bounds are respectively above $\log(1/2)$ and below $\log 3$. For $0 < y \leq 1$, the Gamma integral gives $|\Gamma(1/4 + iy/2)| \leq \Gamma(1/4) < 4 + e^{-1} < 9/2$. Using $e^{\pi/2} < 8$ and $2\pi > 6$ then gives the following full envelopes:

$$\frac{1}{2}y^{-1/2}e^{-\pi y/2} \leq w(y) \quad (y \geq 1), \quad w(y) \leq 32y^{-1/2}e^{-\pi y/2} \quad (y > 0). \quad (\text{EG3})$$

Put $\theta = \pi/2$, $\alpha = 2n - 1/2$, and $d\lambda_n = |y|^\alpha e^{-\theta|y|}dy$. We give the finite polynomial argument at zero, where the lower pointwise envelope cannot be used. The Laguerre derivative identity, from the finite coefficient formula and parameter relation, is

$$(L_j^{(\alpha)})' = - \sum_{i < j} L_i^{(\alpha)}.$$

In the orthonormal basis on the positive half-line its derivative matrix has entries

$$D_{ij} = -\theta \prod_{r=i+1}^j \sqrt{\frac{r}{r+\alpha}}, \quad i < j.$$

For degree at most $2n+2$, every ratio is at most $(2n+2)/(4n+3/2) \leq 12/19 < 16/25$. Each absolute row and column sum is less than 4θ , hence $\|P'\| \leq 4\theta\|P\|$. Integrating the derivatives of $|P|^2 y^\alpha e^{-\theta y}$ and $|P|^2 y^{\alpha-1} e^{-\theta y}$, and writing $B_j = \int |P|^2 y^{\alpha-j} e^{-\theta y} / \|P\|^2$, gives

$$\alpha B_1 \leq 9\theta, \quad (\alpha-1)B_2 \leq \theta B_1 + 8\theta\sqrt{B_2}.$$

Solving this quadratic yields $\|P/y\| \leq 8\theta\|P\|/n$. Boundary terms vanish since $\alpha > 1$. Apply this to $P(y)$ and $P(-y)$, and add their squared norms. Thus the λ_n mass of $|P|^2$ on $|y| < 1$ is at most $64\theta^2/n^2 < 1/4$ of its full norm for $n \geq 36$. Applying (EG3) on the remaining part proves

$$\frac{1}{4}\|P\|_{\lambda_n}^2 \leq \|P\|_{\nu_n}^2 \leq 32\|P\|_{\lambda_n}^2, \quad \deg P \leq 2n+2. \quad (\text{EG4})$$

The factor $1/4$ is weaker than the available $3/8$; using it maintains the supplied constants.

The full auxiliary mass is $2\Gamma(2n+1/2)(2/\pi)^{2n+1/2}$. Its logarithm, minus $2n \log n + \eta_0 n$, where $\eta_0 = 2 \log(4/\pi) - 2$, is

$$\log 4 + 2n \log(1 + 1/(4n)) - \frac{1}{2} + R(2n+1/2).$$

For positive real argument, $0 < R(x) < 1/(12x)$. Bounds $u - u^2/2 \leq \log(1+u) \leq u$ therefore enclose the last three terms between $-1/(16n)$ and $1/(24n)$. Apply (EG4) to the constant polynomial:

$$-\frac{1}{16n} \leq r_d(n) := \log d_n - [2n \log n + \eta_0 n] \leq \log 128 + \frac{1}{24n} < 5. \quad (\text{EG5})$$

The same finite Laguerre expression gives its complete three-term multiplication matrix. Its row and column sums, including the final raising row, are at most $20n/\theta$ through source degree $2n+1$. Transfer this bound with (EG4), and use the already proved inverse-moment inequality for the lower bound. With

$$\kappa = \sqrt{128}, \quad c = (4\pi\kappa)^{-1}, \quad C = 40\kappa/\pi,$$

one obtains

$$cn\|P\|_{\nu_n} \leq \|yP\|_{\nu_n} \leq Cn\|P\|_{\nu_n}, \quad \deg P \leq 2n+1. \quad (\text{EG6})$$

For the lower estimate use $\|P\|^2 = \langle P/y, yP \rangle$ and Cauchy-Schwarz; P/y is an integrable rational function in this identity.

2. Exact evaluation minimum and the equilibrium profile

Let $M = 2h \geq 2$, $n \geq 40$, and $t = 2h/n \leq 3/2$. For the original variable $x = y^2/n^2$, write the even admissible polynomial as $P(y) = R(x)$. The map is a bijection between $R(0) = 1$, $\deg R \leq h$ and the even fibre in (EG2), and

$$\|P\|_{\nu_n}^2 = n^{2n+1} \int_0^\infty x^{n-1/2} w(n\sqrt{x}) |R(x)|^2 dx. \quad (\text{EG7})$$

Both half-lines contribute to (EG7). There is no leading-coefficient condition on R .

Use the proved EIQ equilibrium for the exact potential $V_t(x) = (\pi\sqrt{x} - 2 \log x)/t$. Write its support as $[a, b] = [u^2, v^2]$, and put $p = uv$, $w_0 = (u+v)/2$, $c_0 = (v^2 - u^2)/4$, $W = b - a$. To avoid confusion, w_0 is this endpoint average, while $w(y)$ continues to mean the density (EG1). The endpoints obey

$$uK(\sqrt{1-u^2/v^2}) = 2, \quad vE(\sqrt{1-u^2/v^2}) = 2(1+t).$$

The probability density and Euler identities are

$$\rho_t(x) = \frac{\sqrt{(b-x)(x-a)}}{4\pi tx} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}}, \quad a < x < b, \quad (\text{EG8})$$

$$V_t(x) - 2 \int \log |x-y| d\mu_t(y) \geq \lambda_t \quad (x > 0),$$

with equality on $[a, b]$. The retained EL calculation evaluates

$$\lambda_t = 4 + 4/t - (4/t) \log w_0 - 2 \log c_0, \quad L_t = \int \log x d\mu_t = (4/t + 2) \log w_0 - (2/t) \log p - 2. \quad (\text{EG9})$$

These statements use the actual full-domain EIQ density, whose proof obtains its Cauchy transform $(V'_t - Rh)/2$. Since $h > 0$, its Euler derivative is negative to the left of the interval and positive to the right; this verifies precisely the inequality used on both parts of the integration domain below.

Let $F(z) = 2K(\sqrt{z})/\pi$, $a_j = \binom{2j}{j}^2/16^j$, so $F(z) = \sum a_j z^j$. The unique coordinate $z_t \in (0, 1)$ is specified by

$$t = 4z_t F'(z_t)/F(z_t), \quad J(t) = 2 \log F(z_t) - \frac{t}{2} \log z_t, \quad J(0) = 0. \quad (\text{EG10})$$

The exact Landen map is $\sqrt{z_t} = (v - u)/(v + u)$. Substitution into (EG9), using the two retained original Landen equation sources, proves

$$J(t) = \eta_0 + \frac{t}{2}(\lambda_t + 2L_t), \quad u = \frac{4}{\pi(1 + \sqrt{z})F(z)}, \quad v = \frac{4}{\pi(1 - \sqrt{z})F(z)}. \quad (\text{EG11})$$

No limiting argument is used in this identity. It is the same J as edition025.

The weights $a_j z^j / F(z)$ form an auxiliary probability, and differentiation gives $dt/d \log z = 4 \text{Var}(j) > 0$. Its aspect tends to infinity as $z \uparrow 1$. Differentiating (EG10) gives $J'(t) = -\frac{1}{2} \log z_t > 0$, $J''(t) < 0$. Coefficientwise, $4F' \geq F$ and $4(1 - z)F' \leq F$. Thus $z \leq t(z) \leq z/(1 - z)$, and for $0 < t \leq 1$,

$$\frac{t}{2}(1 - \log t) \leq J(t) \leq \frac{(1 + t) \log(1 + t) - t \log t}{2}. \quad (\text{EG12})$$

In particular $nJ(1/n) \leq 1 + \frac{1}{2} \log n$. Concavity also gives $|J(s) - J(t)| \leq J(|s - t|)$.

3. Harmonic measure gives the finite lower minimum

The harmonic measure of $[a, b]$ at zero has density

$$\omega'_0(x) = \frac{uv}{\pi x \sqrt{(b - x)(x - a)}}, \quad g = \log \frac{u + v}{v - u}.$$

The conformal map $x = (a + b)/2 + (b - a)(\zeta + \zeta^{-1})/4$ sends $|\zeta| > 1$ onto the interval's exterior, and the preimage of zero is $\zeta_0 = -(u + v)/(v - u)$. For $\deg R \leq h$, $R(0) = 1$, the function $\zeta^{-h} R(x(\zeta))$ is analytic at infinity. Poisson-Jensen, or the subharmonic mean inequality there, gives

$$\int \log |R(x)| d\omega_0(x) \geq -hg. \quad (\text{EG13})$$

For the degree-one polynomial $x - s$, $s \in [a, b]$, there is no exterior zero. Its equality case gives $\int \log |x - s| d\omega_0 = \log s - g$, including endpoints by integrable limits. Average the Euler equality first in x against ω_0 , then in s against μ_t . It follows that $\int V_t d\omega_0 + 2g = \lambda_t + 2L_t$. Jensen with the positive density ω'_0 now proves for every admissible R

$$\int_a^b |R|^2 e^{-hV_t} x^{-3/4} dx \geq Z_t e^{-h(\lambda_t + 2L_t)}, \quad Z_t = \pi c_0 \sqrt{p} w_0^{-5/2}. \quad (\text{EG14})$$

Here the entropy constant has been evaluated, rather than bounded by an unspecified quantity. To check it, the same conformal map gives $\int \log x d\omega_0 = 2 \log p - 2 \log w_0$. Applying (EG13)'s root equality at a, b gives the endpoint logarithms $2 \log u - g$, $2 \log v - g$. Substitute these three values into $\int \log(x^{-3/4}/\omega'_0) d\omega_0$; since $e^{-g} = c_0/w_0^2$, the answer is $\log Z_t$.

4. A trial polynomial controls the complete upper minimum

Let $x_1, \dots, x_h$ be the midpoint quantiles of μ_t , let $P_h(x) = \prod_{i=1}^h (x - x_i)$, and put $D_t = W \|\rho_t\|_\infty$. Their empirical distribution has CDF error at most $1/(2h)$. For $0 \leq x \leq B$, $B \geq 2b + 1$, truncate $\log |x - y|$ below $\log(W/h)$. As a function of $y \in [a, b]$ its total variation is at most $2 \log(hB/W)$. Integration by parts against the CDF difference bounds the discrepancy of its average by $h^{-1} \log(hB/W)$. Its integral excess over the untruncated logarithm is at most $2 \|\rho_t\|_\infty \int_0^{W/h} \log((W/h)/s) ds = 2D_t/h$. Since the discrete untruncated logarithm is no larger than its truncation, these observations give

$$\log |P_h(x)| \leq h \int \log |x - y| d\mu_t(y) + \log(hB/W) + 2D_t.$$

The bounded variation of $\log y$ gives $|P_h(0)|^2 \geq (a/b)e^{2hL_t}$. Therefore the admissible polynomial $R = P_h/P_h(0)$, together with the full Euler inequality, has its integral over $(0, B]$ bounded by the first term of

$$U_t(h) = \frac{b}{a} \left[4B^{1/4}(hB/W)^2 e^{4D_t} + 2\sqrt{\frac{2t}{h}} \right]. \quad (\text{EG15})$$

For the second term choose B so that

$$(2/t + 2) \log x - \frac{\pi}{2t} \sqrt{x} \leq -\lambda_t \quad (x \geq B). \quad (\text{EG16})$$

All roots of P_h are positive and less than $B/2$, so $|P_h(x)| \leq x^h$ on this tail. The resulting integrand is bounded by $(b/a)e^{-h(\lambda_t+2L_t)}x^{-3/4}e^{-h\pi\sqrt{x}/(2t)}$. Its integral over $(0, \infty)$, an upper bound for the actual tail, is exactly $2\sqrt{2t/h}$. Thus

$$\int_0^\infty |R|^2 e^{-hV_t} x^{-3/4} dx \leq U_t(h) e^{-h(\lambda_t+2L_t)}. \quad (\text{EG17})$$

This proof includes the original interval next to zero and the unbounded tail.

5. Uniform constants down to the first nonzero degree

The exact rational certificate in the companion checker verifies

$$\sum_{j=0}^{12} (4j - 11/10) a_j (11/20)^j > 3/10000, \quad \sum_{j=1}^{12} j a_j (11/20)^{j-1} > 3/5,$$

and

$$\sum_{j=0}^{10} (4j - 3/2) a_j (16/25)^j > 1/100, \quad \sum_{j=1}^{10} j a_j (16/25)^{j-1} > 3/4.$$

All omitted terms in each comparison are positive. Monotonicity of $t(z)$ shows $z_t < 11/20$ for $t \leq 11/10$, and $z_t < 16/25$ for $t \leq 3/2$.

The integral formula for F gives $1 \leq F(z) \leq (1-z)^{-1/2}$. By (EG11), on the larger aspect interval, $u > 1/3$, $v < 20/3$, and hence

$$1/16 < a < b < 45, \quad w_0 < 7, \quad \sqrt{p} > 1/4.$$

The exact width is

$$W = \frac{64\sqrt{z}}{\pi^2(1-z)^2 F(z)^2} \geq \frac{4\sqrt{z}}{1-z} \geq 4\sqrt{t}. \quad (\text{EG18})$$

The last comparison uses $t \leq z/(1-z)$; it proves in particular the weaker supplied bound $W \geq 4\sqrt{t/3}$. Since $c_0 = W/4$, equation (EG14) gives $Z_t \geq \sqrt{t}/2048$. For example use $c_0 \geq \sqrt{t/3}$, $\sqrt{p} > 1/4$, $w_0 < 7$, $\pi > 3$, $\sqrt{3} < 2$, and $\sqrt{7} < 3$; these give the stronger floor $Z_t > \sqrt{t}/392$.

The density (EG8) gives a uniform bound without multiplying two unnecessary endpoint extremes. Specifically,

$$\max_{a < x < b} \frac{\sqrt{(b-x)(x-a)}}{x} = \frac{W}{2uv}, \quad \int_0^\infty \frac{\sqrt{s}}{(a+s)^2} ds = \frac{\pi}{2u}.$$

They yield

$$D_t \leq \frac{W^2}{16ta\sqrt{b}} = \frac{1 + \sqrt{z}}{\pi(1-z)^3 F'(z)} =: D_*(z). \quad (\text{EG19})$$

Differentiate the positive integral for F' : $F''/F' \leq 3/[2(1-z)]$. Therefore $(\log D_*)' > 0$. At $z = 11/20$, the previous rational lower bound for F' , together with $\sqrt{z} < 3/4$ and $\pi > 3$, gives $D_t < 70000/6561 < 11$. At $z = 16/25$ it gives $D_t < 12500/729 < 18$.

Take $B = 2^{16}$. From (EG9), $w_0 > 1/4$, $c_0 \geq \sqrt{t/3}$, and $t \leq 3/2$,

$$t\lambda_t \leq 4t + 4 + 4\log 4 - t\log(t/3) < 20.$$

At $x = B$, $(2 + 2t)\log x - (\pi/2)\sqrt{x} < -20$, and its derivative is negative thereafter; $5/x < \pi/(4\sqrt{x})$ suffices. This proves (EG16) uniformly. Since $b/a < 720$, $h = tn/2$, and $W^2 \geq 16t/3$, equation (EG15) gives

$$U_t(h) \leq 5760B^{9/4}e^{4D_*n^2}, \quad D_* = 11 \text{ for } t \leq 11/10, \quad D_* = 18 \text{ for } t \leq 3/2. \quad (\text{EG20})$$

Here is an explicit check of this deliberately generous constant. The first term is at most $720 \cdot 4 \cdot 3t/64 B^{9/4} e^{4D_*} n^2 \leq 203 B^{9/4} e^{4D_*} n^2$. The second term is $2880/\sqrt{n}$, at most $B^{9/4} e^{4D_*} n^2$ for $n \geq 40$. Their sum is smaller than (EG20).

Since $n\sqrt{a} > 1$, use the lower envelope (EG3) on the complete equilibrium interval in (EG7); use its upper envelope on the whole positive half-line. Equations (EG14) and (EG17) prove the finite sandwich

$$\boxed{\frac{1}{2} Z_t n^{2n+1/2} e^{-h(\lambda_t+2L_t)} \leq K_{n,2h}(0,0)^{-1} \leq 32 U_t(h) n^{2n+1/2} e^{-h(\lambda_t+2L_t)}} \quad (\text{EG21})$$

Subtract the exact profile (EG11) and insert the complete mass (EG5). For even $M = 2h \geq 2$,

$$-\frac{1}{16n} - \frac{1}{2} \log n - \log(32 U_t(h)) \leq \log[d_n K_{n,M}(0,0)] - nJ(t) \leq 5 - \frac{1}{2} \log n - \log(Z_t/2). \quad (\text{EG22})$$

In the upper bound, $nt = 2h \geq 2$, so the right side is at most $5 + (23/2) \log 2 < 13$. The negative magnitude is at most

$$A(D_*) + \frac{5}{2} \log n, \quad A(D_*) = \log 184320 + 36 \log 2 + 4D_* + 1/640.$$

It dominates 13. At odd degree use exact parity and (EG12); this adds at most $1 + \frac{1}{2} \log n$. The elementary certified estimates $\log 184320 < 97/8$ and $\log 2 < 6932/10000$ imply

$$A(11) + 1 < 83, \quad A(18) + 1 < 111.$$

For completeness, reduce $184320 = 2^{17}(45/32)$, and certify each logarithm with $\log x = 2 \sum_{j=0}^{N-1} r^{2j+1}/(2j+1) + \varepsilon_N$, $r = (x-1)/(x+1)$, $0 < \varepsilon_N < 2r^{2N+1}/[(2N+1)(1-r^2)]$. The companion uses rational arithmetic with $N = 40$. Degree zero is exact, and degree one follows from parity directly. We have proved

$$\boxed{|\log[d_n K_{n,M}(0,0)] - nJ(M/n)| \leq 83 + 3 \log(n+2), \quad 0 \leq M \leq \lfloor 11n/10 \rfloor, \quad n \geq 40,} \quad (\text{EG23})$$

$$\boxed{|\log[d_n K_{n,M}(0,0)] - nJ(M/n)| \leq 111 + 3 \log(n+2), \quad 0 \leq M \leq \lfloor 3n/2 \rfloor, \quad n \geq 40.} \quad (\text{EG24})$$

Thus the incoming constants 110 and 150 both hold. The stronger bounds improve the explicit finite remainder; the profile is unchanged. In particular the actual PP28 remainder is at most $\lfloor 111 + 3 \log(n+2) \rfloor/n$, with no unevaluated zero-statistic term.

6. Complete coefficient jets and both low cutoffs

Now $q \geq 40$, $1 \leq b \leq q/10$, $n = q - b$, and

$$J_{q,b} : \mathcal{P}_M \longrightarrow \mathbb{C}^b, \quad (J_{q,b} P)_r = q^r [y^r] P, \quad M = N - n \geq b - 1.$$

Let $H_{b,N}^{\text{jet}}$ denote its attained minimum metric in ν_n . Multiplication by y^n maps the entire fibre bijectively onto the original high coefficient band in the power quotient modulo y^q , preserving degree N and norm exactly:

$$P \longmapsto y^n P, \quad \|y^n P\|_\sigma = \|P\|_{\nu_n}, \quad [y^n P] \bmod y^q = y^n \sum_{r < b} (J_{q,b} P)_r (y/q)^r. \quad (\text{EG25})$$

This is the full coefficient map, not a map only on diagonal norms.

Set $\alpha = 2n - 1/2$ and

$$\gamma_r = \frac{r!(\alpha+1)_r}{(\theta q)^{2r}}, \quad R_b = (b+1)3^b, \quad \mathcal{L} = 128 R_b^2 \max\{\max_{r \leq b} \gamma_r, (\min_{r \leq b} \gamma_r)^{-1}\}. \quad (\text{EG26})$$

The monic Laguerre basis in $x = y/q$ has coefficient matrix $T_{ir} = (-1)^{r-i} \binom{r}{i} (\alpha + i + 1)_{r-i} / (\theta q)^{r-i}$. Its inverse has the same entries without the sign. Indeed in a product entry the rising factorials combine to a constant $(\alpha + i + 1)_{r-i}$, and the remaining binomial sum is $(1-1)^{r-i}$. Each factor in those entries is less than 2, so both Euclidean operator norms are at most R_b . The orthogonal squared norms divided by the positive half-line mass are exactly γ_r . On the negative half-line, the coefficient map is the diagonal unitary $a_r \mapsto (-1)^r a_r$; it obeys the same bounds. Summing both halves and using (EG4) for both the polynomial and the constant proves

$$\mathcal{L}^{-1} d_n I_b \preceq H_{b,q-1}^{\text{jet}}, \quad H_{b,q}^{\text{jet}} \preceq \mathcal{L} d_n I_b. \quad (\text{EG27})$$

For $N = q$, the last coefficient is freely minimized: the lower estimate survives since the full coefficient norm is no smaller than the prescribed jet norm, and the zero last coefficient is an admissible lift for the upper estimate.

7. Every row of the high covariance and one simultaneous right inverse

For $j \geq 1$, $N_j = q - 1 + j$, $M = j + b - 1$, $L = j - 1$. Let p_r be the orthonormal polynomials for the original ν_n , and let $v_r = J_{q,b} p_r$. The complete covariance is

$$C_M = \sum_{r=0}^M v_r v_r^* = (H_{b,N_j}^{\text{jet}})^{-1}. \quad (\text{EG28})$$

The equality follows by the exact minimizing section $J^*(JJ^*)^{-1}$; its difference from any other lift is orthogonal to its range.

All nonzero roots of the used orthogonal polynomials have modulus at least cn . To verify this directly, push the even measure through $x = y^2$. Even polynomials are orthogonal for its positive pushforward ρ_n , and odd polynomials have the form $yS(y^2)$ with S orthogonal for $x\rho_n$. Their squared nonzero roots are eigenvalues of the corresponding finite multiplication-by- x compression. The characteristic determinant and monic orthogonal polynomial satisfy the same three-term recurrence, proving this identification. Its Rayleigh quotients are $\|yP\|^2/\|P\|^2$; (EG6) puts them between c^2n^2 and C^2n^2 . The additional odd root at zero is retained separately. All source degrees used here are at most $q + 2b \leq 2n + 1$.

Put $B_0 = q/(cn)$, $D_0 = \max(1, Cn/q)$, and define exactly

$$U = \sum_{2h < b} \binom{\lfloor (q+b)/2 \rfloor}{h}^2 B_0^{4h} + B_0^2 \sum_{2h+1 < b} \binom{\lfloor (q+b-1)/2 \rfloor}{h}^2 B_0^{4h},$$

$$V = \sum_{h=0}^{\lfloor (b-1)/2 \rfloor} \binom{\lfloor q/2 \rfloor + h - 1}{h} B_0^{2h}, \quad L_* = bV^2 D_0^{2(b-1)}. \quad (\text{EG29})$$

The letter L_* here is a finite jet constant, distinct from the logarithmic moment L_t . Factor every even orthogonal polynomial as $p(0) \prod (1 - y^2/a_i^2)$. Its $2h$ -th jet coefficient is bounded by $|p(0)| \binom{\lfloor M/2 \rfloor}{h} B_0^{2h}$. For the odd sector, $P = yQ$ and (EG6) give

$$\sup_{P \text{ odd, deg } P \leq M} \frac{|P'(0)|^2}{\|P\|^2} \leq (cn)^{-2} K_{n,M}(0,0).$$

Apply the same paired-root estimate to its remaining coefficients and sum all row norms. It yields $\text{Tr } C_M \leq U K_{n,M}(0,0)$, and positive semidefiniteness gives the full matrix upper bound.

For the opposite direction define $H_L(y) = K_{n,L}(y,0)/K_{n,L}(0,0)$. It has $H_L(0) = 1$, norm squared $1/K_{n,L}(0,0)$, and all its zeros, if any, have modulus at least cn . Reproduction gives $\langle H_L, y^{2r} \rangle = 0$ for $1 \leq r \leq \lfloor L/2 \rfloor$; its pushforward is therefore the degree- $\lfloor L/2 \rfloor$ orthogonal polynomial for $x\rho_n$, proving that root statement by the same compression. Complete homogeneous coefficients then bound the sum of the first b coefficients of $1/H_L(qx)$ by V . For every $a \in \mathbb{C}^b$ put

$$\mathcal{R}a(y) = H_L(y) \left[\frac{\sum_{r < b} a_r x^r}{H_L(qx)} \right]_{x=y/q}. \quad (\text{EG30})$$

This is linear and $J_{q,b} \mathcal{R}a = a$ exactly. Its degree is at most $L + b - 1 \leq M$, including $L = 0$, where $H_L = 1$. Coefficient convolution gives a coefficient-sum bound $\sqrt{b}V\|a\|_2$. Repeated (EG6), with each raising degree included, gives $\|\mathcal{R}a\|^2 \leq L_* \|a\|_2^2 / K_{n,L}(0,0)$. Taking the actual minimum and inverting proves

$$\boxed{\frac{K_{n,L}(0,0)}{L_*} I_b \preceq (H_{b,N_j}^{\text{jet}})^{-1} \preceq U K_{n,M}(0,0) I_b, \quad 1 \leq j \leq q+1.} \quad (\text{EG31})$$

All complex cross-covariances are controlled by the trace estimate and by this simultaneous section.

8. Finite every-vector rate and all reference cutoffs

For $0 \leq h \leq q + b$, applying (EG6) successively to $P, yP, \dots, y^{b-1}P$, with the same weight ν_n , proves $(cn)^{2b} \|P\|_{\nu_n}^2 \leq \|P\|_{\nu_q}^2 \leq (Cn)^{2b} \|P\|_{\nu_n}^2$. Every degree is at most $q + 2b \leq 2n + 1$. Apply the same inequalities to the constant polynomial and to the evaluation minimum; division is done only after both original masses are retained. Thus

$$|\log[d_n K_{n,h}(0,0)] - \log[d_q K_{q,h}(0,0)]| \leq 2b \log(C/c). \quad (\text{EG32})$$

This avoids applying (EG23) at $n = q - b < 40$; it is applied at the original $q \geq 40$.

Define the explicit error

$$\mathcal{J} = \log \mathcal{L} + \max(\log L_*, \log U) + 2b \log(C/c) + 83 + 3 \log(q+2) + qJ((b+1)/q). \quad (\text{EG33})$$

Equations (EG27), (EG31), (EG32), and (EG23), with $M \leq q + b \leq 11q/10$, give for each eigenvalue λ of $(H_{b,N_j}^{\text{jet}})^{-1/2} H_{b,q-1}^{\text{jet}} (H_{b,N_j}^{\text{jet}})^{-1/2}$,

$$|\log \lambda - qJ(j/q)| \leq \mathcal{J}, \quad 0 \leq j \leq q+1. \quad (\text{EG34})$$

At $j = 0$ the matrix is exactly the identity. For $j \geq 1$, use $L = j - 1$, $M = j + b - 1$, and $|J(M/q) - J(j/q)|, |J(L/q) - J(j/q)| \leq J((b+1)/q)$. The same bound holds with low metric $H_{b,q}^{\text{jet}}$ for $j \geq 1$. This states the reference range explicitly; the proof of (EG31) is not applied with $L = -1$.

Equivalently, for every $a \in \mathbb{C}^b$,

$$e^{-qJ(j/q)-\mathcal{J}} a^* H_{b,q-1}^{\text{jet}} a \leq a^* H_{b,N_j}^{\text{jet}} a \leq e^{-qJ(j/q)+\mathcal{J}} a^* H_{b,q-1}^{\text{jet}} a. \quad (\text{EG35})$$

Comparing these two inequalities for any pair $0 \leq i, j \leq q+1$ gives the complete finite pair estimate: every eigenvalue of $(H_{b,N_j}^{\text{jet}})^{-1/2} H_{b,N_i}^{\text{jet}} (H_{b,N_j}^{\text{jet}})^{-1/2}$ has logarithm within $2\mathcal{J}$ of $q[J(j/q) - J(i/q)]$. This includes either low cutoff in either order, without commuting two Hermitian forms.

Finally the displayed finite sums prove $\mathcal{J} = O(b \log(C_1 q/b) + b + \log(q+2))$, for an absolute constant C_1 . Indeed $\binom{N}{r} \leq (eN/r)^r$, the factors B_0, D_0 are bounded independently of q, b , and $r! \geq (r/e)^r$ bounds the inverse γ_r . The endpoint term is at most a constant times $(b+1)[1 + \log(q/(b+1))]$ by (EG12). The exact finite error (EG33), rather than this order notation, is available for every allowed pair.

Sources, scope, and reproducibility

The human references actually used are R. A. Askey and R. Roy, DLMF Chapter5, Gamma Function; T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, DLMF Chapter18, Orthogonal Polynomials, equations 18.5.12, 18.9.13, 18.9.23; and B. C. Carlson, DLMF Chapter19, 19.8.12. The original equation TeX files were obtained and read. The retained programme EIQ/EL proofs provide the full positive equilibrium and exact logarithmic moment; the present finite harmonic-measure/quantile sandwich is independently proved in Sections3–5.

`check_effective_gamma.py` certifies the rational series comparisons and constant arithmetic, the entropy and density identities, and exact full-covariance/right-inverse examples. It uses no native-period surrogate and asserts no individual arithmetic-current sign. The physical action, the original arithmetic roots, and the full invariant constraints enter through the unchanged original source maps when the root task receives (EG25) and (EG34)–(EG35).

Part III

Effective original localization and every fixed subquotient

Effective localization in the full original kernel and word image

Independent derivation, 22 September 2026. The contour calculation, coefficient-to-source comparison, complete polynomial fibres, and relative-spectrum loss are proved here with finite constants. In particular the exceptional-direction contribution improves the incoming LC13. The companion scalar proof supplies its proved bound with constant 83; this value is carried through every original-metric estimate below.

1. Original data, source constants, and the finite scalar input

Fix the stipulated original simple quartet, its admitted period, branch and arithmetic unit. Retain

$$k \equiv 1 \pmod{4}, \quad k \geq 17, \quad q = (k+1)^2, \quad q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48, \quad m = 8k - 16,$$

$$\omega_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad Q_k(y) = \prod_{a,b=0}^k (y - \omega_{ab}), \quad E_k = \mathbb{C}[y]/Q_k. \quad (\text{LX1})$$

The physical coordinate is still $s = k/2 + iy$; multiplication is $M[p] = [yp]$, so its action is $kI/2 + iM$. The onto observation is the complete original Λ_k , and I_K is the unchanged coefficient inclusion of its kernel K_k , of rank m . The complete attained source metric at degree N is G_N , and $H_{K,N} = I_K^* G_N I_K$. The cutoffs are

$$N_j = q - 1 + j, \quad 0 \leq j \leq q + 1, \quad t_j = j/q. \quad (\text{LX2})$$

Thus the four original signs are at $j = 0, 1, q, q + 1$. In particular the degree $2q$ endpoint is retained.

Use the actual comparison source

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad M_\sigma = \sqrt{2\pi}.$$

The original polynomial source comparison, before either restriction or minimization, is

$$\ell_k \|P\|_\sigma^2 \leq \|P\|_{\mu_k}^2 \leq u_k \|P\|_\sigma^2, \quad \deg P \leq 2q, \quad \kappa_k = u_k / \ell_k. \quad (\text{LX3})$$

Here these are the original finite constants, not newly selected matrix widths. Their source formula is $u_k = A_k$, $\ell_k = a_k / [1 + 4(2q + 25)^2]^{25}$ on this simple-quartet domain. In the retained full source proof, writing $\alpha = \pi/2$, $p = 7/2$, $B_h = 50$,

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-50}}{C_\Gamma^{\text{src}} 2^{25}}, \quad A_k = \frac{C_h^{\text{tilt}}}{C_\Gamma^{\text{src}}} k^{9/2} M_\alpha^{k-1}.$$

All constants $c_h, C_h^{\text{tilt}}, \vartheta_h, M_\alpha, c_\Gamma^{\text{src}}, C_\Gamma^{\text{src}}$ retain their original fixed-source definitions. Thus $\log \kappa_k = O_h(k + \log(q + 2))$. To see the polynomial step explicitly, its density bounds give $a_k(1 + y^2)^{-25} d\sigma \leq d\mu_k \leq A_k d\sigma$. The exact Gamma recurrence gives $\|yP\|_\sigma \leq 2(N + 1)\|P\|_\sigma$; expanding $(1 + y^2)^{25}$ and applying that bound through all successively raised degrees gives the factor $[1 + 4(N + 25)^2]^{25}$. Cauchy-Schwarz applied to the reciprocal two weights proves the lower bound. These are the same constants as the retained AMT4-5 full-source proof, and HAR40 retains their absolute mass as well as their ratio.

Let $d\nu_n = |y|^{2n} d\sigma$, $d_n = \nu_n(\mathbb{R})$, and let $K_{n,M}$ be the full degree- M reproducing kernel for this measure. Write J for the original profile (incoming β), with the proved exact dictionary

$$t = 4zF'(z)/F(z), \quad F(z) = \frac{2}{\pi} K(\sqrt{z}), \quad J(t) = 2 \log F(z) - \frac{t}{2} \log z, \quad J(0) = 0. \quad (\text{LX4})$$

It is increasing and concave; $J(1) = C_\partial/2$. These statements, including its exact original-root dictionary, are proved in 025 EP1-17.

The companion *Effective Gamma and jet proof*, EG23, proves the following explicit bound, improving the incoming constant 110:

$$\max_{0 \leq M \leq \lfloor 11q/10 \rfloor} |\log[d_q K_{q,M}(0,0)] - qJ(M/q)| \leq E_G(q) := 83 + 3\log(q+2), \quad q \geq 40. \quad (\text{LX5})$$

Its proof retains the full Gamma measure, both half-lines, the exact mass and all integer degrees, including zero and odd degree. The error is now explicit. In the present original domain $q \geq 324$, so this scalar lower guard always holds.

2. Complete coefficient-to-quotient metric comparison

Put $\rho = k\sqrt{\delta^2 + \gamma^2}/q$ and $\|p\|_{c,q} = \sum_j q^j |y^j| p$. We prove, for $q \geq 40$ and $\rho \leq 1/2$,

$$e^{-10q} \|p\|_{c,q} \leq \|p\|_{G_N^\sigma} \leq e^{2q} \|p\|_{c,q}, \quad \deg p < q, \quad q-1 \leq N \leq 2q. \quad (\text{LX6})$$

First take any polynomial P of degree $d \leq 2q$, and write $p(x) = P(qx)$. The orthonormal basis on $[1, 2]$ is $\sqrt{2j+1} P_j(2x-3)$. Its coefficient sum is $\sqrt{2j+1} P_j(5) \leq \sqrt{2j+1} 10^j$: the alternating coefficients after the shift have absolute sum equal to evaluation at -1 , and the generating function or three-term recurrence gives $P_j(5) \leq 10^j$. Cauchy-Schwarz therefore proves

$$\|p\|_{\text{coefficients},1} \leq (d+1)10^d \|p\|_{L^2[1,2]}. \quad (\text{LX7})$$

The Gamma lower envelope $w(y) \geq \frac{1}{2}y^{-1/2}e^{-\pi y/2}$, $y \geq 1$, gives

$$\|P\|_\sigma^2 \geq \frac{\sqrt{q}}{2\sqrt{2}} e^{-\pi q} \|p\|_{L^2[1,2]}^2.$$

For $q \geq 40, d \leq 2q$, combining this with (LX7) gives $\|P\|_\sigma \geq e^{-8q} \|P\|_{c,q}$. Indeed the negative logarithm of the displayed constant is at most $\pi q/2 + 2q \log 10 + \log(2q+1) - \frac{1}{4} \log q + \frac{3}{4} \log 2 < 8q$.

Every root $\alpha_i = \omega_i/q$ of the scaled monic polynomial has modulus at most ρ . The complete Newton expansion of the remainder of x^j modulo that polynomial has coefficient norm at most

$$\sum_{h=0}^{\min(j,q-1)} \binom{j}{h} \rho^{j-h} (1+\rho)^h \leq (1+2\rho)^j \leq 2^{2q} \quad (j \leq 2q). \quad (\text{LX8})$$

This follows because the divided difference is the complete homogeneous polynomial of degree $j-h$ in the first $h+1$ roots, with $\binom{j}{h}$ terms; the Newton product has norm at most $(1+\rho)^h$. Apply (LX8) to every coefficient of every representative $P \equiv p \pmod{Q_k}$. Then $\|p\|_{c,q} \leq 2^{2q} \|P\|_{c,q}$, so minimizing the preceding lower bound proves the first half of (LX6).

For the upper half retain the canonical degree-below- q representative. The exact Gamma recurrence, including its last raising row, gives $\|y^j\|_\sigma \leq \sqrt{M_\sigma} 2^j j!$. For $j < q$, this is at most $e^{2q} q^j$. Triangle inequality and the attained minimum prove the other half. Thus the finite coefficient transfer applies to complete quotient fibres, not merely to their canonical representatives.

Source nesting and (LX3),(LX6) give the particularly useful global bound

$$0 \leq \log \lambda_a(H_{K,N_j}^{-1/2} H_{K,N_i} H_{K,N_j}^{-1/2}) \leq B_{\text{all}}(k) := 24q + \log \kappa_k \quad (i \leq j). \quad (\text{LX9})$$

The constant 24 is $2(10+2)$; the incoming 40 is unnecessary. The same bound holds on any fixed restriction or attained quotient: apply the form bounds before taking its complete minimum.

3. An explicit circle and the entire residue operator

The actual conductor is $\mathcal{T} = \sum_s a_s e^{\sigma_s D} = D^{v_0} b(D)$, where

$$\mathcal{E}(z) = \sum_s a_s e^{\sigma_s z} = z^{v_0} b(z), \quad b(0) = i^{-v_0} \mu_{v_0}/v_0! \neq 0, \quad B_A = \max(1, \max_s |\sigma_s|), \quad A_b = \frac{(\sum_s |a_s|) B_A^{v_0}}{|\mu_{v_0}|} \geq 1.$$

Taylor's integral remainder proves $|b(z)/b(0)| \leq A_b e^{B_A |z|}$; it is also valid when $v_0 = 0$. For $R \geq 2$, put

$$N_R = \left\lceil \frac{\log A_b + 12 B_A R}{\log 2} \right\rceil, \quad L_R = \log A_b + 6 B_A R, \quad \mathcal{M}_R = |b(0)|^{-1} e^{2L_R} [32(N_R+1)]^{N_R}. \quad (\text{LX10})$$

There is $r \in [R, 2R]$, with no zero on its circle, at most N_R zeros inside it counted with multiplicity, and

$$\max_{|z|=r} |b(z)^{-1}| \leq \mathcal{M}_R. \quad (\text{LX11})$$

Here are the details behind the classical zero-count and harmonic tools in this use. The circular mean of $\log |z-a|$ at radius T equals $\log \max(T, |a|)$, by the mean value property outside its zero and the convergent logarithmic series otherwise. Factoring the zeros and applying the mean value property to the remaining zero-free function gives the Jensen formula. At $T = 12R$, every zero of modulus at most $6R$ contributes at least $\log 2$; this proves the bound N_R , with limits handling a zero on the outer circle. Choose a zero-free $S \in [4R, 6R]$, and factor these zeros by the full disk Blaschke product $B(z) = \prod e^{i\theta_a} S(z-a)/(S^2 - \bar{a}z)$, including repetitions. Choose phases so $B(0) > 0$. The zero-free quotient $h = b/(b(0)B)$ has $\log |h| \leq L_R$ on $|z| = S$. The positive harmonic function $L_R - \log |h|$ has its value at zero at most L_R , because $|h(0)| = 1/|B(0)| \geq 1$. The Poisson kernel ratio is at most $(S+2R)/(S-2R) \leq 3$, so $\log |h(z)| \geq -2L_R$ for $|z| \leq 2R$.

Remove from $[R, 2R]$ the intervals of radius $R/[4(N_R+1)]$ around all zero moduli in $|z| < S$. Their total length is less than $R/2$. At any remaining radius, each Blaschke factor has modulus at least $|z-a|/(S+|z|) \geq 1/[32(N_R+1)]$. Multiplying them proves (LX11). This includes every repeated zero. Finding a numerical such circle for an unspecified period is not claimed.

The operator $D = \partial_y$ on degree-below- q polynomials has $\|D\|_{c,q} \leq 1$. Direct calculation on monomials gives

$$(z-D)^{-1}f = E_{q,z}\lambda_{q,z}(f) - R_z f, \quad E_{q,z} = \sum_{j < q} (zy)^j/j!, \quad \lambda_{q,z}(f) = \sum_{n < q} n! [y^n] f z^{-n-1},$$

$$R_z f = \left[e^{zy} \int_0^y e^{-zt} f(t) dt \right]_{< q}, \quad [y^{n+j}] R_z y^n = \frac{n! z^{j-1}}{(n+j)!} \quad (j \geq 1). \quad (\text{LX12})$$

There is no term of degree at most n in $R_z y^n$. Let

$$A_r = \frac{1}{2\pi i} \oint_{|z|=r} \frac{(z-D)^{-1}}{b(z)} dz, \quad B_r = \frac{1}{2\pi i} \oint_{|z|=r} \frac{R_z}{b(z)} dz,$$

$$L_r f = \sum_{0 < |a| < r} \text{Res}_{z=a} \frac{E_{q,z}\lambda_{q,z}(f)}{b(z)}.$$

The residue at zero is the full nilpotent functional calculus $b(D)^{-1}$; the nonzero residues give

$$b(D)^{-1} = A_r + B_r - L_r, \quad A_r = \sum_{j < q} \alpha_j D^j, \quad |\alpha_j| \leq \mathcal{M}_R r^{-j}, \quad \text{rank } L_r \leq N_R. \quad (\text{LX13})$$

At a pole of order a , its entire range is spanned by $E_{q,z}, \partial_z E_{q,z}, \dots, \partial_z^{a-1} E_{q,z}$ evaluated at that pole: Leibniz's rule puts all remaining derivatives on the scalar functional. This proves the rank count even at multiple poles.

The degree-raising part admits the improved full finite bound

$$\|B_r\|_{c,q} \leq \mathcal{M}_R (6R)^q. \quad (\text{LX14})$$

Indeed (LX12) and $(n+j)!/n! \geq j!$ give $\|R_z\|_{c,q} \leq r^{-1} \sum_{j=1}^{q-1} (qr)^j/j!$. Backward successive ratios in this sum are less than $1/r$, so the sum is at most $r/(r-1)$ times its last term. Using $(q-1)! \geq ((q-1)/e)^{q-1}$ and $(q/(q-1))^{q-1} < e$, it is less than $(er)^q/(r-1) \leq (3r)^q$. The contour length factor is exactly $r\mathcal{M}_R$; the factor r^{-1} just displayed cancels it. Since $r \leq 2R$, (LX14) follows. The finite truncation is essential: its logarithmic cost is $q \log R$, not Rq .

4. The original invariant kernel, all constraints retained

The exact conductor implication is

$$p \in K_k \implies f = \mathcal{T}p = Q_{k-8\eta}, \quad \deg \eta < \Delta - v_0. \quad (\text{LX15})$$

It follows from 024 CK1-7. We use only this implication; every additional invariant row stays in the domain $K_k = \ker \Lambda_k$.

Set $f_0 = D_0^{-v_0} f$, the primitive with its first v_0 coefficients zero. Then $p = b(D)^{-1} f_0 + l(p)$, $\deg l < v_0$. The single cutoff-independent subspace

$$K_{k,R} = \{p \in K_k : l(p) - L_r f_0 = 0\}, \quad c_{\text{loc}} = \min(m, v_0 + N_R) \quad (\text{LX16})$$

has codimension at most c_{loc} , by rank-nullity after restriction to the full invariant kernel. No arithmetic observation row is discarded. On this subspace $p = (A_r + B_r)f_0$.

Take a positive integer b divisible by eight and impose

$$8\Delta + 8 \leq b \leq q/10, \quad q \geq 40, \quad 0 < \rho < 1/2. \quad (\text{LX17})$$

The lower-root polynomial, including every one of its roots, gives

$$\|P_{< q' - h} f\|_{c,q} \leq (1 - \rho)^{-q'} 2^{q'} \rho^h \|f\|_{c,q}. \quad (\text{LX18})$$

To verify this, the coefficient at depth j of its monic reversed polynomial is at most $\binom{q'}{j} \rho^j$. Its full inverse leading block has column sum bounded by $\sum_{j \geq 0} \binom{q' + j - 1}{j} \rho^j = (1 - \rho)^{-q'}$. Multiply these bounds. Also $\|f\|_{c,q} \leq M_T \|p\|_{c,q}$, where $M_T = \sum_s |a_s| e^{|\sigma_s|}$, and $\|f_0\|_{c,q} \leq q^{v_0} \|f\|_{c,q}$.

Define the complete finite errors

$$e_p = q^{v_0} M_T \mathcal{M}_R \left[(2 + (6R)^q) (1 - \rho)^{-q'} 2^{q'} \rho^{b/8} + 2R^{-b/4} \right],$$

$$e_U = (1 - \rho)^{-q} e_p + \frac{4^q \rho^{b/2}}{1 - \rho}, \quad \zeta = e^{12q} (1 + \rho)^q e_U. \quad (\text{LX19})$$

The part of f_0 below degree $q - b/4$ obeys (LX18) with $h = b/8$, since $b/8 + v_0 \geq \Delta$. The remaining part must lose at least $b/4$ degrees through A_r to reach degree below $q - b/2$; the geometric derivative tail is at most $2\mathcal{M}_R R^{-b/4}$. The operator B_r raises degree and therefore uses only the small low input there. Equations (LX13)–(LX14) prove $\|P_{< q - b/2} p\|_{c,q} \leq e_p \|p\|_{c,q}$.

Retain the actual polynomial-part map

$$U_Q F = [Q_k(y) y^{-q} F(y)]_+, \quad U_Q^{-1} p = [y^q p(y) / Q_k(y)]_+. \quad (\text{LX20})$$

On degree-below- q representatives its norm is at most $(1 + \rho)^q$, its inverse at most $(1 - \rho)^{-q}$, and its inverse tail after depth $b/2$ at most $4^q \rho^{b/2} / (1 - \rho)$. The first two assertions are the elementary and complete homogeneous coefficient sums. For the last, use $\binom{q+j-1}{j} \leq 4^q$ for $0 \leq j < q$, and sum the remaining geometric series. Thus the fixed map

$$\mathcal{A}_{k,b,R} : K_{k,R} \longrightarrow y^{q-b} \mathcal{P}_{< b}, \quad \mathcal{A}_{k,b,R} p = P_{\geq q-b} U_Q^{-1} p$$

satisfies, at every original cutoff,

$$\|p - U_Q \mathcal{A}_{k,b,R} p\|_{G_N^\sigma} \leq \zeta \|p\|_{G_N^\sigma}. \quad (\text{LX21})$$

The factor e^{12q} is exactly the upper/lower norm transfer in (LX6). For $\zeta < 1$, this is injective. It evaluates the original vectors in their full original metrics.

5. The whole polynomial fibre and both real integration regions

The map (LX20) is defined on polynomials through degree $2q$. For every band vector f it maps the entire source fibre bijectively:

$$f + y^q \mathcal{P}_{N-q} \longrightarrow U_Q f + Q_k \mathcal{P}_{N-q}. \quad (\text{LX22})$$

Its restriction to the added term is exactly $U_Q(y^q h) = Q_k h$. Nothing is inferred from a truncated list of relation columns.

Here are explicit finite constants for its source norm. Put $\epsilon_0 = 2^{-32}$, $d_* = q + b$, and

$$E_Q = \frac{q(\rho/\epsilon_0)^2}{1 - (\rho/\epsilon_0)^2}, \quad r_Q = \sqrt{M_\sigma} e^{8q} 2^q \rho^{q-b} \epsilon_0^{-b},$$

$$i_F = 2\sqrt{2} M_\sigma q^{-1/2} (d_* + 1)^2 10^{2d_*} e^{\pi q} \epsilon_0^{2(q-b)},$$

$$i_U = 8\sqrt{2} M_\sigma q^{-1/2} (d_* + 1)^2 10^{2d_*} e^{\pi q} (2\epsilon_0 + \rho)^{2q} (2\epsilon_0)^{-2b}. \quad (\text{LX23})$$

Impose the explicit guards

$$\rho \leq \epsilon_0/2, \quad i_F \leq 1/4, \quad i_U \leq 1, \quad r_Q \leq e^{-E_Q/2}/4. \quad (\text{LX24})$$

Then with $C_Q = E_Q + 2 \log 2$,

$$e^{-C_Q} \|F\|_\sigma^2 \leq \|U_Q F\|_\sigma^2 \leq e^{C_Q} \|F\|_\sigma^2, \quad F \in y^{q-b} \mathcal{P}_{N-q+b}, \quad q-1 \leq N \leq 2q. \quad (\text{LX25})$$

For the proof write $F(qx) = x^{q-b} h(x)$, $\deg h \leq d_*$. On $|x| \geq \epsilon_0$, pairing all opposite roots of the original Q_k gives $|\log |Q_k(qx)/(qx)^q|| \leq E_Q$, because $|\log |1-z|| \leq |z|/(1-|z|)$. The negative Laurent part of $q^{-q} Q_k(qx)x^{-b} h(x)$ has coefficient sum at most $2^q \rho^{q-b} \|h\|_1$. Every negative power there lies between $-b$ and -1 , so its complete exterior norm is at most $\sqrt{M_\sigma} 2^q \rho^{q-b} \epsilon_0^{-b} \|h\|_1$. The pre-minimum e^{-8q} coefficient bound in Section 2 makes this at most $r_Q \|F\|_\sigma$.

For the inner region, (LX7) gives $\sup_{|x| \leq 1} |h(x)| \leq (d_* + 1) 10^{d_*} \|h\|_{L^2[1,2]}$. The contribution of $[q, 2q]$ to $\|F\|_\sigma^2$ is at least $\sqrt{q} e^{-\pi q} \|h\|_2^2 / (2\sqrt{2})$. Therefore the entire inner norm of F is at most $i_F \|F\|_\sigma^2$. Cauchy's polynomial-part formula on $|z| = 2\epsilon_0$ gives

$$|U_Q F(qx)| \leq 2(2\epsilon_0 + \rho)^q (2\epsilon_0)^{-b} \sup_{|z|=2\epsilon_0} |h(z)| \quad (|x| \leq \epsilon_0).$$

This bounds the entire inner norm of $U_Q F$ by $i_U \|F\|_\sigma^2$. The two norm ratios are consequently bounded below by $e^{-E_Q/2} \sqrt{1-i_F} - r_Q > \frac{1}{2} e^{-E_Q/2}$, and above by $\sqrt{(e^{E_Q/2} + r_Q)^2 + i_U} < 2e^{E_Q/2}$. Squaring proves (LX25). No Laurent expression is integrated through zero. Equation (LX22) passes this bound to complete attained minima.

For every finite degree-below- q representative, the arithmetic action defect is exactly

$$MU_Q - U_Q M_0 = -[1](c_{q-1}, c_{q-2}, \dots, c_0), \quad Q_k(y) = \sum_{j=0}^q c_j y^j. \quad (\text{LX26})$$

For column y^j , polynomial-part multiplication leaves the constant $-c_{q-j-1}$; all positive powers cancel. At $j = q-1$, use $y^q \equiv -\sum_{a < q} c_a y^a \pmod{Q_k}$ to obtain the same constant. Thus U_Q is the proved fibre map with this retained rank-one action defect. It does not replace the physical multiplication operator.

6. Full power-band constants and the native estimate

Here the complete jet bounds from 024 PJ6-40 are made finite. Put $n = q - b$, $\theta = \pi/2$, $\alpha = 2n - 1/2$,

$$c = (4\pi\sqrt{128})^{-1}, \quad C = 40\sqrt{128}/\pi, \quad B_0 = q/(cn), \quad D_0 = \max(1, Cn/q).$$

The full two-half-line Gamma/Laguerre comparison is $\frac{1}{4} \|P\|_{\lambda_n}^2 \leq \|P\|_{\nu_n}^2 \leq 32 \|P\|_{\lambda_n}^2$ for $n \geq 36$, $\deg P \leq 2n + 2$, where $d\lambda_n = |y|^{2n-1/2} e^{-\pi|y|/2} dy$. The near-zero lower contribution is recovered by $\|P/y\|_{\lambda_n} \leq 4\pi \|P\|_{\lambda_n}/n$, giving missing mass at most $16\pi^2/n^2 \leq 1/4$. The full Laguerre derivative matrix and last multiplication row give $cn \|P\|_{\nu_n} \leq \|yP\|_{\nu_n} \leq Cn \|P\|_{\nu_n}$ through source degree $2n + 1$. These are the original PJ6-16 calculations with their literal measure mass retained.

Define

$$g_j = \frac{j!(\alpha+1)_j}{(\theta q)^{2j}}, \quad \mathcal{L} = 128[(b+1)3^b]^2 \max\{\max_{j \leq b} g_j, (\min_{j \leq b} g_j)^{-1}\},$$

$$U = \sum_{2h < b} \binom{\lfloor (q+b)/2 \rfloor}{h}^2 B_0^{4h} + B_0^2 \sum_{2h+1 < b} \binom{\lfloor (q+b-1)/2 \rfloor}{h}^2 B_0^{4h},$$

$$V = \sum_{h=0}^{\lfloor (b-1)/2 \rfloor} \binom{\lfloor q/2 \rfloor + h - 1}{h} B_0^{2h}, \quad L = bV^2 D_0^{2(b-1)},$$

$$\mathcal{J}_{q,b} = \log \mathcal{L} + \max(\log L, \log U) + 2b \log(C/c) + E_G(q) + qJ((b+1)/q). \quad (\text{LX27})$$

The jet map is $P \mapsto (q^a[y^a]P)_{a < b}$, with its entire degree- $j + b - 1$ affine fibre in ν_n . Every low-to- N_j logarithmic relative eigenvalue of that full jet metric is within $\mathcal{J}_{q,b}$ of $qJ(t_j)$.

For completeness the finite argument is as follows. The monic Laguerre coefficient matrix has entries $(-1)^{j-i} \binom{j}{i} (\alpha + i + 1)_{j-i} / (\theta q)^{j-i}$, and its inverse has the same magnitudes. Multiplication proves the inverse formula by the binomial alternating sum. Every factor is at most 2 here, so both operator norms are at most $(b+1)3^b$; its exact squared monic norms relative to the half-line

mass are g_j . Applying both half-lines and their full mass comparison gives the low metric between $d_n \mathcal{L}^{-1}I$ and $d_n \mathcal{L}I$, including the extra lift degree at $j = 1$.

For the upper high covariance, the nonzero roots of each orthogonal polynomial have modulus at least cn : their squares are Rayleigh values of multiplication by y^2 , and the preceding complete raising bound applies to every test polynomial. Pair the even roots. Each divided coefficient is bounded by its constant coefficient times $\binom{\lfloor M/2 \rfloor}{h}(cn)^{-2h}$. For odd polynomials, its derivative-at-zero norm is at most $K_{n,M}(0,0)/(cn)^2$, by the exact bijection $P \mapsto yP$ on the odd subspace. Summing all squared rows gives trace bound $UK_{n,M}(0,0)$, and a positive covariance is no larger than its trace times identity.

For the lower high covariance let $H(y) = K_{n,j-1}(y,0)/K_{n,j-1}(0,0)$. Reproduction gives $H(0) = 1$ and $\|H\|^2 = 1/K_{n,j-1}(0,0)$; the nonzero roots have the same gap. Its reciprocal coefficients through degree $b-1$ have total bounded by V . Multiplying $H(y)$ by the first b Taylor coefficients of the requested jet divided by $H(qx)$ yields a simultaneous right inverse of the entire jet, with degree at most $j+b-2$. Its squared norm is bounded by $L\|a\|_2^2/K_{n,j-1}(0,0)$, including every coefficient combination. This proves both full covariance bounds.

Finally, repeated raising applied to a polynomial and to the constant gives $|\log[d_n K_{n,h}(0,0)] - \log[d_q K_{q,h}(0,0)]| \leq 2b \log(C/c)$. All raising source degrees are at most $q+2b-1 \leq 2n+1$. Concavity gives $|J(x+h) - J(x)| \leq J(h)$, so the full integer shifts between $j-1, j, j+b-1$ are bounded by the last term in (LX27). At $j=0$ the relative metric is exactly identity. This proves the claimed all-vector finite jet estimate for all $q+2$ original cutoffs.

Set

$$E_{\text{loc}}(k, b, R) = \mathcal{J}_{q,b} + 2C_Q + 2 \log \frac{1+\zeta}{1-\zeta} + \log \kappa_k, \quad \zeta \leq 1/4. \quad (\text{LX28})$$

Equations (LX21)–(LX25), followed by (LX3), show on the fixed subspace $K_{k,R}$

$$\left| \log \frac{\|p\|_{G_{N_0}}^2}{\|p\|_{G_{N_j}}^2} - qJ(t_j) \right| \leq E_{\text{loc}} \quad (p \neq 0). \quad (\text{LX29})$$

The two triangle-inequality endpoint errors cost exactly the logarithm shown, the two complete source comparisons cost $2C_Q$, and the arithmetic source ratio costs $\log \kappa_k$.

7. A sharper exceptional-direction penalty

Let I_G be any fixed frame inclusion of $K_{k,R}$ in the full original kernel frame. For $i \leq j$ the matrix

$$V_j = H_{K,N_j}^{1/2} I_G (I_G^* H_{K,N_j} I_G)^{-1/2}$$

satisfies $V_j^* V_j = I$, and direct multiplication gives

$$V_j^* (H_{K,N_j}^{-1/2} H_{K,N_i} H_{K,N_j}^{-1/2}) V_j = (I_G^* H_{K,N_j} I_G)^{-1/2} I_G^* H_{K,N_i} I_G (I_G^* H_{K,N_j} I_G)^{-1/2}. \quad (\text{LX30})$$

This is the exact isometric compression relating the two presentations. If a full spectral subspace above the good interval had dimension greater than the codimension c , it would intersect $\text{im } V_j$, contradicting its upper Rayleigh bound. The same applies below the good interval. Thus at most $c \leq c_{\text{loc}}$ eigenvalues lie above, and at most c below.

Write $d = q[J(t_j) - J(t_i)]$. By the elementary elliptic coefficient inequality $J(t) \leq [(1+t) \log(1+t) - t \log t]/2$, valid for all $t > 0$, we have $0 \leq d < q$ in the present range. Equation (LX9) puts every actual logarithm in $[0, B_{\text{all}}]$. Upper exceptional deviations total at most $c(B_{\text{all}} - d)$; lower ones total at most cd . Their sum is at most cB_{all} , retaining the complete exceptional quotient. Consequently

$$\Omega_{0j} := \sum_{a=1}^m |\log \lambda_a - qJ(t_j)| \leq mE_{\text{loc}} + c_{\text{loc}} B_{\text{all}}. \quad (\text{LX31})$$

Subtracting (LX29) at the two endpoints gives the all-pair estimate

$$\Omega_{ij} := \sum_{a=1}^m |\log \lambda_a - d| \leq 2mE_{\text{loc}} + c_{\text{loc}} B_{\text{all}}, \quad 0 \leq i \leq j \leq q+1. \quad (\text{LX32})$$

This improves the incoming $2c[B_{\text{all}} + d]$. Define the useful explicit native discrepancy

$$e_{\text{native}}(k, b, R) = \frac{mE_{\text{loc}} + c_{\text{loc}} B_{\text{all}}}{mq}. \quad (\text{LX33})$$

The same quantity bounds uniformly the difference between the actual logarithmic determinant profile divided by m_q and $J(t_j)$. This follows by summing the signed logarithms and then the absolute deviations, not by a rank allocation.

8. One growing radius and one growing band

Put

$$L_k = \log(k + e), \quad R_k = e^{2\sqrt{L_k}}, \quad b_k = 8 \left\lceil \frac{4q}{\sqrt{L_k}} \right\rceil. \quad (\text{LX34})$$

Use this choice only when (LX17), (LX24), and $\zeta \leq 1/4$ hold. All are inequalities in finite displayed data. We verify that they hold eventually, with the original source fixed.

First $b_k/q = 32/\sqrt{L_k} + O(q^{-1})$, $\Delta = O(k)$, and $\rho = O_h(k^{-1})$. Thus the band and root guards eventually hold. Also $\log \mathcal{M}_{R_k} = O_A(R_k \log(R_k + 2)) = o(q)$. The high derivative term in (LX19), after the full coefficient transfer, has logarithm at most

$$12q - \frac{b_k}{4} \log R_k + o(q) \leq -4q + o(q).$$

The low-degree term has logarithm

$$q \log(6R_k) + \frac{b_k}{8} \log \rho + O_{h,A}(q) + o(q) = -2q\sqrt{L_k} + O_{h,A}(q).$$

All factors $q^{v_0}, M_T, \mathcal{M}_R, (1 - \rho)^{-q-q'}, 2^{q'}$ are included here. The inverse-triangular tail has the still smaller leading exponent $-16q\sqrt{L_k}$. Hence $\zeta \leq e^{-q}$ eventually.

For (LX24), $E_Q = O_h(1)$, $\log r_Q = -(q - b) \log k + O_h(q + b) = -q \log k + o(q \log k)$. The logarithms of i_F, i_U , divided by q , have limiting upper bound at most $\pi + 2 \log 10 + 2 \log(2\epsilon_0) < 0$. Therefore all four guards hold. Their sufficient threshold can be extremely large: already $b \leq q/10$ asks approximately $\sqrt{\log k} \geq 320$. The conclusion is an effective eventual rate, not an assertion of useful numerical enclosures at small native k .

The finite binomial sums and Laguerre factors in (LX27), using $\binom{n}{h} \leq (en/h)^h$, give

$$\mathcal{J}_{q,b} = O(b \log(C_1 q/b) + b + \log(q + 2)) + E_G(q).$$

Thus the verified scalar bound $E_G(q) = O(\log(q + 2))$ gives

$$e_{\text{native}} = O_{h,A} \left(\frac{\log L_k}{\sqrt{L_k}} + \frac{e^{2\sqrt{L_k}}}{k} + \frac{\log \kappa_k}{q} \right). \quad (\text{LX35})$$

The first term includes the original endpoint modulus $J((b + 1)/q)$. The residue dimension contributes the second, because $N_R = O_A(R)$ and $m \asymp k$. No fixed-codimension limit is silently exchanged with a moving contour.

9. Every positive word direction

Retain $Q_k = D_k O_k$, $\deg D_k = q'$, $\deg O_k = \Delta$, the original word $T = D_k(M)$, and $W = D_k \mathcal{P}_{<\Delta}$. Its kernel has rank q' ; the original boundary map π_{∂} has that same kernel. The exact identity $\text{rem}_{Q_k}(Dp) = D \text{rem}_{O_k} p$ retains its complete root factors.

There is a sharper direct full-image tail than the two-stage incoming WE1. For $p = Da \in W$,

$$U_Q^{-1} p = [y^q a / O_k]_+, \quad \|P_{<q-b} U_Q^{-1} p\|_{c,q} \leq \epsilon_W \|p\|_{c,q},$$

$$\epsilon_W = (1 - \rho)^{-q'} 2^{q+\Delta} \rho^{b-\Delta}, \quad \zeta_W = e^{12q} (1 + \rho)^q \epsilon_W. \quad (\text{LX36})$$

Indeed the inverse leading block of D bounds the scaled coefficients of a by $(1 - \rho)^{-q'} \|p\|_{c,q}$. The reciprocal coefficients of O_k have bound $\binom{\Delta+j-1}{j} \rho^j$, and the depths needed to lie below $q - b$ are at least $b - \Delta$. Their total count through depth $q - 1$ is $\binom{q+\Delta-1}{q-1} \leq 2^{q+\Delta}$. This proves (LX36) on every image vector. Its connecting maps are also proved in 025 PH3-8.

With $\zeta_W \leq 1/4$, (LX25), (LX27) give on the entire fixed image

$$e^{-qJ(t_j)-a_W} G_{W,N_0} \preceq G_{W,N_j} \preceq e^{-qJ(t_j)+a_W} G_{W,N_0}, \quad a_W = \mathcal{J}_{q,b} + 2C_Q + 2 \log \frac{1 + \zeta_W}{1 - \zeta_W} + \log \kappa_k. \quad (\text{LX37})$$

There is no finite-rank subtraction for this statement. For example the incoming band $b = 8\lceil 16q/\log(1/\rho) \rceil$, together with all the same finite guards, gives $\zeta_W \leq e^{-q}$ eventually and $a_W = O_h(q \log \log k / \log k + \log \kappa_k + 1)$. The original full image, not only its determinant, is controlled.

For an entirely explicit low-energy band put $R_* = k\sqrt{\delta^2 + \gamma^2}$, $\epsilon_* = 2^{-32}$, $T_* = \epsilon_* q$, and require $R_* \leq T_*/2$, $\Delta \leq q/10$, $q' \geq 36$. Let

$$K_D = 2\sqrt{2}M_\sigma \Delta^2 q^{-1/2} 10^{2\Delta} e^{\pi q} (\epsilon_* + \rho)^{2q'}, \quad E_D = \frac{2q'(R_*/T_*)}{1 - R_*/T_*} + \log(1 + K_D). \quad (\text{LX38})$$

For every $\deg h < \Delta$, $e^{-E_D} \|y^{q'} h\|_\sigma^2 \leq \|Dh\|_\sigma^2 \leq e^{E_D} \|y^{q'} h\|_\sigma^2$. On the exterior use all q' factors and $|\log |1-z|| \leq |z|/(1-|z|)$. On the interior the degree- $\Delta-1$ Legendre evaluation bound gives $\Delta^2 10^{2\Delta}/q$ times $\int_q^{2q} |h|^2$; the latter contributes at least $q^{2q'} e^{-\pi q} / (2\sqrt{2}q)$ times itself to the reference outer norm. Both inner integrals are therefore at most K_D times that reference outer norm. Combining these estimates proves (LX38), with both regions retained. Here $E_D = O_h(k)$.

The scaled boundary Vandermonde $V_{ij} = (\omega_i/q)^j$ obeys

$$\|V\| \leq \Delta, \quad \|V^{-1}\| \leq V_- := \Delta((q + R_*)/(2\delta))^{\Delta-1}. \quad (\text{LX39})$$

These are respectively its Frobenius bound and the complete Lagrange coefficient bound; every pair of distinct original roots has separation at least 2δ . If a domain boundary value is b , the word image is Dh with $h(\omega_i) = b_i$, so this is the actual word coordinate map, with no omission of the factors $D(\omega_i)$.

The full boundary quotient satisfies

$$(A_\partial)^{-1} I \preceq G_N^{\partial, \sigma} \preceq B_\partial I, \quad A_\partial = \frac{\Delta e^2}{M_\sigma} (2q+1)^{3/2} (4q+1)^{R_*+1}, \quad B_\partial = \Delta M_\sigma ((2\Delta + R_*)/(2\delta))^{2(\Delta-1)}. \quad (\text{LX40})$$

For the lower bound the exact Gamma generating function is $(1+t^2)^{-1/4} e^{z \arctan t} = \sum c_n(z) t^n$, with orthonormal multiplier squared $n!/(M_\sigma (1/2)_n) \leq (2n+1)/M_\sigma$. Cauchy's estimate at $|t| = N/(N+1)$, the complete complex bound $|\arctan t| \leq \frac{1}{2} \log(2N+1)$, and summation give evaluation kernel at most $e^2 (N+1)^{3/2} (2N+1)^{R_*+1}/M_\sigma$. The trace of the full boundary covariance gives its operator bound. For the upper bound, each degree- $\Delta-1$ Lagrange lift has norm bounded by multiplying all factors with $\|(y-z)P\|_\sigma \leq [2(L+1) + |z|] \|P\|_\sigma$; Cauchy-Schwarz over the Δ lifts gives (LX40). These rederive 018 IC12-13.

Let $\mathcal{L}_{q,\Delta}$ be the full low coefficient constant in (LX27) with $n = q - \Delta = q'$. Put

$$c_0 = 2q \log q + [2 \log(4/\pi) - 2]q,$$

$$e_0 = E_D + \log \mathcal{L}_{q,\Delta} + \log \kappa_k + 2 \log \Delta + 2 \log \max(1, V_-) + \log \max(1, A_\partial) + \log \max(1, B_\partial) + |\log d_{q'} - c_0|. \quad (\text{LX41})$$

The word image form just proved, divided by the actual attained boundary metric, has all its Δ positive energies in $[e^{c_0 - e_0}, e^{c_0 + e_0}]$. The other q' energies are exactly zero. The numerator and denominator source transfers together cost $\log \kappa_k$. The scalar Gamma mass bound gives $|\log d_{q'} - c_0| \leq 2\Delta(\log q + 1) + |2 \log(4/\pi) - 2|\Delta + 5$; hence $e_0 = O_h(k \log q)$.

The full boundary metric changes by at most $\omega_k = \log \kappa_k + \log \max(1, A_\partial) + \log \max(1, B_\partial)$. Applying the fixed image form comparison (LX37) and the exact boundary minimum to every positive direction yields

$$|\log \lambda_a^+(T^{\dagger G_{N_j}} T) - [c_0 - qJ(t_j)]| \leq e_{\text{word}} := e_0 + a_W + \omega_k, \quad a \leq \Delta, \quad 0 \leq j \leq q+1. \quad (\text{LX42})$$

On the displayed guards, the verified scalar remainder gives $e_{\text{word}} = O_h(q \log \log k / \log k + k \log q) = o(q)$. All positive energies, complete boundary quotients and exact zero directions remain present. This supplies the original phase/heat receivers; it does not evaluate the separate phase-bearing arithmetic-current numerator.

10. The same finite improvement on every fixed subquotient

Let $I_S : \mathbb{C}^s \hookrightarrow \mathbb{C}^m$ be any fixed coefficient inclusion, and let $R : \mathbb{C}^s \twoheadrightarrow \mathbb{C}^r$ be any fixed onto map. At every original cutoff define the complete restriction and attained quotient

$$H_{S,N} = I_S^* H_{K,N} I_S, \quad Q_N = (R H_{S,N}^{-1} R^*)^{-1}.$$

This includes every specified S_k/R_k by choosing an onto coordinate map with kernel R_k . The relative restriction is the exact isometric compression in (LX30), with I_G replaced by I_S . For the quotient put

$$A_S = H_{S,N_j}^{-1/2} H_{S,N_i} H_{S,N_j}^{-1/2}, \quad W_j = H_{S,N_j}^{-1/2} R^* Q_{N_j}^{1/2}.$$

Then direct multiplication proves

$$W_j^* W_j = I_r, \quad W_j^* A_S^{-1} W_j = Q_{N_j}^{1/2} Q_{N_i}^{-1} Q_{N_j}^{1/2} = (Q_{N_j}^{-1/2} Q_{N_i} Q_{N_j}^{-1/2})^{-1}. \quad (\text{LX43})$$

Thus inversion, isometric compression and inversion give exactly the relative quotient. The spectral dimension argument from Section 7 applies at each step. Restriction cannot increase the number of eigenvalues above or below a fixed threshold; inversion reverses the two thresholds; the second inversion restores them. Hence the quotient still has at most $\min(c_{\text{loc}}, r)$ logarithms on either side of the good band. The full nesting bound $[0, B_{\text{all}}]$ is preserved because the quotient takes the full minimum of the same pair of comparable forms.

With $d = q[J(t_j) - J(t_i)]$, the separate exceptional deviations are again at most cd and $c(B_{\text{all}} - d)$. Therefore

$$\sum_{a=1}^r |\log \lambda_a(Q_{N_j}^{-1/2} Q_{N_i} Q_{N_j}^{-1/2}) - d| \leq r E_{ij} + \min(c_{\text{loc}}, r) B_{\text{all}}, \quad E_{0j} = E_{\text{loc}}, \quad E_{ij} = 2E_{\text{loc}} \ (i > 0). \quad (\text{LX44})$$

These are the original fixed coefficient subquotients, with their cutoff-dependent minimizing sections retained through (LX43). In particular a fixed-rank quotient keeps its finite exceptional term; it is not assigned a vanishing error merely because the full kernel rank grows.

11. Provenance and scope

The new finite contour and original-metric proof is LX6-26. The sharper exceptional penalty is LX31-33; the growing choice is LX34-35; the full word control is LX36-42; the complete fixed-subquotient improvement is LX43-44. The input's LC1-15, SQ1 and WE1-9 are supported with these strengthened finite discrepancies and with the proved scalar constant 83 in LX5. The positive Gamma envelopes and scalar estimate are proved in the companion EG1-24; no numerical native period or actual native Gram matrix is claimed to have been evaluated here.

Human formula provenance is retained: R. A. Askey and R. Roy for the Gamma equations; T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt for the original Legendre and Meixner-Pollaczek formulas DLMF18.12.E11, 18.22.E8, 18.23.E7. The retained original equation TeX was read, and the source ledger records its exact bytes, reading coverage and receiving equations. The zero-count, harmonic and contour steps are proved above from their explicit kernels, so the finite constants do not depend on an unnamed external asymptotic theorem.

Part IV

Original observed current: complete incoming proof

Original objects, source status, and the claim being proved

Fix the stipulated simple quartet, the original admitted period ϖ and branch, and the complete physical arithmetic unit. Keep

$$k \equiv 1 \pmod{4}, \quad k \geq 17, \quad q = (k+1)^2, \quad q_0 = (k-7)^2, \quad \Delta = q - q_0 = 16k - 48,$$

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta].$$

The original coefficient algebra, arithmetic operator and physical coordinate are

$$E = \mathbb{C}[y]/Q_k, \quad M[p] = [yp], \quad S = k/2 + iy, \quad \mathcal{A} = kI/2 + iM.$$

The source is the unchanged even measure

$$d\mu_k(y) = w_h^{*k}(y) dy, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}.$$

For $q-1 \leq N \leq 2q$, let G_N be the entire degree- N quotient minimum on E . The unchanged onto observation and its actual minimum section are

$$\Lambda : E \rightarrow B, \quad K = \ker \Lambda, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}. \quad (\text{OCP1})$$

The projection $P_{B,N} = L_N \Lambda$ is orthogonal in G_N . Its complement is $P_{K,N}$. Every adjoint below is either an explicitly written metric adjoint or the coordinate conjugate transpose $*$.

The reference measure and its complete monic norms are

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad h_n = \sqrt{2\pi} n! (1/2)_n. \quad (\text{OCP2})$$

Its monic polynomials π_n satisfy

$$\pi_{n+1} = y\pi_n - n(n-1/2)\pi_{n-1}, \quad \sum_{n \geq 0} \pi_n(y) t^n / n! = (1+t^2)^{-1/4} e^{y \arctan t}. \quad (\text{OCP3})$$

These are the Meixner–Pollaczek identities in the stated coordinate and mass; the ordinary identities are DLMF 18.22.8 and 18.23.7 [1]. No monic norm or source measure is normalized to probability in the proofs.

Let p_n be the real monic polynomial for μ_k , with squared norm ω_n . Use the complete comparison, through degree $2q+2$,

$$\ell \|P\|_\sigma^2 \leq \|P\|_{\mu_k}^2 \leq u \|P\|_\sigma^2, \quad \log(u/\ell) = O_h(k \log^2(q+2)). \quad (\text{OCP4})$$

The extra raising degree is used only to bound p_{N+1} and its conductor image. The actual source metric remains the degree- N minimum. The exact source identities and original monic-profile estimate are retained inputs [4, 5]. The current's rank-two formula itself is rederived below; it is not claimed as a new identity. The quantitative new step is the full two-column inequality in Section 4.

Two exact polynomial minima and the original current

For the complete relation measure $|Q_k|^2 d\mu_k$, let ν_j^μ be the squared norm of its real monic polynomial U_j . The polynomial $Q_k U_j$ is monic of degree $q+j$ and orthogonal to all preceding original relation rows. Put

$$b_n = [p_n], \quad E_N = \|b_N\|_{G_N}^2, \quad F_N = \|b_{N+1}\|_{G_N}^2.$$

Lemma 0.1. *The complete original quotient norms are*

$$E_N = \begin{cases} \omega_N, & N = q-1, \\ \omega_N(1 - \chi_N), & N \geq q, \end{cases} \quad \chi_N = \omega_N / \nu_{N-q}^\mu, \\ F_N = \nu_{N+1-q}^\mu - \omega_{N+1}, \quad \langle b_N, b_{N+1} \rangle_{G_N} = 0. \quad (\text{OCP5})$$

The current is exactly

$$M = C_N + \frac{b_{N+1} b_N^* G_N}{\omega_N}, \quad C_N^{\dagger G_N} = C_N. \quad (\text{OCP6})$$

The rank-one term squares to zero and has norm

$$\varepsilon_N = \frac{\sqrt{E_N F_N}}{\omega_N}. \quad (\text{OCP7})$$

Proof. Projection of p_N onto the complete relation space uses only its last orthogonal monic row: its coefficient is ω_N / ν_{N-q}^μ . This proves the first formula, including the first cutoff where there is no relation row. The minimum representative of b_{N+1} is

$$p_{N+1} - Q_k U_{N+1-q} \in \mathcal{P}_N.$$

It is orthogonal to every relation of degree at most N . Its squared norm is $\nu_{N+1-q}^\mu - \omega_{N+1}$. The two classes have opposite parity in the even source and even relation quotient, proving orthogonality.

For the minimum representative P of any class, project yP onto $\mathcal{P}_N$. Its remaining polynomial is $[y^N]P p_{N+1}$. The further compression to the minimum-section space is selfadjoint, because multiplication by real y is selfadjoint in the original source. Since

$$[y^N]P = \langle p_N, P \rangle_{\mu_k} / \omega_N = \langle b_N, [P] \rangle_{G_N} / \omega_N,$$

this proves (OCP6). Orthogonality proves its square-zero assertion and (OCP7). $\square$

Write $n_B = \dim B$ and define the actual observed centered current, in its original metric's orthonormal coordinates,

$$\mathcal{W}_{B,N} = Q_{B,N}^{-1/2} L_N^* (\mathcal{A}^* G_N + G_N \mathcal{A} - k G_N) L_N Q_{B,N}^{-1/2}. \quad (\text{OCP8})$$

Let

$$e_N = b_N / \sqrt{E_N}, \quad f_N = b_{N+1} / \sqrt{F_N}, \quad U_N = [Q_{B,N}^{1/2} \Lambda e_N, Q_{B,N}^{1/2} \Lambda f_N].$$

Its complete two-column Gram is

$$\mathcal{C}_N = U_N^* U_N = \begin{pmatrix} a_N & r_N \\ \bar{r}_N & b_N^\circ \end{pmatrix}, \quad 0 \preceq \mathcal{C}_N \preceq I_2. \quad (\text{OCP9})$$

The superscript on b_N° distinguishes the scalar norm from the quotient vector b_N . Define

$$J_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Then

$$\mathcal{W}_{B,N} = \varepsilon_N U_N J_2 U_N^*. \quad (\text{OCP10})$$

Indeed $\mathcal{A}^* G + G \mathcal{A} - k G = i(GM - M^* G)$. This fixes the imaginary phase in the real- y convention. In the physical monic convention $b_n^S = i^n b_n$, the corresponding rank-two matrix has the usual symmetric plus form.

The original conductor and its two lower test classes

Retain the complete original conductor

$$\mathcal{T}p = \sum_{u,v=0}^8 a_{uv} p(y + \sigma_{uv}), \quad \sigma_{uv} = (2v - 8)\gamma - i(2u - 8)\delta.$$

Its first nonzero moment has original order $v_0 \leq 80$ and coefficient $\mu_{v_0} \neq 0$. On a monic degree- n polynomial its leading coefficient after lowering degree is

$$\kappa_n = i^{-v_0} \mu_{v_0} \binom{n}{v_0}. \quad (\text{OCP11})$$

Let $Q_0 = Q_{k-8}$, $E_0 = \mathbb{C}[y]/Q_0$, and let $C_A : E \rightarrow E_0$ be the induced map. The exact original identities are

$$C_A[p] = [\mathcal{T}p]_{Q_0}, \quad K \subset \ker C_A. \quad (\text{OCP12})$$

No equality between these kernels is used. In particular no period-dependent choice of a generic kernel is made.

One finite forward bound from the full generating function is

$$\|\mathcal{T}\|_{\sigma, \mathcal{P}_n \rightarrow \mathcal{P}_{n-v_0}} \leq \mathcal{F}_n := A_1 e^{1+2\pi\gamma} (n+1)(2n+1)^{4\delta+1/2}, \quad A_1 = \sum |a_{uv}|. \quad (\text{OCP13})$$

Translation multiplies (OCP3) by $e^{\sigma_{uv} \arctan t}$. On $|t| = n/(n+1)$, its modulus is at most $e^{2\pi\gamma} (2n+1)^{4\delta}$. Cauchy's estimate costs at most e , the full orthonormal coefficient ratio at most $\sqrt{2n+1}$, and summing all $n+1$ diagonals proves (OCP13). The original source version is the IVO forward bound [5].

For every class $x \in E$, minimize over its complete original observed fibre. Every candidate has degree at most N and conductor value $C_A x$. Consequently

$$\|\Lambda x\|_{Q_{B,N}}^2 \geq \frac{\ell}{\mathcal{F}_N^2} \|C_A x\|_{G_D^-}^2, \quad D = N - v_0, \quad (\text{OCP14})$$

where G_D^- is the full Gamma quotient metric on E_0 at degree D . All original upper relations map into the lower relation ideal, which is why (OCP14) holds for every representative.

Let ν_j^- denote the complete monic norm for $Q_0^2 d\sigma$ and set

$$t = D - q_0, \quad d_N = \frac{h_D}{\nu_t^-}, \quad e_N^\circ = \frac{h_{D+1}}{\nu_{t+1}^-}. \quad (\text{OCP15})$$

Here $D \geq q_0$ throughout the stated window when $k \geq 17$. Both ratios are strictly between zero and one: equality would make an ordinary real-rooted orthogonal polynomial divisible by Q_0 , whose roots are nonreal.

The lower quotient classes $a = [\pi_D]$, $b = [\pi_{D+1}]$ are orthogonal and have norms

$$A_D^2 = h_D(1 - d_N), \quad B_D^2 = h_{D+1} \frac{1 - e_N^\circ}{e_N^\circ}. \quad (\text{OCP16})$$

For every $R \in \mathcal{P}_{D-1}$,

$$\left| \left\langle \frac{a}{A_D}, [R] \right\rangle_{G_D^-} \right| \leq \sqrt{\frac{d_N}{1 - d_N}} \|R\|_\sigma. \quad (\text{OCP17})$$

To prove this, the projection of π_D onto the whole relation space is $(h_D/\nu_t^-)Q_0 U_t^-$. Its squared norm is $h_D d_N$. The original π_D is orthogonal to R ; subtracting this exact relation projection and dividing by A_D proves (OCP17).

A finite area bound, retaining the complex cross term

Write

$$\mathcal{T}p_n = \kappa_n \pi_{n-v_0} + R_n, \quad \deg R_n < n - v_0.$$

By the full source comparison and orthogonality of the leading Gamma component,

$$\|R_n\|_\sigma \leq \|\mathcal{T}p_n\|_\sigma \leq P_n := \mathcal{F}_n \sqrt{u/\ell} \sqrt{h_n}. \quad (\text{OCP18})$$

Define the entirely specified positive scalar

$$\beta_n = \frac{P_n}{|\kappa_n| \sqrt{h_{n-v_0}}}.$$

Let

$$\begin{aligned} x_N &= \frac{\beta_N \sqrt{d_N}}{1 - d_N}, \\ y_N &= \beta_{N+1} \sqrt{\frac{e_N^\circ}{1 - e_N^\circ}}, \\ z_N &= \beta_N \beta_{N+1} \sqrt{\frac{e_N^\circ}{(1 - d_N)(1 - e_N^\circ)}}, \\ \zeta_N &= (1 - x_N)(1 - y_N) - z_N. \end{aligned}$$

The symbol ζ_N here is a finite positive margin under the guard below, not a zeta zero or an observation pairing.

Theorem 0.2 (Full observed two-column area). *Suppose*

$$x_N + y_N + z_N \leq \frac{1}{2}. \quad (\text{OCP19})$$

Let $\nu_r^{+, \sigma}$ be the complete original $Q_k^2 d\sigma$ monic norm, where $r = N + 1 - q$, and put

$$\xi_N = \frac{h_{N+1}}{\nu_r^{+, \sigma}}, \quad L_n = |\kappa_n|^2 \frac{h_{n-v_0}}{h_n} = \frac{|\mu_{v_0}|^2}{(v_0!)^2} \prod_{j=0}^{v_0-1} \frac{n-j}{n-j-1/2}. \quad (\text{OCP20})$$

Then $\zeta_N \geq 1/2$ and

$$\boxed{\Theta_N I_2 \preceq \mathcal{C}_N \preceq I_2, \quad \det \mathcal{C}_N \geq \Theta_N > 0,} \quad (\text{OCP21})$$

where

$$\boxed{\Theta_N = \min \left\{ 1, \left(\frac{\ell}{u} \right)^2 \mathcal{F}_N^{-4} L_N L_{N+1} (1 - d_N) (1 - e_N^\circ) \frac{\xi_N}{e_N^\circ} \zeta_N^2 \right\}.} \quad (\text{OCP22})$$

All minima in this formula are the complete monic minima of the stated measures.

Proof. Test the two conductor images against the orthonormal lower classes $a/A_D, b/B_D$, giving the complex matrix Y . By (OCP17)–(OCP18), its entries obey

$$\begin{aligned} |Y_{11} - \kappa_N A_D| &\leq \sqrt{d_N/(1 - d_N)} P_N, & |Y_{21}| &\leq P_N, \\ |Y_{12}| &\leq P_{N+1}, & |Y_{22} - \kappa_{N+1} B_D| &\leq P_{N+1}. \end{aligned}$$

The off-diagonal leading term in Y_{12} is zero by the exact parity orthogonality of a and b . No phase of R_N or of κ_n is changed. The triangle inequality for the two products in the determinant gives

$$|\det Y| \geq |\kappa_N \kappa_{N+1}| A_D B_D \{(1 - x_N)(1 - y_N) - z_N\}. \quad (\text{OCP23})$$

The guard ensures the factors used in this lower bound are positive.

For $X = [b_N, b_{N+1}]$, the form inequality (OCP14), and then orthogonal projection onto the two lower test directions, give

$$X^* \Lambda^* Q_{B,N} \Lambda X \succeq \frac{\ell}{\mathcal{F}_N^2} (C_A X)^* G_D^- (C_A X) \succeq \frac{\ell}{\mathcal{F}_N^2} Y^* Y.$$

Both are two-dimensional positive forms. Hence

$$\det \mathcal{C}_N \geq \left(\frac{\ell}{\mathcal{F}_N^2} \right)^2 \frac{|\kappa_N \kappa_{N+1}|^2 A_D^2 B_D^2 \zeta_N^2}{E_N F_N}.$$

The full monic comparisons give $E_N \leq u h_N$ and $F_N \leq u \nu_r^{+, \sigma}$. Substitution and cancellation of the identical Gamma masses yield (OCP22). Since $\mathcal{C}_N \preceq I_2$, its smaller eigenvalue is at least its determinant, proving (OCP21). $\square$

The first source direction and the whole kernel current

The first normalized source direction is even better preserved than (OCP21) alone states. Put

$$\alpha_N = \|P_{K,N}e_N\|_{G_N}^2.$$

Every degree- N minimum representative of a class in K has conductor image divisible by Q_0 . If its leading coefficient is c , that image has leading coefficient $\kappa_N c$, so its full lower source norm is at least $|\kappa_N c| \sqrt{\nu_t^-}$. Applying (OCP13)–(OCP14) to the leading-coefficient functional on K proves

$$\alpha_N \leq \frac{\omega_N^2 \mathcal{F}_N^2}{E_N \ell |\kappa_N|^2 \nu_t^-}. \quad (\text{OCP24})$$

This uses the identity $\langle b_N, x \rangle_{G_N} = \omega_N [y^N]$ (minimum representative of x) on the entire kernel.

For an entirely Gamma-based upper certificate, assume additionally, when $N \geq q$, that

$$\frac{u}{\ell} \frac{h_N}{\nu_{N-q}^{+, \sigma}} \leq \frac{1}{2}. \quad (\text{OCP25})$$

At $N = q - 1$ this extra guard is empty. Then (OCP5), the monic comparison and (OCP20) imply

$$\alpha_N \leq \mathbf{a}_N := \min \left\{ 1, 2 \frac{u}{\ell} \frac{\mathcal{F}_N^2}{L_N} d_N \right\}. \quad (\text{OCP26})$$

The finite formula (OCP24) is valid without (OCP25).

The full normalized source pair is orthonormal, so

$$a_N = 1 - \alpha_N, \quad |r_N|^2 \leq \alpha_N \|P_{K,N}f_N\|^2 \leq \alpha_N. \quad (\text{OCP27})$$

The complete kernel-current compression obeys

$$\|\mathcal{W}_{K,N}\| \leq 2\varepsilon_N \sqrt{\mathbf{a}_N}, \quad |\text{Tr } \mathcal{W}_{B,N}| \leq 2\varepsilon_N \sqrt{\mathbf{a}_N}. \quad (\text{OCP28})$$

For the first inequality write the original current as $i\varepsilon_N(f_N e_N^* - e_N f_N^*)$ in a source isometry and project both factors into K . For the second use (OCP27) and the exact trace in Section 7. The source total trace is zero, so the two diagonal traces are opposite. The full off-diagonal block $i\varepsilon_N(f_K e_B^* - e_K f_B^*)$ is retained; no invariance of K is inferred.

Evaluate all growing-degree errors

The sole asymptotic input beyond the original full-source comparison is the retained monic-profile theorem:

$$\log \frac{\nu_j^{+, \sigma}}{h_{q+j}} = 2q\psi(j/q) + O_h(k \log^2(q+2)), \quad 0 \leq j \leq q+1. \quad (\text{OCP29})$$

Its analogue for Q_0 holds with q, k replaced by $q_0, k-8$. This is the reference version of the source's H4/V4 calculation; its actual arithmetic version follows by the same complete monic comparison [4, 6]. It retains its written analytic proof status. The finite results preceding this section do not require (OCP29).

We explicitly extend the needed lower index by $O(k)$ instead of silently using the old degree range. For the lower relation measure $Q_0^2 d\sigma$, let $a_j^- = \sqrt{\nu_j^- / \nu_{j-1}^-}$. The digamma series (DLMF 5.7.6, [2]) and the nonreal root distance give

$$|(\log \sigma)'| \leq 2 + \pi/2, \quad |(\log(Q_0^2 \sigma))'| \leq B_0 := 2q_0/\delta + 2 + \pi/2.$$

For the first bound, in the series for $\Im\psi(1/4 + ib)$ the initial term has absolute value at most two, and the remaining decreasing sum is bounded by $\pi/2$. Integration by parts against consecutive monic relation polynomials gives

$$j\nu_{j-1}^- \leq B_0 \sqrt{\nu_j^- \nu_{j-1}^-}.$$

The trial yU_{j-1}^- and the full Gamma multiplication bound give the other side:

$$\boxed{\frac{j}{B_0} \leq a_j^- \leq 2(q_0 + j)}. \quad (\text{OCP30})$$

All boundary terms vanish by exponential decay. The Gamma recurrence (OCP3) proves $\|yP\|_\sigma \leq 2(n+1)\|P\|_\sigma$ for degree at most n .

For $j = q_0 + O(k)$, a consecutive change in $\log(\nu_j^-/h_{q_0+j})$ is therefore $O_h(\log q)$. At most $O(k)$ extra changes reach all indices $t, t+1$ used above, including the final $N = 2q$ column. Since ψ is smooth near one, (OCP29) extends to them with the same $O_h(k \log^2 q)$ bound.

The exact reference profile may be written using its complementary elliptic ratio $r_s \in (0, 1)$ and modulus $\kappa_s = \sqrt{1 - r_s^2}$:

$$1 + s = \frac{E(\kappa_s)}{r_s K(\kappa_s)}, \quad \psi(s) = s \log \frac{\kappa_s}{E(\kappa_s)} + \log \frac{1 + r_s}{E(\kappa_s)}. \quad (\text{OCP31})$$

It is positive and decreasing on the required interval, with

$$a_0 = \psi(0) = \log(4/\pi), \quad a_1 = \psi(1) = \log \frac{\kappa_1(1 + r_1)}{E(\kappa_1)^2} > 0.$$

The endpoint expansion gives $|\psi'(s)| \leq C(1 + |\log s|)$ near zero; on compact positive intervals it is bounded. These are the same profile and elliptic conventions as the source, with the modulus rather than its square used in K, E [3, 4].

Let $s_N = (N + 1 - q)/q$. The two indices of the lower profile differ by one, and

$$q_0 \psi \left(\frac{r + \Delta - v_0}{q_0} \right) - q \psi(r/q) = O_h(k \log(q + 2)) \quad (\text{OCP32})$$

uniformly for $0 \leq r \leq q + 1$. To prove this, the two arguments differ by $O(k/q)$; integrate the logarithmic derivative bound over that interval. The factor $q - q_0 = O(k)$ contributes only $O(k)$.

Equations (OCP18), (OCP20) and (OCP29)–(OCP32) yield, uniformly in the original window,

$$\log \beta_n = O_{h,\varpi}(k \log^2(q + 2)), \quad \log d_N, \log e_N^\circ = -2q\psi(s_N) + O_{h,\varpi}(k \log^2(q + 2)).$$

Consequently the finite guards (OCP19) and (OCP25) hold eventually at every cutoff. Crucially, in (OCP22) the ratio ξ_N/e_N° has logarithm

$$2q_0 \psi \left(\frac{r + \Delta - v_0}{q_0} \right) - 2q\psi(r/q) + O_h(k \log^2 q) = O_{h,\varpi}(k \log^2 q).$$

Thus the two large monic factors have been compared before being bounded separately. We obtain

$$\boxed{0 \leq -\log \Theta_N = O_{h,\varpi}(k \log^2(q + 2))}, \quad (\text{OCP33})$$

$$\boxed{\alpha_N \leq \exp[-2q\psi(s_N) + O_{h,\varpi}(k \log^2(q + 2))]} \quad (\text{OCP34})$$

The inherited exact allowance formula (OCP5)–(OCP7) and (OCP29) also give

$$\log \varepsilon_N = q\psi(s_N) + O_h(k \log^2(q + 2)). \quad (\text{OCP35})$$

All implicit constants are fixed with the original packet and period; no uniformity near a vanishing conductor moment is claimed.

Two observed signs and the complete kernel-current bound

Let $r_N = x_N^\circ + iy_N^\circ$ in (OCP9). The nonzero spectrum of (OCP10) is that of $\varepsilon_N J_2 \mathcal{C}_N$, whose characteristic polynomial is

$$\lambda^2 + 2\varepsilon_N y_N^\circ \lambda - \varepsilon_N^2 \det \mathcal{C}_N.$$

The two eigenvalues are therefore exactly

$$\boxed{\lambda_{\pm, N} = \varepsilon_N \left[-\Im r_N \pm \sqrt{a_N b_N^\circ - (\Re r_N)^2} \right]}. \quad (\text{OCP36})$$

They have opposite signs under (OCP19), and every remaining observed eigenvalue is zero. Orthogonal compression of the full current bounds their absolute values by ε_N . Their product is $-\varepsilon_N^2 \det \mathcal{C}_N$, so

$$\varepsilon_N \Theta_N \leq \lambda_{+,N}, -\lambda_{-,N} \leq \varepsilon_N. \quad (\text{OCP37})$$

The exact inertia is $(1, 1, n_B - 2)$, with all zero directions retained.

Theorem 0.3 (Original observed current and hidden current). *For fixed admitted data and sufficiently large k , uniformly in $q - 1 \leq N \leq 2q$,*

$$\begin{aligned} \log \lambda_{+,N} &= q\psi(s_N) + O_{h,\varpi}(k \log^2 q), \\ \log(-\lambda_{-,N}) &= q\psi(s_N) + O_{h,\varpi}(k \log^2 q). \end{aligned} \quad (\text{OCP38})$$

In the entire original kernel,

$$\|\mathscr{W}_{K,N}\| + |\text{Tr } \mathscr{W}_{B,N}| \leq \exp[O_{h,\varpi}(k \log^2(q+2))]. \quad (\text{OCP39})$$

The two observed signs are asymptotically balanced:

$$\left| \frac{\lambda_{+,N}}{-\lambda_{-,N}} - 1 \right| \leq \frac{2\sqrt{a_N}}{\Theta_N} = \exp[-q\psi(s_N) + O_{h,\varpi}(k \log^2 q)]. \quad (\text{OCP40})$$

Proof. Equations (OCP33), (OCP35), and (OCP37) prove (OCP38). Equations (OCP28), (OCP34), and (OCP35) prove (OCP39). Finally $\lambda_+ + \lambda_- = -2\varepsilon_N \Im r_N$ and $|r_N| \leq \sqrt{a_N}$; divide by (OCP37) to prove (OCP40). $\square$

This is the centered Hermitian current, not the full kernel action and not its energy. In particular, the previously proved growing rank of ΛMI_K is compatible with (OCP39). The complete mixed current is

$$i\varepsilon_N(f_K e_B^* - e_K f_B^*).$$

Its difference from the first rank-one term has norm at most $\varepsilon_N \sqrt{a_N} = e^{o(q)}$. This retains both leakage directions rather than assigning invariance to the kernel.

The latest PEN section isomorphisms equip the canonical section spaces with the actual pull-backs of restricted G_N . The current on their hidden domains is the restriction of $i(G_N M - M^* G_N)$. Equation (OCP39) therefore gives the same $e^{o(q)}$ operator-norm bound on every such hidden section domain. It does not replace the full ambient short minimum by the hidden minimum; PEN18-27 and its terminal correction remain unchanged [7].

Two prescribed original observed classes with opposite current values

The current signs can be attained without choosing its native eigenvectors. In the physical monic convention, define the cutoff-labelled original observed values

$$\mathbf{b}_{\pm,N} = \Lambda \left(\frac{b_N^S}{\sqrt{E_N}} \pm \frac{b_{N+1}^S}{\sqrt{F_N}} \right), \quad b_n^S = i^n b_n. \quad (\text{OCP41})$$

They differ by the common unit phase i^N from the real-coordinate values $\Lambda(e_N \pm if_N)$. Their degree- N source representatives use exactly the minimum polynomial of Section 2, including the cancellation of the degree- $N+1$ monic terms. No source cutoff is increased.

Let

$$\Phi_N(b) = b^* L_N^* (\mathscr{A}^* G_N + G_N \mathscr{A} - k G_N) L_N b, \quad j_{\pm,N} = \frac{\Phi_N(\mathbf{b}_{\pm,N})}{\|\mathbf{b}_{\pm,N}\|_{Q_{B,N}}^2}.$$

The complete complex formula is

$$j_{\pm,N} = \varepsilon_N \left[-2\Im r_N \pm \frac{2 \det \mathcal{C}_N}{a_N + b_N^\circ \mp 2\Im r_N} \right]. \quad (\text{OCP42})$$

It follows by substituting $(1, \pm i)^T$ into $\mathcal{C}_N J_2 \mathcal{C}_N$. In particular the cross phase is not replaced by the determinant.

If, in addition to (OCP19) and (OCP25),

$$\sqrt{a_N} \leq \Theta_N/4, \quad (\text{OCP43})$$

then these two values are nonzero and

$$\boxed{\frac{\varepsilon_N \Theta_N}{2} \leq j_{+,N}, -j_{-,N} \leq \varepsilon_N.} \quad (\text{OCP44})$$

To prove it, their squared observed norms are $a_N + b_N^\circ \mp 2\mathfrak{S}r_N$, lying in $(0, 2]$ because $0 \prec \mathcal{C}_N \preceq I$. Thus the positive fraction in (OCP42) is at least $\det \mathcal{C}_N$, whereas $2|\mathfrak{S}r_N| \leq \Theta_N/2$. The upper bound is the full compressed operator norm bound.

Condition (OCP43) holds eventually by (OCP33)–(OCP34), uniformly over all original cutoffs. Therefore

$$\boxed{\log j_{+,N} = q\psi(s_N) + O_{h,\varpi}(k \log^2 q), \quad \log(-j_{-,N}) = q\psi(s_N) + O_{h,\varpi}(k \log^2 q).} \quad (\text{OCP45})$$

These are actual signed values on specified original observed classes. They are a cutoff-labelled source family, not the separately fixed terminal arithmetic eigenclass v_λ . Nothing in (OCP45) assigns a sign to $\Phi_N(\Lambda v_\lambda)$.

Evaluated return and scalar allowance

For the original four signs at $q-1, q, 2q-1, 2q$, (OCP38) and (OCP45) give

$$\boxed{\mathcal{R} \log \lambda_{+,N} = \mathcal{R} \log(-\lambda_{-,N}) = 2(a_0 - a_1)q + O_{h,\varpi}(k \log^2 q).} \quad (\text{OCP46})$$

The same formula holds with λ_+ and $-\lambda_-$ replaced by j_+ and $-j_-$. These are returns of cutoff-labelled operator spectra or cutoff-labelled classes, respectively. They are not a fixed-class determinant return.

At $N = q-1$, $s_N = 0$; at $N = q$, $s_N = 1/q$; and the upper two arguments are $1, 1+1/q$. The endpoint derivative bound makes their small displacement cost only $O(\log q)$, already included in (OCP46).

A new outward integer calculation encloses

$$a_0 = 0.2415644752704904446910368915632944245\dots, \\ 0.0665189520202739196651 < a_1 < 0.0665189520202739196824,$$

and proves

$$\boxed{0.35009104650043305001 < 2(a_0 - a_1) < 0.35009104650043305006.} \quad (\text{OCP47})$$

The checker uses the actual root equation $E = 2rK$, 4096 positive series terms with geometric tails, rational Machin bounds for π , integer square-root enclosures, and an outward logarithm series. This certifies a universal reference coefficient, not a hypothetical zero or native period.

For any scalar $b \geq 0$, the inequality $\mathscr{W}_{B,N} + bI \succeq 0$ is equivalent to $b \geq -\lambda_{-,N}$. Thus the least scalar positive correction is

$$\boxed{b_N^{\text{opt}} = -\lambda_{-,N} = \exp[q\psi(s_N) + O_{h,\varpi}(k \log^2 q)].} \quad (\text{OCP48})$$

Uniformly through the original window, it exceeds every fixed polynomial in q, k eventually. In particular the original polynomial benchmark $\delta q(k+1)/2$ cannot dominate this canonical observed current. This is a conclusion about that stated metric and observation. A different tensor-section metric, a proper-source construction, or a geometric comparison is not declared equivalent to it.

A new short-time law for the actual arithmetic propagator

The current estimate also controls the actual first-order arithmetic evolution, rather than the heat of its positive energy. Define, in the original observed metric,

$$\mathscr{U}_{B,N}(t) = \Lambda e^{itM} L_N, \quad \mathscr{V}_{B,N}(t) = e^{itM_{B,N}}, \quad M_{B,N} = \Lambda M L_N. \quad (\text{OCP49})$$

These are different operators: the first measures the full original evolution, while the second evolves the actual minimum-section compression. They are not identified at finite time.

In the original observed isometry,

$$M_{B,N} = C_{B,N} + R_{B,N}, \quad C_{B,N} = J_B^\dagger C_N J_B = C_{B,N}^\dagger, \quad R_{B,N} = \varepsilon_N v_N u_N^*,$$

where u_N, v_N are the two columns of U_N . Put

$$\varepsilon_{B,N} = \|R_{B,N}\| = \varepsilon_N \sqrt{a_N b_N^\circ}, \quad z_{B,N} = \text{Tr } R_{B,N} = \varepsilon_N r_N. \quad (\text{OCP50})$$

Thus $R_{B,N}^2 = z_{B,N} R_{B,N}$, and the previous finite inequalities give

$$\varepsilon_N \sqrt{\Theta_N} \leq \varepsilon_{B,N} \leq \varepsilon_N, \quad |z_{B,N}| \leq \varepsilon_N \sqrt{a_N}. \quad (\text{OCP51})$$

The full polynomial multiplication inequality gives the finite bound

$$\|C_N\|, \|C_{B,N}\| \leq \mathfrak{c}_N := 2(N+1)\sqrt{u/\ell}, \quad \log \mathfrak{c}_N = O_{h,\varpi}(k \log^2 q). \quad (\text{OCP52})$$

It follows by applying the original source comparison to yP and P , and the exact Gamma recurrence to P through degree N . The source cutoff in (OCP49) is still N .

Lemma 0.4 (Rank-one propagator estimate with the noncommuting part retained). *Let C be any bounded operator, let R have rank one and norm ϵ , and write $R^2 = zR$. For $T \geq 0$ set*

$$\delta_T = \|C\| e^{T|z|} (T + \epsilon T^2/6).$$

If $\delta_T < 1$, then for $0 \leq t \leq T$,

$$\left| \frac{\|e^{it(C+R)} - I - itR\|}{1 + t\epsilon} \right| \leq e^{T|z|} \left[\frac{\delta_T}{1 - \delta_T} + \frac{T|z|}{2} \right]. \quad (\text{OCP53})$$

No commutation of C and R is assumed.

Proof. Exactly,

$$e^{itR} = I + f_t(z)R, \quad f_t(z) = \frac{e^{itz} - 1}{z}, \quad f_t(0) = it,$$

so $\|e^{itR}\| \leq e^{T|z|}(1 + t\epsilon)$ and $|f_t(z) - it| \leq t^2|z|e^{T|z|}/2$. Duhamel's identity gives

$$e^{it(C+R)} - e^{itR} = i \int_0^t e^{i(t-s)R} C e^{is(C+R)} ds.$$

Let $M_T = \sup_{0 \leq s \leq T} \|e^{is(C+R)}\|/(1 + s\epsilon)$. The exact integral of the norm-ratio factor is

$$\int_0^t \frac{(1 + (t-s)\epsilon)(1 + s\epsilon)}{1 + t\epsilon} ds = t + \frac{\epsilon^2 t^3}{6(1 + t\epsilon)} \leq T + \epsilon T^2/6.$$

Hence $M_T \leq e^{T|z|} + \delta_T M_T$ and the relative Duhamel error is at most $\delta_T e^{T|z|}/(1 - \delta_T)$. Adding the exact rank-one exponential remainder proves (OCP53). $\square$

Use (OCP53) first on the full source, where $z = 0$, and then on the observed compression, where $z = z_{B,N}$. The first error, after observation, must be multiplied by

$$\frac{1 + t\epsilon_N}{1 + t\epsilon_{B,N}} \leq \Theta_N^{-1/2}; \quad (\text{OCP54})$$

this factor is retained. The main term in both cases is exactly $I + itR_{B,N}$.

For every fixed $b > a_0/2$, take the same real time at all cutoffs,

$$t_k(b) = e^{-bq}.$$

Equations (OCP33)–(OCP35), (OCP51)–(OCP54) give, for a fixed $c_b > 0$ smaller than $\min\{b, 2b - a_0\}$,

$$\sup_N \frac{\|\mathcal{U}_{B,N}(t_k) - I - it_k R_{B,N}\| + \|\mathcal{V}_{B,N}(t_k) - I - it_k R_{B,N}\|}{1 + t_k \varepsilon_{B,N}} \leq e^{-c_b q + O_{h,\varpi,b}(k \log^2 q)}. \quad (\text{OCP55})$$

In particular their difference has the same relative bound. This comparison includes the actual hidden-source coupling through Duhamel's complete integral; it is not a claim that observation commutes with evolution.

A rank-one operator R/ϵ of norm one and trace ζ is unitarily equivalent on its nontrivial two-dimensional span to

$$\begin{pmatrix} \zeta & \sqrt{1-|\zeta|^2} \\ 0 & 0 \end{pmatrix}.$$

Its distance from the unit nilpotent matrix is at most $2|\zeta|$. Here $|z_{B,N}|/\epsilon_{B,N} \leq \sqrt{a_N/\Theta_N}$ is exponentially small. The nilpotent shear has exact norm

$$\left\| I + ix \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\| = \frac{\sqrt{x^2 + 4} + x}{2}, \quad x \geq 0. \quad (\text{OCP56})$$

Combining it with (OCP55) proves

$$\|\mathcal{U}_{B,N}(t_k(b))\| = \|\mathcal{V}_{B,N}(t_k(b))\| (1 + o(1)) = \frac{\sqrt{4 + x_{k,N}^2} + x_{k,N}}{2} (1 + o(1)), \quad x_{k,N} = t_k(b)\epsilon_{B,N}. \quad (\text{OCP57})$$

Both relative errors are uniform over the original cutoff window. Their rates follow from (OCP55) and the explicit $\sqrt{a_N/\Theta_N}$ error. The equations do not require an absolute evaluation of the finite $\epsilon_{B,N}$ to obtain the following logarithmic value:

$$\begin{aligned} \frac{1}{q} \log \|\mathcal{U}_{B,N}(e^{-bq})\| &= \frac{1}{q} \log \|e^{ie^{-bq}M_{B,N}}\| + o(1), \\ \frac{1}{q} \log \|\mathcal{U}_{B,N}(e^{-bq})\| &= (\psi(s_N) - b)_+ + o_{h,\varpi,b}(1), \quad b > a_0/2. \end{aligned} \quad (\text{OCP58})$$

It includes equality $\psi(s_N) = b$, because $\log[(\sqrt{4+x^2}+x)/2]$ differs from $\log(1+x)$ by at most a fixed constant.

In particular $a_1 < 1/8 < a_0$ and $1/8 > a_0/2$. At this completely specified common time,

$$\begin{aligned} \mathcal{R} \log \left\| \Lambda e^{ie^{-q/8}M} L_N \right\|_{Q_{B,N}} &= (2 \log(4/\pi) - 1/4)q + O_{h,\varpi}(k \log^2 q), \\ \mathcal{R} \log \left\| e^{ie^{-q/8}M_{B,N}} \right\|_{Q_{B,N}} &= (2 \log(4/\pi) - 1/4)q + O_{h,\varpi}(k \log^2 q). \end{aligned} \quad (\text{OCP59})$$

The coefficient is

$$2 \log(4/\pi) - 1/4 = 0.2331289505409808893820737831265888490 \dots$$

The original uncentered physical evolution is $e^{t\mathcal{A}} = e^{kt/2}e^{itM}$. The common scalar adds $kt/2$ to each logarithmic norm and cancels exactly under the four signs. Thus (OCP59) is also the return of the actual physical scaling evolution, with its central factor retained.

This growth is not eigenvalue growth of the original arithmetic spectrum: every eigenvalue of $e^{it_k M}$ tends to one since $|\omega_{ab}| = O_h(k)$. The intrinsic observed generator has

$$\|M_{B,N}^2\| \leq \mathfrak{c}_N^2 + 2\mathfrak{c}_N\epsilon_{B,N} + |z_{B,N}|\epsilon_{B,N} \leq e^{a_0 q + o(q)}.$$

Hence its evolution eigenvalues also tend uniformly to one when $b > a_0/2$, while its operator norm has the positive exponential rate in (OCP58) where $\psi(s_N) > b$. The original metric nonnormality, rather than an inserted spectrum, produces the evaluated transient.

Verification and the precise remaining boundary

The new checker reconstructs complete finite Gamma-polynomial source Grams, entire relation minima, nontrivial conductor maps, complex observations, and original quotient projections. Its auxiliary fixtures include conductor orders zero and one, output ranks two and three, nonunitary changes of coefficient frame, and arbitrary retained common source masses. Two auxiliary large-gap examples satisfy the finite determinant-dominance guard and check the full numerical value of the rational lower certificate. The native-shaped small fixtures test the finite identities without

asserting the eventual guard there. Separate-column positivity and deletion of the real-coordinate imaginary phase are rejected controls.

The finite checks are not the analytic proof. The asymptotic theorem uses the original source comparison and monic-profile theorem. The latter is neither inferred from a heat determinant nor promoted from a finite auxiliary grid. The bounded additional lower-degree range is proved in (OCP30).

The prior responses already recorded formal rank-two spectra in unknown projected components and asserted related exponential behavior. This paper supplies an independent full-source two-column determinant proof, its explicit positivity guard, the kernel-current bound, and prescribed current-bearing observed classes. It does not present the inherited rank-two characteristic formula as a new discovery.

The latest consulted publication is commit 2b1445b21cc2fcd7e8b7ff34c6aea7139fb6afb5, edition SZ-20260922-024. Its kernel determinant, uniform measured-resolvent and measured frequency-phase results remain unchanged. Those frequency phases are not the first-order arithmetic current calculated here. The separate marked terminal-eigenclass current and the independent proper-source target minimum remain separate calculations; neither is assigned a value by the operator signature.

References

- [1] Koornwinder, T. H., Wong, R., Koekoek, R., & Swarttouw, R. F. (n.d.). Orthogonal polynomials. In *NIST Digital Library of Mathematical Functions*, Chapter 18, equations 18.22.8 and 18.23.7. <https://dlmf.nist.gov/18.22.E8>; <https://dlmf.nist.gov/18.23.E7>.
- [2] Askey, R. A., & Roy, R. (n.d.). Gamma function. In *NIST Digital Library of Mathematical Functions*, Chapter 5, equation 5.7.6. <https://dlmf.nist.gov/5.7>.
- [3] Carlson, B. C. (n.d.). Elliptic integrals. In *NIST Digital Library of Mathematical Functions*, Chapter 19, section 19.4. <https://dlmf.nist.gov/19.4>.
- [4] Zeta research continuation. (2026). *The original exponential-time heat transition and the cost of weak source activation*, H1-H9. Complete supplied file COMPLETE_PROOFS(4).md. The original y convention, complete monic norms and retained monic-profile theorem are the inputs cited here.
- [5] Zeta research continuation. (2026). *The original observation preserves an entire inverse-power determinant at the kq scale*, equations 8-13. Complete supplied file Pastedmarkdown(20260921-031454).md. The exact conductor, first nonzero moment and forward norm are retained.
- [6] Zeta research continuation. (2026). *Displaced lower-root covariance and source-order continuation*. Complete supplied file Pastedmarkdown(20260919-191918).md, monic estimates in section 1. The lower source order remains one.
- [7] Split-Zero research programme. (2026). *The original kernel pencil, canonical section maps, and the coupled physical current*, PEN14-PEN27. Commit 9104d852c6bc5b79e6fc4588c403128a40b7fda3, file workbenches/splitzero-tandem/continuations/20260922-arithmetic-probe-mass/PENCIL_PROOFS.md.
- [8] Split-Zero research programme. (2026). *Original kernel coefficient and measured frequency return*, SZ-20260922-024. Commit 2b1445b21cc2fcd7e8b7ff34c6aea7139fb6afb5, file workbenches/splitzero-tandem/continuations/20260922-kernel-frequency/RESULTS_20260922_024.md.

Part V

Signed-time evolution and full hidden-source defect

Signed-time evolution, extreme singular values, and the original hidden-source defect

22 September 2026. Independent continuation of the supplied *Original observed arithmetic-current plane*, OCP49–59. The finite arguments below are complete. The growing-degree application uses the source comparison, monic profile, two-column area and leakage estimates proved in that supplied manuscript; the exact required estimates are listed in CE20. No current of a separately marked terminal eigenclass is evaluated here.

1. The original spaces and their exact isometry

Retain the original simple-quartet domain $k \equiv 1 \pmod{4}$, $k \geq 17$, $q = (k+1)^2$, $q_0 = (k-7)^2$, $0 < \delta < 1/2$, $\gamma > 2$, the admitted period and branch. Let

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/Q_k, \quad M[p] = [yp], \quad \mathcal{A} = kI/2 + iM. \quad (\text{CE1})$$

For every original cutoff $q-1 \leq N \leq 2q$, let $G = G_N$ be its complete source minimum, $\Lambda : E \rightarrow B$ the original onto map, $K = \ker \Lambda$, and

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q_B, \quad J = L Q_B^{-1/2}. \quad (\text{CE2})$$

Here E has inner product $x^* G y$, conjugate-linear in the first argument; the domain of J has the usual inner product. Direct multiplication gives

$$\Lambda L = I_B, \quad L^* G L = Q_B, \quad J^\dagger J = I_B, \quad J^\dagger = Q_B^{1/2} \Lambda, \quad P = J J^\dagger = L \Lambda. \quad (\text{CE3})$$

Thus P is the orthogonal projection with kernel exactly K . This is an isometry of the actual metric, retaining all original minimum relations. In particular

$$U(t) = J^\dagger e^{itM} J = Q_B^{1/2} (\Lambda e^{itM} L) Q_B^{-1/2}, \quad V(t) = e^{itM_B}, \quad M_B = J^\dagger M J. \quad (\text{CE4})$$

Their usual norms are precisely the original Q_B -operator norms.

Let p_n be the real monic polynomials of the unchanged even source, with full squared norms ω_n . Put $b_n = [p_n]$, $E_N = \|b_N\|_G^2$, $F_N = \|b_{N+1}\|_G^2$. The complete quotient representatives and parity give the original identity OCP5–7:

$$e = b_N / \sqrt{E_N}, \quad f = b_{N+1} / \sqrt{F_N}, \quad \langle e, f \rangle_G = 0, \quad \|e\|_G = \|f\|_G = 1, \quad (\text{CE5})$$

$$M = C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad \epsilon = \sqrt{E_N F_N} / \omega_N.$$

For completeness, the minimum representative of b_{N+1} is $p_{N+1} - Q_k U_{N+1-q}$, where U_j is the monic polynomial for $Q_k^2 d\mu_k$; its degree is at most N . Projecting multiplication by y on the actual minimum-section space leaves the degree- $N+1$ term $[y^N] P p_{N+1}$. Its coefficient is $\langle p_N, P \rangle / \omega_N$, yielding CE5. The remaining compressed multiplication is selfadjoint. Opposite parity yields $R^2 = 0$, with the entire degree- N source unchanged.

Define

$$u = J^\dagger e, \quad v = J^\dagger f, \quad a = \|u\|^2, \quad d = \|v\|^2, \quad r = u^* v, \quad \alpha = \|(I - P)e\|^2 = 1 - a. \quad (\text{CE6})$$

Compression gives

$$C_B = J^\dagger C J = C_B^*, \quad R_B = \epsilon v u^*, \quad \epsilon_B = \|R_B\| = \epsilon \sqrt{ad}, \quad z = \text{tr } R_B = \epsilon r, \quad R_B^2 = z R_B. \quad (\text{CE7})$$

The Gram $\mathcal{C} = \begin{pmatrix} a & r \\ \bar{r} & d \end{pmatrix}$ obeys $0 \preceq \mathcal{C} \preceq I_2$. Since $e \perp f$, $r = -\langle (I - P)e, (I - P)f \rangle$, so $|r| \leq \sqrt{\alpha}$. If the complete two-column certificate supplies $\det \mathcal{C} \geq \Theta > 0$, then

$$\epsilon\sqrt{\Theta} \leq \epsilon_B \leq \epsilon, \quad |z| \leq \epsilon\sqrt{\alpha}, \quad |z|/\epsilon_B \leq \sqrt{\alpha/\Theta}. \quad (\text{CE8})$$

No separate norm lower bounds can replace this determinant certificate.

2. A finite noncommuting propagator estimate at both signed times

For a bounded C , a rank-one R with $\|R\| = \varepsilon$ and $R^2 = zR$, set, for $T \geq 0$,

$$D(T; \varepsilon, z, c) = ce^{T|z|} \left[T + \frac{\varepsilon^2 T^3}{6(1 + T\varepsilon)} \right], \quad c \geq \|C\|. \quad (\text{CE9})$$

Whenever $D < 1$, every $|t| \leq T$ satisfies

$$\frac{\|e^{it(C+R)} - I - itR\|}{1 + |t|\varepsilon} \leq e^{T|z|} \left[\frac{D}{1 - D} + \frac{T|z|}{2} \right]. \quad (\text{CE10})$$

Proof. It suffices first to take $t \geq 0$. The identity $R^j = z^{j-1}R$ for $j \geq 1$ yields

$$e^{itR} = I + f_t(z)R, \quad f_t(z) = (e^{itz} - 1)/z, \quad f_t(0) = it, \quad |f_t(z) - it| \leq \frac{1}{2}t^2|z|e^{T|z|}. \quad (\text{CE11})$$

Differentiate $e^{i(t-s)R}e^{is(C+R)}$ and integrate, obtaining exactly

$$e^{it(C+R)} - e^{itR} = i \int_0^t e^{i(t-s)R} C e^{is(C+R)} ds. \quad (\text{CE12})$$

No interchange of C and R occurs. Let $M_T = \sup_{0 \leq s \leq T} \|e^{is(C+R)}\|/(1 + s\varepsilon)$, finite by continuity. The exact scalar integral is

$$\int_0^t \frac{(1 + (t-s)\varepsilon)(1 + s\varepsilon)}{1 + t\varepsilon} ds = t + \frac{\varepsilon^2 t^3}{6(1 + t\varepsilon)}. \quad (\text{CE13})$$

The right side is increasing in $t \geq 0$. Hence $M_T \leq e^{T|z|} + DM_T$; substitute $M_T \leq e^{T|z|}/(1 - D)$ in CE12, then add CE11. For negative t , apply the same proof to $-C, -R$, whose rank-one trace is $-z$ and whose norms are unchanged. This proves CE10 for both signs. The coarser bound $D \leq ce^{T|z|}(T + \varepsilon T^2/6)$ recovers OCP53.

Apply CE10 to the full space and its actual compression. Put $c = 2(N+1)\sqrt{u_0/\ell_0}$, where $\ell_0\|P\|_\sigma^2 \leq \|P\|_{\mu_k}^2 \leq u_0\|P\|_\sigma^2$ holds through degree $N+1$, retaining the absolute source masses. The exact Gamma recurrence has off-diagonal orthonormal coefficients $\sqrt{n(n-1/2)} \leq n$, so multiplication on degree at most N has norm at most $2(N+1)$. Taking the ratio of these two actual source inequalities, then compressing orthogonally, proves $\|C\|, \|C_B\| \leq c$.

Let $D_F = D(T; \varepsilon, 0, c)$, $D_B = D(T; \varepsilon_B, z, c)$, and suppose both are less than one. Define

$$A_F = \frac{1 + T\varepsilon}{1 + T\varepsilon_B} \frac{D_F}{1 - D_F}, \quad A_B = e^{T|z|} \left[\frac{D_B}{1 - D_B} + \frac{T|z|}{2} \right]. \quad (\text{CE14})$$

For each signed time $t = \pm T$, contraction by $J^\dagger, J$ gives the finite inequalities

$$\|U(t) - I - itR_B\| \leq (1 + T\varepsilon_B)A_F, \quad \|V(t) - I - itR_B\| \leq (1 + T\varepsilon_B)A_B. \quad (\text{CE15})$$

The factor $(1 + T\varepsilon)/(1 + T\varepsilon_B) \leq \Theta^{-1/2}$ is retained; it need not be one. In particular

$$\|U(t) - V(t)\| \leq (1 + T\varepsilon_B)(A_F + A_B). \quad (\text{CE16})$$

3. Exact finite shear, and the forward/backward singular receiver

Set $x = T\varepsilon_B$, $h = |z|/\varepsilon_B \leq 1$, and

$$S(x) = \frac{\sqrt{x^2 + 4} + x}{2}. \quad (\text{CE17})$$

On the span of u, v , a unitary frame places R_B/ε_B into $\begin{pmatrix} \zeta & \sqrt{1-|\zeta|^2} \\ 0 & 0 \end{pmatrix}$, where $\zeta = z/\varepsilon_B$. To construct the frame, take its first vector along the range of R_B , and choose the phase of the second to make the off-diagonal entry nonnegative. The rank-one norm identity fixes that entry.

Its distance from $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is at most $2h$: the diagonal difference is h , and $1 - \sqrt{1 - h^2} \leq h$. Directly computing the two eigenvalues of the shear's Gram gives $S(x)$ and $S(x)^{-1}$ as its singular values. Consequently

$$\left| \frac{\|U(\pm T)\|}{S(x)} - 1 \right| \leq \eta_F := 2A_F + 2h, \quad \left| \frac{\|V(\pm T)\|}{S(x)} - 1 \right| \leq \eta_B := 2A_B + 2h. \quad (\text{CE18})$$

Indeed $S(x) \geq \max\{1, x\}$, so $(1+x)/S(x) \leq 2$ and $x/S(x) \leq 1$.

The exact inverse is $V(T)^{-1} = V(-T)$, because both are exponentials of the same original compressed generator. If $\eta_B < 1$, CE18 proves

$$\begin{aligned} (1 - \eta_B)S(x) &\leq \sigma_{\max}(V(T)) \leq (1 + \eta_B)S(x), \\ \frac{1}{(1 + \eta_B)S(x)} &\leq \sigma_{\min}(V(T)) \leq \frac{1}{(1 - \eta_B)S(x)}, \\ (1 - \eta_B)^2 S(x)^2 &\leq \text{cond}_{Q_B}(e^{iT\Lambda ML}) \leq (1 + \eta_B)^2 S(x)^2. \end{aligned} \quad (\text{CE19})$$

Here condition number means the product of the operator norm and inverse operator norm in the original metric. The finite error is explicit. No inverse identity for the observed full operator $U(T)$ has been used; CE26 below gives its exact inverse defect.

One may retain the small complex trace without any shear replacement: on the two-dimensional active span the squared singular values of $I + itR_B$ are the roots of

$$w^2 - (2 + t^2 \epsilon_B^2 - 2t\Im z)w + |1 + itz|^2 = 0.$$

All other singular values are one. This follows from the Frobenius norm and determinant of its displayed two-by-two matrix and preserves the sign of $t\Im z$.

4. Application to every original cutoff, with the required input displayed

The original two-column and profile proofs OCP29-35 supply, uniformly for $q - 1 \leq N \leq 2q$, with $s_N = (N + 1 - q)/q$,

$$\begin{aligned} \log \epsilon_N &= q\psi(s_N) + O(k \log^2(q + 2)), \quad 0 \leq -\log \Theta_N = O(k \log^2(q + 2)), \\ \alpha_N &\leq \exp[-2q\psi(s_N) + O(k \log^2(q + 2))], \quad \log c_N = O(k \log^2(q + 2)), \\ \det \mathcal{C}_N &\geq \Theta_N > 0. \end{aligned} \quad (\text{CE20})$$

These are the precise inputs to the following propagation, including the full complex area estimate. The present note proves their consequences, not an independent replacement for the conductor argument. The original elliptic profile is

$$1 + s = \frac{E(\sqrt{1 - r_s^2})}{r_s K(\sqrt{1 - r_s^2})}, \quad \psi(s) = s \log \frac{\sqrt{1 - r_s^2}}{E(\sqrt{1 - r_s^2})} + \log \frac{1 + r_s}{E(\sqrt{1 - r_s^2})},$$

with its endpoint limit at $s = 0$. Its endpoint values are $a_0 = \log(4/\pi)$ and $a_1 = \psi(1) > 0$. It is decreasing, has logarithmically bounded derivative near zero, and bounded derivative near one, as retained in OCP31. Write $L_k = k \log^2(q + 2) = o(q)$. To propagate sharper inherited source bounds without concealing them in this coarser allowance, define the actual finite cost

$$\mathfrak{E}_k = 1 + \max_N \left\{ |\log \epsilon_N - q\psi(s_N)|, -\log \Theta_N, \log^+ c_N, \log^+(\alpha_N e^{2q\psi(s_N)}) \right\}. \quad (\text{CE20a})$$

Here $\log^+ x = \max\{0, \log x\}$ for $x > 0$, with $\log^+ 0 = 0$. This is an explicitly specified scalar of the original finite data, not a new assumption. CE20 proves $\mathfrak{E}_k = O(L_k)$. In particular, for the original source cost $\mathcal{L}_k = \log(u_0/\ell_0)$, the multiplication contribution is exactly $\log c_N = \log(2(N + 1)) + \mathcal{L}_k/2$. Every occurrence of $O(L_k)$ in the propagation estimates CE21-23 can be replaced by $O(\mathfrak{E}_k + \log q)$. Thus a proved sharper upper bound on the same source and area scalars transfers directly; no evolution argument is repeated.

For each fixed $b > a_0/2$, take the same physical time $T = e^{-bq}$ at every cutoff. CE8 and CE20 give

$$\log \epsilon_{B,N} = q\psi(s_N) + O(L_k), \quad |z_N| = e^{O(L_k)}, \quad h_N \leq e^{-q\psi(s_N) + O(L_k)}.$$

Moreover $D_F, D_B \leq e^{-bq + O(L_k)} + e^{(a_0 - 2b)q + O(L_k)}$, and the full-to-observed factor in CE14 is $e^{O(L_k)}$. Therefore for every positive $c_b < \min\{b, 2b - a_0, a_1\}$, all finite guards eventually hold and

$$\eta_F + \eta_B \leq e^{-c_b q + O_{h,\varpi,b}(L_k)}. \quad (\text{CE21})$$

Since $\log S(e^w) - \max\{w, 0\}$ is bounded uniformly on the real line, CE18-19 prove the new original-metric extreme singular laws

$$\begin{aligned} \log \|U(e^{-bq})\| &= q(\psi(s_N) - b)_+ + O_{h,\varpi,b}(L_k), \\ \log \sigma_{\max}(V(e^{-bq})) &= q(\psi(s_N) - b)_+ + O_{h,\varpi,b}(L_k), \\ \log \sigma_{\min}(V(e^{-bq})) &= -q(\psi(s_N) - b)_+ + O_{h,\varpi,b}(L_k), \\ \log \text{cond}_{Q_{B,N}}(e^{ie^{-bq}\Lambda M L_N}) &= 2q(\psi(s_N) - b)_+ + O_{h,\varpi,b}(L_k). \end{aligned} \quad (\text{CE22})$$

The exact inverse, rather than a perturbation claim about the smallest singular value of a large shear, proves the third line. It also covers the transition $\psi(s_N) = b$.

The four original signs are $+, +, -, -$ at $N = q-1, q, 2q-1, 2q$. At $b = 1/8$, the certified inequalities $a_1 < 1/8 < a_0$ and $a_0/2 < 1/8$ give

$$\begin{aligned} \mathcal{R} \log \sigma_{\max}(V(e^{-q/8})) &= [2 \log(4/\pi) - 1/4]q + O_{h,\varpi}(L_k), \\ \mathcal{R} \log \sigma_{\min}(V(e^{-q/8})) &= -[2 \log(4/\pi) - 1/4]q + O_{h,\varpi}(L_k), \\ \mathcal{R} \log \text{cond}_{Q_{B,N}}(e^{ie^{-q/8}\Lambda M L_N}) &= [4 \log(4/\pi) - 1/2]q + O_{h,\varpi}(L_k). \end{aligned} \quad (\text{CE23})$$

The profile arguments are exactly $0, 1/q, 1, 1 + 1/q$; no cutoff is merged with its neighbour. Integrating the endpoint derivative bound makes the lower displacement $O(\log q)$, and the upper displacement costs $O(1)$, both already inside $O(L_k)$. The condition-number coefficient is twice the incoming norm coefficient, approximately $0.4662579010819617787641475662531776980$.

For the uncentered physical action $\mathcal{A} = kI/2 + iM$, the scalar $e^{kT/2}$ multiplies both observed evolutions. It cancels in the condition number at each cutoff and cancels exactly from the four-cutoff logarithmic norm and singular-value returns. The arithmetic unit is therefore retained.

5. The complete hidden-source defect and its connecting equations

The second-order difference is exactly

$$U(t) - V(t) = -\frac{1}{2}t^2\mathcal{D} + O(t^3), \quad \mathcal{D} = J^\dagger M(I - P)MJ. \quad (\text{CE24})$$

The statement is a finite-dimensional Taylor expansion at each fixed original cutoff. Its remainder is not being promoted to a uniform growing-degree estimate. Expanding the original $C + R$ gives

$$\mathcal{D} = J^\dagger C(I - P)CJ + J^\dagger C(I - P)RJ + J^\dagger R(I - P)CJ - zR_B. \quad (\text{CE25})$$

Indeed $J^\dagger R(I - P)RJ = J^\dagger R^2J - R_B^2 = -zR_B$. This retains the hidden excursion and calculates its apparent quadratic rank-one contribution exactly. In the original data,

$$\|\mathcal{D}\| \leq c^2 + c\epsilon[\sqrt{a(1-d)} + \sqrt{d(1-a)}] + |z|\epsilon_B.$$

The two mixed bounds follow by substituting $R = \epsilon f e^\dagger$, with $\|(I - P)f\| = \sqrt{1-d}$, $\|(I - P)e\| = \sqrt{1-a}$. Under CE20 this entire upper bound is $\exp[q\psi(s_N) + O(L_k)]$, rather than the unproved estimate obtained by treating two factors of ϵ independently.

There is also an exact all-time inverse defect:

$$U(-t)U(t) = I_B - J^\dagger e^{-itM}(I - P)e^{itM}J. \quad (\text{CE26})$$

It follows by inserting $P = JJ^\dagger = I - (I - P)$ between the two exponentials. In particular $U(-t)U(t) = I - t^2\mathcal{D} + O(t^3)$, consistently with CE24. This identifies the exact map preventing the intrinsic inverse identity from being assigned to the measured full evolution.

Write $M_{BK} = J^\dagger M|_K$, $M_{KB} = (I - P)MJ$, $M_{KK} = (I - P)M|_K$. The coupled source equations yield the exact Volterra identity, for $t \geq 0$,

$$U(t) - V(t) = - \int_0^t \int_0^s e^{i(t-s)M_B} M_{BK} e^{i(s-r)M_{KK}} M_{KB} U(r) dr ds. \quad (\text{CE27})$$

To prove it, put $H(t) = (I - P)e^{itM}J$. Then $U' = iM_B U + iM_{BK}H$, $H' = iM_{KB}U + iM_{KK}H$, with $U(0) = I$, $H(0) = 0$. Solve the second equation by differentiating its variation-of-constants expression and substitute into the first. All finite-dimensional integrals converge. CE27 retains every original kernel coordinate and all mixed actions; its value at the diagonal origin is exactly CE24.

The hidden propagator itself has the finite bound

$$\|e^{itM_{KK}}\| \leq e^{|t|\epsilon\sqrt{\alpha(1-d)}}. \quad (\text{CE28})$$

For a solution $x' = iM_{KK}x$, its squared norm has derivative bounded in absolute value by $2\|R_{KK}\|\|x\|^2$, since C_{KK} is selfadjoint and $\|R_{KK}\| = \epsilon\sqrt{\alpha(1-d)}$. Integrating this scalar differential inequality proves CE28 for both signs. The complete memory integrand has the original finite bound

$$\|M_{BK}e^{itM_{KK}}M_{KB}\| \leq e^{|t|\epsilon\sqrt{\alpha(1-d)}}(c + \epsilon\sqrt{d\alpha})(c + \epsilon\sqrt{a(1-d)}). \quad (\text{CE29})$$

This follows by writing both original mixed blocks explicitly. CE28–29 do not assume that the kernel is invariant. They give the actual hidden object and its maps which control the failure of the measured full operator to be a group.

6. Provenance and exact scope

The supplied full TeX *Original observed arithmetic-current plane*, OCP1–7, OCP19–35 and OCP49–59, defines the original data and contains the conductor proof supplying CE20. The present derivation independently proves CE9–19 and CE24–29; CE22–23 are new applications of the signed-time bounds. Root integration must retain the full supplied proof alongside this note and its original author citations. Its two-column certificate must be checked on the actual original source; this note has not checked that part in place of the root calculation.

The retained kernel-pencil proof, PEN14–27, describes the original canonical sections. The present isometry CE2–3 and full-kernel excursion CE24–29 preserve those sections and the entire minimum.

Human source used for the coefficient recurrence: T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, *Orthogonal Polynomials*, NIST DLMF, 18.22.8 and 18.23.7. Their original TeX is retained in `human_sources/18.22.E8.tex` and `human_sources/18.23.E7.tex`. Substituting $\lambda = 1/4$, $\phi = \pi/2$, $x = y/2$ and multiplying P_n by $n!$ gives $\pi_{n+1} = y\pi_n - n(n - 1/2)\pi_{n-1}$ and the generating function used in the source. The complete mass and norms remain the source's $h_n = \sqrt{2\pi}n!(1/2)_n$. The elliptic profile retains B. C. Carlson's DLMF Chapter 19 conventions cited by the source. The elementary exponential, Duhamel and singular-value calculations used here are all proved above.

Neither the positive/negative operator signature nor the large singular values determine the current of the separately marked arithmetic eigenclass. That remains a different, actual matrix evaluation on its prescribed vector; no sign has been assigned to it by this continuation.

Part VI

Effective cutoff: complete incoming proof

Status, original objects, and principal results

The repository advanced during this calculation. Edition SZ-20260922-025, read at commit 76473e0564b32f7566e191829bf67477c8950f0b, now contains the complete original cutoff profile, its fixed subquotient consequences, elliptic derivatives and endpoint correction, and phase/heat receivers. Those conclusions are not presented here as new. This continuation strengthens their scalar and localization remainders to explicit finite bounds, obtains an effective single-sequence native rate, and uses that rate to evaluate shrinking-window source responses and their negative mesoscopic curvature. It also gives a separately certified integrated profile. The full derivations are retained so that the quantitative upgrade can be audited rather than inferred from a summary.

The decisive new scalar estimate is (CU7a): it replaces the unevaluated zero-statistic remainder in the newer PP24–PP28 calculation by $150 + 3 \log(n+2)$ on its entire ratio range. The sharper constant 110 applies on the subrange used in this proof's jet reduction. Section 8 records the exact integration with the newly arrived edition and proves the additional shrinking-window results.

The baseline repository proof is KokunoYumeto/zeta-function-research-reader, commit 2b1445b21cc2fcd7e8b7ff34c6aea7139fb6afb5. Its COMBINED_KERNEL_AND_RECEIVERS.md (CK1–19), complete POWER_JET_PROOF.md (PJ1–70), and ELLIPTIC_LOG_MOMENT_PROOF.md (EL1–14) are inherited inputs. The independent EIQ1–33 equilibrium proof was also read from the original uploaded cumulative source, lines 4364–4812; that exact byte range is included. A failed direct fetch of the much larger GitHub source bank is not counted as reading it. The original current phases are not evaluated by this continuation.

Retain

$$k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad m = 8k - 16, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\ Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E_k = \mathbb{C}[y]/Q_k, \quad M[p] = [yp]. \quad (\text{CU1})$$

The original physical action is $kI/2 + iM$. The full observation is Λ_k , its kernel is K_k , and I_K is its fixed coefficient frame. At original polynomial cutoff N , the attained metric is G_N , including the complete native measure and all relation polynomials. Put

$$H_{K,N} = I_K^* G_N I_K, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}.$$

A matrix star is conjugate transpose in the displayed coefficient frames. Every physical adjoint uses its specified metric. The full source comparison, for every polynomial through degree $2q$, is

$$\ell_k \|P\|_\sigma^2 \leq \|P\|_{\mu_k}^2 \leq u_k \|P\|_\sigma^2, \quad \kappa_k = u_k / \ell_k, \quad \log \kappa_k = o(q), \\ d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}. \quad (\text{CU2})$$

For quantitative orders one may use the actual stronger finite provider $\log \kappa_k = O_h(k + \log q)$. The proof states explicitly where even this finite source comparison is needed.

Index the whole original window by

$$N_j = q - 1 + j, \quad 0 \leq j \leq q + 1, \quad t_j = j/q. \quad (\text{CU3})$$

The original four signs are at $j = 0, 1, q, q + 1$, not at a newly chosen window. We will use the full interval $0 \leq t \leq 1$ for limits, and a harmless extension to $11/10$ to include all finite band shifts.

Define

$$F(z) = \frac{2}{\pi} K(\sqrt{z}) = \sum_{n=0}^{\infty} \frac{\binom{2n}{n}^2}{16^n} z^n, \quad 0 \leq z < 1.$$

For $t > 0$, let z_t be the unique solution of

$$t = 4z_t F'(z_t) / F(z_t).$$

The profile is

$$\boxed{\beta(t) = 2 \log F(z_t) - \frac{t}{2} \log z_t, \quad \beta(0) = 0.} \quad (\text{CU4})$$

At the original endpoint, $2\beta(1) = C_\theta$, as proved by the retained EL dictionary.

The principal native result is

$$\boxed{\sup_{0 \leq j \leq q+1} \left| \frac{\log \det H_{K, N_0} - \log \det H_{K, N_j}}{mq} - \beta(t_j) \right| \rightarrow 0.} \quad (\text{CU5})$$

A stronger result below controls the mean absolute displacement of **every** logarithmic relative eigenvalue, and applies to fixed coefficient subquotients. It supplies an explicit finite error and an eventual rate

$$O_{h,A} \left(\frac{\log \log(k+e)}{\sqrt{\log(k+e)}} + \frac{e^{2\sqrt{\log(k+e)}}}{k} + \frac{\log \kappa_k}{q} \right). \quad (\text{CU6})$$

The rate is asserted only once the displayed finite guards hold; the slowly shrinking band used to prove it has a very large sufficient threshold. The scalar Gamma theorem itself has the much sharper uniform bound

$$\boxed{|\log[d_q K_{q,M}(0,0)] - q\beta(M/q)| \leq 110 + 3 \log(q+2), \quad q \geq 40, \quad 0 \leq M \leq \lfloor 11q/10 \rfloor,} \quad (\text{CU7})$$

where $d_q = \int |y|^{2q} d\sigma$, and $K_{q,M}$ is the original evaluation kernel in that full measure. The large numerical constant is a deliberately uniform simplification; the parameter-dependent finite sandwich in (HK10) is substantially sharper. The same proof also yields the full range used by the new PP28:

$$\boxed{|\log[d_q K_{q,M}(0,0)] - q\beta(M/q)| \leq 150 + 3 \log(q+2), \quad 0 \leq M \leq \lfloor 3q/2 \rfloor, \quad q \geq 40.} \quad (\text{CU7a})$$

The cutoff symbols in this scalar theorem are integers, not physical arithmetic indices. Application to the original source is through the maps proved in Sections 3-4.

1. The exact scalar profile

1.1 Equilibrium dictionary, with every scale retained

For $t > 0$, use the retained equilibrium probability for

$$V_t(x) = \frac{\pi\sqrt{x} - 2 \log x}{t}, \quad x > 0.$$

Its parameters in EIQ and EL are exactly $(\alpha, \beta_{\text{pot}}) = (2/t, \pi/t)$. Write its support as $[a_t, b_t] = [u_t^2, v_t^2]$, and set

$$r_t = u_t/v_t, \quad \varkappa_t = \sqrt{1 - r_t^2}, \quad w_t = (u_t + v_t)/2, \quad p_t = u_t v_t, \quad c_t = (v_t^2 - u_t^2)/4.$$

The endpoints solve

$$u_t K(\varkappa_t) = 2, \quad v_t E(\varkappa_t) = 2(1+t). \quad (\text{PR1})$$

The exact EIQ density, obtained from its positive Cauchy-transform representation, is

$$\rho_t(x) = \frac{\sqrt{(b_t - x)(x - a_t)}}{4\pi t x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a_t+s)(b_t+s)}}. \quad (\text{PR2})$$

It is zero outside the support, positive inside it, and has mass one. EIQ proves the full Euler inequality, on all of $(0, \infty)$:

$$V_t(x) - 2 \int \log |x - y| d\mu_t(y) \geq \lambda_t,$$

with equality precisely on the support. EL gives

$$\lambda_t = 4 + 4/t - (4/t) \log w_t - 2 \log c_t,$$

$$L_t := \int \log x d\mu_t = (4/t + 2) \log w_t - (2/t) \log p_t - 2. \quad (\text{PR3})$$

Therefore

$$\boxed{\beta(t) = 2 \log(4/\pi) - 2 + \frac{t}{2}(\lambda_t + 2L_t)} \\ = 2 \log(4/\pi) + 2(1+t) \log w_t - 2 \log p_t - t \log c_t. \quad (\text{PR4})$$

This exact parameter dictionary, not an extrapolation from $t = 1$, will enter the norm minimum.

1.2 Hypergeometric coordinate and strict concavity

Put $z = ((1-r_t)/(1+r_t))^2$. The Landen identities (DLMF 19.8.12), or their power-series verification in the defining elliptic integrals, give

$$K(\varkappa_t) = (1 + \sqrt{z})K(\sqrt{z}), \\ E(\varkappa_t) = \frac{2E(\sqrt{z}) - (1-z)K(\sqrt{z})}{1 + \sqrt{z}}.$$

Substitution into $E(\varkappa_t)/(r_t K(\varkappa_t)) = 1 + t$ gives

$$t = 2 \left[\frac{E(\sqrt{z})}{(1-z)K(\sqrt{z})} - 1 \right] = 4zF'(z)/F(z). \quad (\text{PR5})$$

The last equality follows by differentiating the defining integral. Substitution into (PR4) gives exactly (CU4).

Give $n \geq 0$ the probability weight $a_n z^n / F(z)$, $a_n = \binom{2n}{n}^2 / 16^n$. Then $t = 4\mathbb{E}_z n$ and

$$\frac{dt}{d \log z} = 4 \text{Var}_z(n) > 0.$$

The aspect runs from zero to infinity, because $F(z)$ diverges logarithmically while $F'(z)$ diverges at order $(1-z)^{-1}$. Alternatively, the endpoint uniqueness in EIQ gives the same onto statement. Differentiating (CU4), the terms involving z'_t cancel, proving

$$\boxed{\beta'(t) = -\frac{1}{2} \log z_t > 0, \quad \beta''(t) = -\frac{1}{2z_t t'(z_t)} < 0.} \quad (\text{PR6})$$

Thus β is strictly increasing and strictly concave.

The hypergeometric differential equation $z(1-z)F'' + (1-2z)F' - F/4 = 0$ gives the exact Riccati form

$$4z(1-z)t'(z) = 4z(t+1) - (1-z)t^2. \quad (\text{PR7})$$

Equivalently,

$$\beta'' = -\frac{2(e^{2\beta'} - 1)}{4(t+1) - t^2(e^{2\beta'} - 1)}.$$

These are scalar equations on the explicitly defined profile; no original arithmetic eigenvalue is replaced by z_t .

Coefficientwise,

$$4(1-z)F'(z) = \sum_{n \geq 0} \frac{a_n}{n+1} z^n \leq F(z), \quad 4F'(z) \geq F(z).$$

Therefore

$$z \leq t(z) \leq \frac{z}{1-z}, \quad \frac{t}{1+t} \leq z_t \leq t.$$

For $0 < t \leq 1$, integrating (PR6) yields

$$\boxed{\frac{t}{2}(1 - \log t) \leq \beta(t) \leq \frac{(1+t) \log(1+t) - t \log t}{2}.} \quad (\text{PR8})$$

In particular the error after the first term is between zero and $t^2/4$. The exact local series begins

$$\beta(t) = \frac{t}{2}(1 - \log t) + \frac{7}{32}t^2 - \frac{43}{768}t^3 + \frac{497}{24576}t^4 - \frac{1381}{163840}t^5 + O(t^6). \quad (\text{PR9})$$

1.3 Certified scalar values and a new integrated value

The integer-interval calculation in `code/certify_profile.py` uses no floating point. All displayed decimals in this table are outward rational intervals.

Quantity	Lower endpoint	Upper endpoint
$\beta(1/4)$	0.31115542601070381606034394785043	0.31115542601070381606034394785044
$\beta(1/2)$	0.47202413882565744657617131470452	0.47202413882565744657617131470453
$\beta(3/4)$	0.58720530893796571051071192663424	0.58720530893796571051071192663425
$\beta(1)$	0.67714099391264606644199488757692	0.67714099391264606644199488757693
$\beta'(1)$	0.32298207997956527877403877885087	0.32298207997956527877403877885088

The full data additionally include aspects $1/100$, $1/10$, $11/10$, and $3/2$. The endpoint agrees with the independently retained interval for $C_\partial/2$.

The integrated profile has the exact one-dimensional representation

$$\int_0^1 \beta(t) dt = \beta(1) + \frac{1}{4} \log z_1 - \frac{1}{4} \int_0^{z_1} \frac{t(z)^2}{z} dz. \quad (\text{PR10})$$

Integrate $d\beta = -\frac{1}{2} \log z dt$ by parts twice. The terms at zero vanish because $t(z) = z + O(z^2)$.

On $|z| = R < 1$, the integral for F has $\Re F(z) \geq [2(1+R)]^{-1/2}$, while $|F'(z)| \leq [2(1-R)^{3/2}]^{-1}$. Consequently

$$|t(z)|^2 \leq \frac{8R^2(1+R)}{(1-R)^3}.$$

At $R = 4/5$ this is 1152. If the Taylor integral in (PR10) is retained through degree L , its omitted absolute contribution before the factor $1/4$ is at most

$$\frac{1152}{L+1} \frac{(z_1/R)^{L+1}}{1 - z_1/R}. \quad (\text{PR11})$$

The calculation uses exact rational division for the coefficients of $4zF'/F$, degree $L = 224$, and this symmetric tail. It proves

$$0.436944268921521509525169751272 < \int_0^1 \beta(t) dt < 0.436944268921521509525169751273. \quad (\text{PR12})$$

Also,

$$0.697083979657197470191113447667 < \frac{\beta(1/2)}{\beta(1)} < 0.697083979657197470191113447668. \quad (\text{PR13})$$

These certify the universal scalar profile and its integrals, not the finite native convergence error.

2. Effective evaluation of the full Gamma kernel

2.1 Explicit Gamma envelopes and mass

Write $w(y) = |\Gamma(1/4 + iy/2)|^2/(2\pi)$. We use the concrete constants

$$\frac{1}{2}y^{-1/2}e^{-\pi y/2} \leq w(y) \quad (y \geq 1), \quad w(y) \leq 32y^{-1/2}e^{-\pi y/2} \quad (y > 0). \quad (\text{HK1})$$

Here is a proof rather than a qualitative use of Stirling's formula. For $z = 1/4 + iy/2$, $y \geq 1$, the first log-Gamma remainder has magnitude at most $\sec^2(\arg z/2)/(12|z|) \leq 1/(3y)$ (DLMF 5.11(ii)). Taking real parts gives

$$\log(\sqrt{y}e^{\pi y/2}w(y)) = \frac{1}{2} \log 2 - \frac{1}{4} \log(1 + 1/(4y^2)) + y \arctan(1/(2y)) - \frac{1}{2} + 2\Re R(z).$$

It lies between $\frac{1}{2} \log 2 - 37/48$ and $\frac{1}{2} \log 2 + 2/3$. This is strictly between $\log(1/2)$ and $\log 3$. For $0 < y \leq 1$, the Gamma integral gives $|\Gamma(1/4 + iy/2)| \leq \Gamma(1/4) < 9/2$; using $e^{\pi/2} < 8$, $2\pi > 6$, yields the stated upper constant 32.

Put $d\lambda_n = |y|^{2n-1/2} e^{-\pi|y|/2} dy$. The elementary Laguerre derivative bound used in PJ3 implies, on every polynomial of degree at most $2n + 2$, for $n \geq 36$,

$$\frac{1}{4} \|P\|_{\lambda_n}^2 \leq \|P\|_{\nu_n}^2 \leq 32 \|P\|_{\lambda_n}^2, \quad d\nu_n = |y|^{2n} d\sigma. \quad (\text{HK2})$$

For clarity, the lower half follows by excluding $|y| < 1$ in (HK1): integration by parts in the Laguerre weight gives $\|P/y\|_{\lambda_n} \leq [8(\pi/2)/n] \|P\|_{\lambda_n}$. Consequently the mass on $|y| < 1$ is at most $64(\pi/2)^2/n^2 \leq 1/4$ of the full squared norm. The rational expression P/y is integrable in this weight and is not asserted to be a polynomial. The full PJ3 proof, including its highest allowed degree and both half-lines, is retained as an input to (HK2).

Its mass comparison is particularly explicit. With

$$d_n = \nu_n(\mathbb{R}), \quad \eta_0 = 2 \log(4/\pi) - 2,$$

Stirling's real remainder for $\Gamma(2n + 1/2)$, combined with (HK2), gives

$$-\frac{1}{16n} \leq \log d_n - [2n \log n + \eta_0 n] \leq \log 128 + \frac{1}{24n} < 5. \quad (\text{HK3})$$

Indeed the full mass of λ_n is $2\Gamma(2n + 1/2)(2/\pi)^{2n+1/2}$; its own remainder is between $\log 4 - 1/(16n)$ and $\log 4 + 1/(24n)$. The factors 2 and the Gamma mass have not been dropped.

2.2 Exact change of variable for evaluation, not a monic minimum

Let $M = 2h \geq 2$, $t = M/q$. Symmetry makes the minimum representative of evaluation at zero even. Write it as $P(y) = R(y^2/q^2)$, where $\deg R \leq h$, $R(0) = 1$. Then

$$\|P\|_{\nu_q}^2 = q^{2q+1} \int_0^\infty x^{q-1/2} w(q\sqrt{x}) |R(x)|^2 dx. \quad (\text{HK4})$$

There is no monic factor q^{2M} in this map: it is the evaluation minimum. The two half-lines contribute exactly the density in (HK4).

The comparison integrand is

$$q^{2q+1/2} e^{-hV_t(x)} x^{-3/4} |R(x)|^2.$$

The upper Gamma bound is global. The lower Gamma bound holds on $[a_t, b_t]$ when $qu_t \geq 1$.

2.3 Harmonic-measure lower bound, including its entropy constant

Let ω_0 be harmonic measure of $[a, b]$ at the exterior point 0:

$$d\omega_0(x) = \frac{\sqrt{ab}}{\pi x \sqrt{(b-x)(x-a)}} dx, \quad a = u^2, \quad b = v^2.$$

Put $g = \log[(u+v)/(v-u)]$. For every polynomial R of degree at most h , with $R(0) = 1$,

$$\int \log |R| d\omega_0 \geq -hg. \quad (\text{HK5})$$

To prove this, map $|\zeta| > 1$ onto the slit exterior by $x = (a+b)/2 + (b-a)(\zeta + \zeta^{-1})/4$. The preimage of zero is $\zeta_0 = -(u+v)/(v-u)$. The analytic function $\zeta^{-h} R(x(\zeta))$ is bounded at infinity. Poisson-Jensen at ζ_0 gives (HK5). For a root $s \in [a, b]$, the same function for $R(x) = x - s$ has no zero in the exterior, so equality gives $\int \log |x - s| d\omega_0 = \log s - g$. Boundary zeros are handled by their integrable logarithmic limits.

Integrating the original Euler equality against ω_0 therefore gives $\int V_t d\omega_0 + 2g = \lambda_t + 2L_t$. Jensen's inequality on the same complete interval now proves

$$\int_a^b |R|^2 e^{-hV_t} x^{-3/4} dx \geq Z_t e^{-h(\lambda_t + 2L_t)},$$

where the entropy constant is exactly

$$Z_t = \exp \int \log \frac{x^{-3/4}}{\omega'_0(x)} d\omega_0 = \pi c_t \sqrt{p_t} w_t^{-5/2}. \quad (\text{HK6})$$

For verification, $\int \log x d\omega_0 = 2 \log p_t - 2 \log w_t$, and the two endpoint logarithms are $2 \log u_t - g$, $2 \log v_t - g$. Substitution in the density of ω_0 gives (HK6).

2.4 Complete quantile trial upper bound

Let $W_t = b_t - a_t$, $D_t = W_t \|\rho_t\|_\infty$, and let $x_1, \dots, x_h$ be the midpoint quantiles of the actual equilibrium probability. Put $P_h(x) = \prod_i (x - x_i)$ and use the admissible trial $R = P_h/P_h(0)$.

Its empirical CDF differs from μ_t 's by at most $1/(2h)$. Truncate the logarithm at distance W_t/h , not at a fixed absolute distance. On $0 \leq x \leq B_*$, with $B_* \geq 2b_t + 1$, the truncated logarithm has variation at most $2 \log(hB_*/W_t)$. The difference between its continuous truncated and untruncated integral is at most $2D_t/h$. Consequently

$$\log |P_h(x)| \leq h \int \log |x - y| d\mu_t(y) + \log(hB_*/W_t) + 2D_t.$$

Also $|P_h(0)|^2 \geq (a_t/b_t)e^{2hL_t}$, by the monotone variation of $\log x$ on the support.

Choose B_* so that, for all $x \geq B_*$,

$$(2/t + 2) \log x - \frac{\pi}{2t} \sqrt{x} \leq -\lambda_t.$$

The full Euler inequality gives the integral bound on $(0, B_*]$; on the remaining tail use $|P_h(x)| \leq x^h$. The result, including both regions, is

$$\int_0^\infty |R|^2 e^{-hV_t} x^{-3/4} dx \leq U_t(h) e^{-h(\lambda_t + 2L_t)},$$

$$U_t(h) = \frac{b_t}{a_t} \left[4B_*^{1/4} (hB_*/W_t)^2 e^{4D_t} + 2\sqrt{\frac{2\pi}{h(\pi/t)}} \right]. \quad (\text{HK7})$$

This is a linear-space evaluation bound; no zero distribution of the unknown orthogonal polynomial was used.

2.5 Uniform finite remainder, including the small-aspect boundary

On $0 < t \leq 11/10$, the positive coefficient series gives the sharper uniform bound

$$z_t < 11/20.$$

Indeed the exact rational inequalities

$$-\frac{11}{10} + \sum_{j=1}^{12} (4j - 11/10) a_j (11/20)^j > \frac{3}{10000}, \quad \sum_{j=1}^{12} j a_j (11/20)^{j-1} > \frac{3}{5}$$

prove respectively $t(11/20) > 11/10$ and $F'(11/20) > 3/5$; every omitted coefficient is positive. The endpoint formulas give the conservative bounds

$$\frac{1}{16} < a_t < b_t < 45, \quad 4\sqrt{t/3} \leq W_t \leq 45\sqrt{t}.$$

The density bound has a substantially smaller uniform constant than these separate endpoint bounds suggest. The elementary maximum

$$\max_{a < x < b} \frac{\sqrt{(b-x)(x-a)}}{x} = \frac{b-a}{2\sqrt{ab}}$$

is attained at $x = 2ab/(a+b)$, as direct differentiation verifies. Also $\int_0^\infty s^{1/2}(a+s)^{-2} ds = \pi/(2\sqrt{a})$. Applying both estimates inside the complete density (PR2) gives

$$D_t \leq \frac{W_t^2}{16ta_t\sqrt{b_t}} = \frac{1 + \sqrt{z}}{\pi(1-z)^3 F'(z)} =: D_{\text{bd}}(z), \quad z = z_t.$$

For the last identity, the exact endpoint formulas are

$$u_t = \frac{4}{\pi(1 + \sqrt{z})F(z)}, \quad v_t = \frac{4}{\pi(1 - \sqrt{z})F(z)}, \quad t = 4zF'/F.$$

Differentiating the positive integral for F' shows $F''/F' \leq 3/[2(1-z)]$. Therefore

$$\frac{d}{dz} \log D_{\text{bd}}(z) = \frac{1}{2\sqrt{z}(1+\sqrt{z})} + \frac{3}{1-z} - \frac{F''}{F'} > 0.$$

At $z = 11/20$, use $\sqrt{z} < 3/4$, $\pi > 3$, and the displayed rational lower bound for F' . It follows that

$$D_t < \frac{70000}{6561} < 11, \quad Z_t \geq \sqrt{t}/2048. \quad (\text{HK8})$$

One uniform choice is $B_* = 2^{16}$. Indeed $t\lambda_t \leq 20$ by (PR3) and the endpoint bounds, while $(2+2t)\log x - (\pi/2)\sqrt{x} < -20$ at $x = 2^{16}$ and decreases thereafter. With $h = tq/2$, (HK7) yields

$$U_t(h) \leq 5760B_*^{9/4} e^{44} q^2. \quad (\text{HK9})$$

For $q \geq 40$, $q\sqrt{a_t} > 1$, so the Gamma lower bound applies.

Combining (HK1), (HK4), (HK6), and (HK7) gives the finite sandwich

$$\frac{1}{2} Z_t q^{2q+1/2} e^{-h(\lambda_t+2L_t)} \leq K_{q,2h}(0,0)^{-1} \leq 32U_t(h) q^{2q+1/2} e^{-h(\lambda_t+2L_t)}. \quad (\text{HK10})$$

This is the primary finite estimate. It is valid for arbitrary fixed positive aspect when the explicit lower-envelope guard and tail choice hold.

For $0 < t \leq 11/10$, (HK3), (HK8)–(HK10), and (PR4) prove (CU7) for even M . For odd M , parity gives exactly $K_{q,M}(0,0) = K_{q,M-1}(0,0)$. Concavity and (PR8) give

$$q[\beta(M/q) - \beta((M-1)/q)] \leq q\beta(1/q) \leq 1 + \frac{1}{2} \log q.$$

The even error is at most a fixed constant plus $(5/2)\log q$; the displayed constant 110 covers all fixed terms, leaving (CU7). At $M = 0$, $d_q K_{q,0}(0,0) = 1$ exactly.

To obtain the larger range in (CU7a), use $0 < t \leq 3/2$. At $z = 16/25$, the first ten nonconstant coefficients of $4zF' - (3/2)F$, including its constant term $-3/2$, exceed $1/100$; the first ten coefficients of F' exceed $3/4$. All omitted coefficients are positive. Hence $z_t < 16/25$. The same monotonic density expression gives

$$D_t < \frac{9/5}{3(9/25)^3(3/4)} = \frac{12500}{729} < 18.$$

The bounds $1/16 < a_t < b_t < 45$, $W_t \geq 4\sqrt{t/3}$, and $Z_t \geq \sqrt{t}/2048$ persist. These can be checked from the same elementary endpoints: $1 \leq F(z) \leq (1-z)^{-1/2} \leq 5/3$ and $z \leq 16/25$ give $u > 1/3$ and $v < 20/3$. Also

$$W_t = \frac{64\sqrt{z}}{\pi^2(1-z)^2 F(z)^2} \geq \frac{4\sqrt{z}}{1-z} \geq 4\sqrt{t},$$

since $t \leq z/(1-z)$. The weaker width displayed above is therefore valid. In (HK6), use $c_t \geq \sqrt{t/3}$, $\sqrt{p_t} > 1/4$, and $w_t < 7$ to obtain the stated entropy floor. Finally

$$t\lambda_t \leq 4t + 4 + 4\log 4 - t\log(t/3) < 20 \quad (0 < t \leq 3/2),$$

using $-t\log(t/3) \leq 3/e$. The upper width may be replaced by $60\sqrt{t}$, which does not worsen the quantile error because its denominator uses the lower width bound. Also $t\lambda_t \leq 20$ and the same $B_* = 2^{16}$ remain valid. Thus (HK9) holds with e^{72} in place of e^{44} . The even logarithmic remainder is bounded by $72+9+36+7/2+1+(5/2)\log q$; the remaining odd-degree shift costs at most $1+(1/2)\log q$. The constant 150 covers these terms for every $q \geq 40$, proving (CU7a).

This proves a uniform $O(\log q)$ remainder without consuming the monic zero-statistic limit in PJ or PP. The inherited equilibrium Euler theorem and its exact elliptic dictionary remain required. For the newer paper's notation $F_n(M) = \log[d_n K_{n,M}(0,0)]$, its exact PP28 remainder now satisfies

$$\epsilon_n^G = \max_{0 \leq M \leq \lfloor 3n/2 \rfloor} |F_n(M)/n - J(M/n)| \leq \frac{150 + 3\log(n+2)}{n}. \quad (\text{HK11})$$

Here its J is precisely our β , by EP8 and (CU4); no variable or source metric is changed.

3. Every vector in a growing power band

This section makes the band error explicit. Its constants can also be replaced by the sharper finite PJ constants without changing any map.

Let $q \geq 40$, $1 \leq b \leq q/10$, $n = q - b$. In this section b is an integer band width, not an equilibrium endpoint. The map

$$J_{q,b}P = (q^j[y^j]P)_{j=0}^{b-1}$$

defines the full attained jet metric $H_{b,N}^{\text{jet}}$ on $\mathcal{P}_{N-n}$ in $d\nu_n$. Multiplication by y^n identifies its full affine fibres with the high coefficient band modulo y^q , with no change of source degree.

The explicit constants in the multiplication bound PJ16 may now be chosen as

$$\kappa_\Gamma = \sqrt{128}, \quad c_\Gamma = (4\pi\kappa_\Gamma)^{-1}, \quad C_\Gamma = 40\kappa_\Gamma/\pi. \quad (\text{JB1})$$

They follow from (HK1)–(HK2), the complete Laguerre derivative estimate, and its final raising row. In particular, $c_\Gamma n \|P\|_{\nu_n} \leq \|yP\|_{\nu_n} \leq C_\Gamma n \|P\|_{\nu_n}$ through the stated PJ degree guard.

3.1 A finite low metric constant

Put $\alpha = 2n - 1/2$, $\vartheta = \pi/2$, and

$$\gamma_j = \frac{j!(\alpha+1)_j}{(\vartheta q)^{2j}}, \quad R_b = (b+1)3^b,$$

$$\mathcal{L}_{q,b} = 128R_b^2 \max \left\{ \max_{0 \leq j \leq b} \gamma_j, \left(\min_{0 \leq j \leq b} \gamma_j \right)^{-1} \right\}. \quad (\text{JB2})$$

Then at $N = q - 1$ and $N = q$,

$$\mathcal{L}_{q,b}^{-1} d_n I_b \preceq H_{b,N}^{\text{jet}} \preceq \mathcal{L}_{q,b} d_n I_b. \quad (\text{JB3})$$

To prove it, in the scaled coefficient variable the monic Laguerre change of basis has entries

$$T_{ij} = (-1)^{j-i} \binom{j}{i} \frac{(\alpha + i + 1)_{j-i}}{(\vartheta q)^{j-i}}, \quad i \leq j.$$

Its inverse has the same entry magnitudes; multiplication of the two finite triangular matrices verifies that identity. Since each factor is at most 2 in this range, both operator norms are at most R_b . The orthogonal squared norms divided by their original half-line mass are precisely γ_j . Apply (HK2) on both half-lines; its ratio is 128. This bounds the complete degree- b coefficient Gram. A prescribed jet through degree $b - 1$ has coefficient norm at least the prescribed norm for every lift; choosing its degree- $b - 1$ representative supplies the upper minimum. This proves both cutoffs in (JB3).

3.2 Full high covariance, not its diagonal only

For $j \geq 1$, put

$$N_j = q - 1 + j, \quad M = j + b - 1, \quad L = j - 1, \quad B_0 = \frac{q}{c_\Gamma n}, \quad D_0 = \max(1, C_\Gamma n/q).$$

For a common bound on all $0 \leq j \leq q + 1$, set

$$U_{q,b} = \sum_{2h < b} \binom{\lfloor (q+b)/2 \rfloor}{h}^2 B_0^{4h} + B_0^2 \sum_{2h+1 < b} \binom{\lfloor (q+b-1)/2 \rfloor}{h}^2 B_0^{4h},$$

$$V_{q,b} = \sum_{h=0}^{\lfloor (b-1)/2 \rfloor} \binom{\lfloor q/2 \rfloor + h - 1}{h} B_0^{2h}, \quad L_{q,b} = bV_{q,b}^2 D_0^{2(b-1)}. \quad (\text{JB4})$$

Then

$$\boxed{\frac{\mathbf{K}_{n,L}(0,0)}{L_{q,b}} I_b \preceq (H_{b,N_j}^{\text{jet}})^{-1} \preceq U_{q,b} \mathbf{K}_{n,M}(0,0) I_b.} \quad (\text{JB5})$$

All sums are finite and all entries in these constants are specified.

For the upper covariance, pair the actual orthogonal-polynomial zeros. The even $2h$ -th divided coefficient is bounded by its constant coefficient times $\binom{\lfloor M/2 \rfloor}{h} (c_\Gamma n)^{-2h}$. The odd first-derivative

evaluation has squared kernel at most $(c_\Gamma n)^{-2} K_{n,M}(0,0)$, by the full comparison between the positive pushforward measures $d\rho_n$ and $x d\rho_n$. The remaining odd coefficients obey the same paired-root bound. After the retained factors q^{2h} and q^{2h+1} , summing row norms bounds the whole covariance by the first matrix in (JB4). It is not an entrywise-to-Loewner inference without a trace bound.

For the lower covariance, put $H_L(y) = K_{n,L}(y,0)/K_{n,L}(0,0)$. Its zeros, if present, have modulus at least $c_\Gamma n$, since yH_L is the next odd orthogonal polynomial. The Taylor coefficients of H_L^{-1} , through degree $b-1$, are bounded by the complete homogeneous coefficients in $V_{q,b}$. Multiply H_L by the full truncated inverse times the prescribed jet. This produces a polynomial of degree at most $L+b-1 < M+1$ with exactly that jet. Applying the complete multiplication bound at most $b-1$ times gives squared norm at most $L_{q,b} \|a\|_2^2 / K_{n,L}(0,0)$. This simultaneous right inverse proves the lower bound in (JB5), including all complex jet combinations.

Changing the weight ν_n to $\nu_q = y^{2b} \nu_n$ gives

$$|\log[d_n K_{n,h}(0,0)] - \log[d_q K_{q,h}(0,0)]| \leq 2b \log(C_\Gamma/c_\Gamma) \quad (\text{JB6})$$

for the present degree range. Apply the full raising bound repeatedly to the same polynomial and to the constant; the mass factors cancel only in the displayed product.

Combining (CU7), (JB3), (JB5), and (JB6), every relative eigenvalue λ of the low jet metric against the cutoff N_j satisfies

$$|\log \lambda - q\beta(t_j)| \leq \mathcal{J}_{q,b}, \quad (\text{JB7})$$

where one fully finite choice is

$$\begin{aligned} \mathcal{J}_{q,b} = & \log \mathcal{L}_{q,b} + \max(\log L_{q,b}, \log U_{q,b}) + 2b \log(C_\Gamma/c_\Gamma) \\ & + 110 + 3 \log(q+2) + q\beta((b+1)/q). \end{aligned} \quad (\text{JB8})$$

The last term is the exact concavity bound for the integer degree shifts. In particular

$$\mathcal{J}_{q,b} = O(b \log(Cq/b) + b + \log(q+2))$$

with the fixed additive constant displayed in (JB8). No division of this error by an unjustified fixed rank is made.

4. Quantitative localization of the original conductor kernel

The inherited CK construction places a finite-codimension part of the original kernel in a high power band. Here its radius and every tail can be chosen with explicit dependence on k . This turns the endpoint argument into a uniform, quantitative cutoff calculation.

4.1 A zero-free circle with a certified modulus bound

Write the exact centred symbol as

$$\mathcal{E}(z) = \sum_s a_s e^{\sigma_s z} = z^{v_0} b(z), \quad b(0) = i^{-v_0} \mu_{v_0} / v_0! \neq 0.$$

Put

$$B_A = \max(1, \max_s |\sigma_s|), \quad A_b = \frac{(\sum_s |a_s|) B_A^{v_0}}{|\mu_{v_0}|} \geq 1.$$

Taylor's remainder gives the global bound $|b(z)/b(0)| \leq A_b e^{B_A |z|}$. For $R \geq 2$, define

$$N_R = \left\lceil \frac{\log A_b + 12 B_A R}{\log 2} \right\rceil, \quad L_R = \log A_b + 6 B_A R,$$

$$\mathcal{M}_R = |b(0)|^{-1} e^{2L_R} [32(N_R + 1)]^{N_R}. \quad (\text{LC1})$$

There is a radius $r \in [R, 2R]$ containing no zero on its circle, with at most N_R interior zeros counting multiplicity, such that

$$\max_{|z|=r} |b(z)^{-1}| \leq \mathcal{M}_R. \quad (\text{LC2})$$

Here is the proof. Jensen's formula on radius $12R$ bounds the number of zeros in $6R$ by N_R . Choose a zero-free outer radius $S \in [4R, 6R]$, and divide $b/b(0)$ by the finite disk Blaschke product of its zeros in S , with unit phases chosen so that their values at zero are positive (not set equal to one). The quotient is zero-free. Its log modulus is at most L_R on the boundary. The positive harmonic function $L_R - \log |b/(b(0)B)|$ has value at zero at most L_R , so Harnack on $|z| \leq 2R$ bounds it by $3L_R$. Hence the quotient modulus is at least e^{-2L_R} .

Choose $r \in [R, 2R]$ at distance at least $R/[4(N_R + 1)]$ from the moduli of all zeros in S . The excluded intervals occupy less than half the radius interval. On that circle every Blaschke factor has modulus at least $[32(N_R + 1)]^{-1}$. Their complete product proves (LC2). This is also constructive after certified isolation of the finite zeros; no such isolation is claimed to have been executed for an unspecified native period.

4.2 The exact finite-rank subtraction and the retained action defect

Use the auxiliary coefficient norm $\|p\|_{c,q} = \sum_j q^j |[y^j]p|$. Under the finite guard $q \geq 40$, $\rho_k = k\sqrt{\delta^2 + \gamma^2}/q \leq 1/2$, one may use

$$\boxed{e^{-10q} \|p\|_{c,q} \leq \|p\|_{G_N^c} \leq e^{2q} \|p\|_{c,q}, \quad \deg p < q.} \quad (\text{LC3})$$

For the upper bound, $\|y^j\|_\sigma \leq \sqrt{\sqrt{2\pi} 2^j j!}$ follows from the full Gamma multiplication recurrence, including each raising row. For the lower polynomial bound, expand $p(qx)$ in the complete Legendre basis on $[1, 2]$. The coefficient sum of $P_j(2x - 3)$ is $P_j(5) \leq 10^j$. The lower Gamma envelope on $[q, 2q]$ yields an exponent at most $8q$ for every degree through $2q$. The complete Newton remainder modulo Q_k has coefficient norm at most 2^{2q} : each actual root has scaled modulus at most $1/2$, and the complete homogeneous divided differences give the sum $(1 + 2\rho_k)^j$. This proves (LC3) for every original affine representative, rather than just its canonical coefficient representative.

Let $D = \partial_y$ on degree below q . Its norm in $\|\cdot\|_{c,q}$ is at most one. On the radius in (LC2), the exact CK6-7 identity is

$$\begin{aligned} b(D)^{-1} &= A_r + B_r - L_r, \\ A_r &= \sum_{j < q} a_j D^j, \quad |a_j| \leq \mathcal{M}_R r^{-j}, \quad \|B_r\| \leq r \mathcal{M}_R q e^{rq}, \quad \text{rank } L_r \leq N_R. \end{aligned} \quad (\text{LC4})$$

The full residue at each nonzero symbol pole is retained. The range of an order- a pole is spanned by the a derivatives of the actual truncated exponential. The operator B_r raises degree strictly.

The published bound $r \mathcal{M}_R q e^{rq}$ for this degree-raising term can be improved substantially when r grows. The exact coefficient from y^n to y^{n+j} in $R_z y^n$ is $n! z^{j-1} / (n+j)!$. Thus, in the original auxiliary coefficient norm,

$$\|R_z\|_{c,q} \leq \frac{1}{r} \sum_{j=1}^{q-1} \frac{(qr)^j}{j!} \quad (|z| = r).$$

The last terms dominate this finite sum for $r \geq 2$: successive ratios backwards are smaller than $1/r$. Using $(q-1)! \geq ((q-1)/e)^{q-1}$ and $(q/(q-1))^{q-1} < e$, one obtains

$$\sum_{j=1}^{q-1} \frac{(qr)^j}{j!} \leq \frac{r}{r-1} \frac{(qr)^{q-1}}{(q-1)!} < \frac{(er)^q}{r-1} \leq (3r)^q.$$

The contour factor is exactly $r \mathcal{M}_R$. Therefore, since $r \leq 2R$,

$$\boxed{\|B_r\|_{c,q} \leq \mathcal{M}_R (6R)^q.} \quad (\text{LC4a})$$

This uses the entire finite truncated exponential, not an infinite-tail approximation or a discarded pole. Its logarithm grows as $q \log R$ rather than Rq . It is the further estimate that permits the faster single-sequence rate below.

For $p \in K_k$, the original conductor gives $f = \mathcal{T}p = Q_{k-8}\eta$. Let $f_0 = D_0^{-v_0} f$ with all coefficients below v_0 zero. Then $p = b(D)^{-1} f_0 + l(p)$, with $\deg l < v_0$. Define the actual subspace

$$K_{k,R} = \{p \in K_k : l(p) - L_r f_0 = 0\}. \quad (\text{LC5})$$

It is independent of cutoff and has codimension at most $c_R = v_0 + N_R$. On it, $p = (A_r + B_r) f_0$.

Choose an integer b divisible by eight, with

$$8\Delta + 8 \leq b \leq q/10, \quad \Delta = q - (k - 7)^2, \quad \rho = \rho_k < 1/2. \quad (\text{LC6})$$

Put $M_T = \sum_s |a_s| e^{|\sigma_s|}$, and define

$$\begin{aligned} e_p &= q^{v_0} M_T \mathcal{M}_R \left[(2 + (6R)^q)(1 - \rho)^{-q'} 2^{q'} \rho^{b/8} + 2R^{-b/4} \right], \\ e_U &= (1 - \rho)^{-q} e_p + \frac{4^q \rho^{b/2}}{1 - \rho}, \\ \zeta_{k,b,R} &= e^{12q} (1 + \rho)^q e_U. \end{aligned} \quad (\text{LC7})$$

These are explicit finite quantities in the original fixed symbol and roots.

The lower polynomial's complete coefficient tail gives

$$\|P_{< q' - h} f\|_{c,q} \leq (1 - \rho)^{-q'} 2^{q'} \rho^h \|f\|_{c,q}.$$

Because $b \geq 8\Delta + 8$, the part of f_0 below $q - b/4$ is bounded by the term with $\rho^{b/8}$ in (LC7). The high part must lose at least $b/4$ degrees to reach below $q - b/2$ through A_r , giving $2\mathcal{M}_R R^{-b/4}$. The degree-raising B_r uses only the small low input. This proves the first line of (LC7) as a relative coefficient bound on $P_{< q - b/2} p$.

Retain the actual triangular map

$$U_Q F = [Q_k(y) y^{-q} F(y)]_+. \quad (\text{LC8})$$

Its norm is at most $(1 + \rho)^q$, its inverse norm at most $(1 - \rho)^{-q}$, and its inverse coefficient tail beyond depth $b/2$ is at most $4^q \rho^{b/2} / (1 - \rho)$. These are complete elementary and homogeneous symmetric-function bounds. Thus

$$A_{k,b,R} p = P_{\geq q-b} U_Q^{-1} p$$

satisfies

$$\|p - U_Q A_{k,b,R} p\|_{G_N^q} \leq \zeta_{k,b,R} \|p\|_{G_N^q} \quad (\text{LC9})$$

at every cutoff. Equation (LC3) is what transfers the coefficient tail to this relative metric estimate. If $\zeta < 1$, the map is injective on $K_{k,R}$.

The source's complete CK10-11 comparison gives a fixed C_Q such that, on every band affine fibre,

$$e^{-C_Q} \|F\|_\sigma^2 \leq \|U_Q F\|_\sigma^2 \leq e^{C_Q} \|F\|_\sigma^2. \quad (\text{LC10})$$

Its mechanism is the complete CK10-11 comparison, not a formal quotient identification. Here is one explicit finite choice of its constant. Put $\epsilon_0 = 2^{-32}$, $d_* = q + b$, $M_\sigma = \sqrt{2\pi}$, and

$$\begin{aligned} E_Q &= \frac{q(\rho/\epsilon_0)^2}{1 - (\rho/\epsilon_0)^2}, \quad r_Q = \sqrt{M_\sigma} e^{8q} 2^q \rho^{q-b} \epsilon_0^{-b}, \\ i_F &= 2\sqrt{2} M_\sigma q^{-1/2} (d_* + 1)^2 10^{2d_*} e^{\pi q} \epsilon_0^{2(q-b)}, \\ i_U &= 8\sqrt{2} M_\sigma q^{-1/2} (d_* + 1)^2 10^{2d_*} e^{\pi q} (2\epsilon_0 + \rho)^{2q} (2\epsilon_0)^{-2b}. \end{aligned}$$

On the finite guards

$$\rho \leq \epsilon_0/2, \quad i_F \leq 1/4, \quad i_U \leq 1, \quad r_Q \leq e^{-E_Q/2}/4,$$

one may take

$$C_Q = E_Q + 2 \log 2. \quad (\text{LC10a})$$

To prove it, write $F(qx) = x^{q-b} h(x)$. Outside $|x| < \epsilon_0$, pairing all actual opposite roots gives the squared logarithmic multiplier bound E_Q . The negative Laurent part of $q^{-q} Q_k(qx) x^{-b} h(x)$ has co-efficient sum at most $2^q \rho^{q-b} \|h\|_1$. Its full exterior norm relative to $\|F\|_\sigma$ is therefore at most r_Q , by the pre-quotient coefficient lower bound used in (LC3). The complete Legendre estimate for h on $[1, 2]$ bounds the inner part of F by $i_F \|F\|_\sigma^2$. Cauchy's polynomial-part formula on $|x| = 2\epsilon_0$ bounds the inner part of $U_Q F$ by $i_U \|F\|_\sigma^2$. Hence

$$\begin{aligned} \frac{\|U_Q F\|_\sigma}{\|F\|_\sigma} &\geq e^{-E_Q/2} \sqrt{1 - i_F} - r_Q > \frac{1}{2} e^{-E_Q/2}, \\ \frac{\|U_Q F\|_\sigma}{\|F\|_\sigma} &\leq \sqrt{(e^{E_Q/2} + r_Q)^2 + i_U} < 2e^{E_Q/2}. \end{aligned}$$

This proves (LC10a). Both inner integrals and the exterior Laurent remainder are explicit. The guards hold eventually uniformly for $b \leq q/10$; $E_Q = O_h(1)$. No root is removed.

The arithmetic action itself is not identified with the power model. If $Q_k = \sum c_j y^j$, its exact defect remains

$$\boxed{MU_Q - U_Q M_0 = -[1](c_{q-1}, c_{q-2}, \dots, c_0),} \quad (\text{LC11})$$

where M_0 is multiplication modulo y^q . This is verified by polynomial parts on every monomial. All metric statements below use (LC10), not an unproved action intertwiner.

4.3 Finite relative spectral estimate and an effective choice

Suppose $\zeta_{k,b,R} \leq 1/4$. Set

$$A_{k,b,R}^{\text{err}} = \mathcal{J}_{q,b} + 2C_Q + 2 \log \frac{1 + \zeta_{k,b,R}}{1 - \zeta_{k,b,R}} + \log \kappa_k, \\ B_k^{\text{whole}} = 40q + \log \kappa_k, \quad c_R = \min(m, v_0 + N_R). \quad (\text{LC12})$$

The full source nesting and (LC3) imply all relative metric logarithms lie in $[0, B_k^{\text{whole}}]$. On the fixed core $K_{k,R}$, (JB7), (LC9)-(LC10), and the native comparison (CU2) put every relative logarithm within $A_{k,b,R}^{\text{err}}$ of $q\beta(t_j)$. Generalized eigenvalue interlacing therefore gives

$$\boxed{\sum_{i=1}^m |\log \lambda_i(H_{K,N_j}, H_{K,N_0}) - q\beta(t_j)| \leq m A_{k,b,R}^{\text{err}} + 2c_R [B_k^{\text{whole}} + q\beta(11/10)].} \quad (\text{LC13})$$

Here $\lambda_i(H_{K,N_j}, H_{K,N_0})$ denotes the spectrum of $H_{K,N_j}^{-1/2} H_{K,N_0} H_{K,N_j}^{-1/2}$. At most c_R directions can lie above the core interval and at most c_R below it. This is the origin of the factor two; no favorable sign is assigned to the exceptional quotient.

One effective eventual choice is

$$L_k = \log(k + e), \quad R_k = e^{2\sqrt{L_k}}, \quad b_k = 8 \left\lceil \frac{32q}{8\sqrt{L_k}} \right\rceil. \quad (\text{LC14})$$

Use it only when (LC6), the complete two-region guards in (LC10a), and $\zeta \leq 1/4$ hold. All are explicit finite inequalities. They hold eventually. In fact

$$\log \mathcal{M}_R = O_A(R \log(R + 2)) = o(q), \quad b_k/(q) = 32/\sqrt{L_k} + O(1/q).$$

The high-derivative term has exponent $-(b_k/4) \log R_k \leq -16q$ before the e^{12q} coefficient-to-metric cost. The low-degree tail in (LC7) now has exponent at most

$$q \log(6R_k) - \frac{b_k}{8} \log(1/\rho_k) + O_h(q + k + \log q) = -2q\sqrt{L_k} + O_h(q).$$

Here $\log(1/\rho_k) = L_k - O_h(1)$; every original lower-root coefficient factor and the primitive's q^{v_0} remain inside the indicated explicit (LC7) constants. The inverse-triangular tail is smaller still. Thus $\zeta \leq e^{-q}$ eventually.

The complete exceptional codimension is now $c_R = O_A(e^{2\sqrt{L_k}}) = k^{o(1)}$, while (JB8) gives

$$\frac{A_{k,b,R}^{\text{err}}}{q} = O_{h,A} \left(\frac{\log L_k}{\sqrt{L_k}} + \frac{\log \kappa_k}{q} \right).$$

Consequently the single-sequence native error is

$$\boxed{\frac{R_{K,k}}{mq} = O_{h,A} \left(\frac{\log L_k}{\sqrt{L_k}} + \frac{e^{2\sqrt{L_k}}}{k} + \frac{\log \kappa_k}{q} \right).} \quad (\text{LC15})$$

This improves the rate obtained by using the whole infinite exponential e^{Rq} in the residue bound. Its sufficient threshold is still large, since the chosen band must satisfy $b_k \leq q/10$ and every two-region/source guard. It is not advertised as a practical small- k native enclosure. No derivative is taken of an uncontrolled asymptotic remainder.

5. Original subquotients, observation layers, and the loss distribution

5.1 A simultaneous subquotient theorem

Let $S_k \subseteq K_k$ and $R_k \subseteq S_k$ be fixed coefficient subspaces as the cutoff varies. The metric on S_k/R_k is its full attained metric, not a raw coordinate block; let its rank be r_k . With the notation of (LC12),

$$\sum_{i=1}^{r_k} |\log \lambda_i - q\beta(t_j)| \leq r_k A_{k,b,R}^{\text{err}} + 2 \min(c_R, r_k) [B_k^{\text{whole}} + q\beta(11/10)]. \quad (\text{SQ1})$$

Restriction first intersects the original core with S_k , increasing its codimension by at most c_R . For the quotient, the metric is the inverse compression of the inverse restricted metric. Generalized min-max therefore preserves the same upper and lower exceptional counts. This proves (SQ1) and retains every complete minimum.

For every $r_k \rightarrow \infty$,

$$\sup_j \frac{1}{r_k q} \sum_i |\log \lambda_i - q\beta(t_j)| \rightarrow 0. \quad (\text{SQ2})$$

For this assertion one may first fix an arbitrarily narrow band, then choose its fixed zero-free circle, and only then let k grow. The core codimension is then fixed and divided by r_k . Letting the band width tend to zero completes the proof. It requires no specified growth rate beyond divergence of the subquotient rank. For an explicit rate, use (SQ1) with the actual ranks and (LC14)-(LC15).

For arbitrary $0 \leq r_k \leq m$, even bounded ranks,

$$\log \frac{\det H_{S/R, N_0}}{\det H_{S/R, N_j}} = r_k q \beta(t_j) + o_{h,A}(kq)$$

uniformly in the subquotient. A universal per-direction limit for a fixed exceptional line is not asserted.

5.2 Every actual arithmetic observation layer

Keep the original filtration $K_s = \bigcap_{h=0}^s \ker(\Lambda M^h)$, and its complete layer covariance from AP14-16:

$$D_{s,N} = F_s H_{s-1,N}^{-1} F_s^*, \quad b_{s,k} = \dim K_{s-1} - \dim K_s.$$

Its inverse is the actual attained metric on K_{s-1}/K_s . Therefore

$$\log \frac{\det D_{s,N_j}}{\det D_{s,N_0}} = b_{s,k} q \beta(t_j) + o_{h,A}(kq), \quad (\text{SQ3})$$

with an $o(b_{s,k}q)$ error when $b_{s,k} \rightarrow \infty$. In particular, the first layer has $b_{1,k} \geq k - 18$ on the original period domain, so its full trajectory is determined in the stronger normalization. At the original four cutoffs,

$$-\mathcal{R} \log \det D_{s,N} = b_{s,k} C_{\partial} q + o(kq).$$

The ranks still come from the actual conductor and complete invariant observation. They have not been set from a generic chain model.

The earlier two-response secant also inherits this full trajectory. In the original observed metric use the correctly retained combination

$$Z_N = 12(2I + \mathcal{Y}_N(2\sigma/3))^{-1} - 4(I + \mathcal{Y}_N(\sigma/2))^{-1} - 2I.$$

The current source corrects the outer factor in its derivation; this displayed formula is unchanged. For its original selected coefficient $p_N = e_{m-h_k}(Z_N)$, the proved per-cutoff comparison to $\det \Gamma_N$ has logarithmic width $O_{h,A}(q \log q)$, with $h_k = O_{h,A}(1)$. The boundary metric changes by $O(k \log q)$ per direction. Hence

$$\sup_j \left| \frac{\log p_{N_j} - \log p_{N_0}}{mq} - \beta(t_j) \right| \rightarrow 0. \quad (\text{SQ4})$$

This is a statement about that selected positive characteristic coefficient, not the zero full secant determinant or the independent complex-current numerator.

5.3 A measure of actual cutoff loss

Define the positive discrete measure on $[0, 1]$

$$\mu_k = \frac{1}{mq} \sum_{j=0}^{q-1} \log \frac{\det H_{K, N_j}}{\det H_{K, N_{j+1}}} \delta_{(j+1)/q}. \quad (\text{SQ5})$$

Its atoms are nonnegative by source nesting. The full cutoff theorem gives

$$\mu_k \Rightarrow \beta'(t)dt, \quad \mu_k([0, 1]) \rightarrow \beta(1) = C_\partial/2. \quad (\text{SQ6})$$

No derivative of an $o(q)$ error is taken: the proof is uniform convergence of the cumulative loss, followed by integration against continuous test functions.

If the normalized cumulative error is at most $e_k < \beta(1)$, the probability obtained by dividing μ_k by its actual total mass has one-dimensional Wasserstein error at most

$$W_1(\mu_k/\mu_k([0, 1]), \beta'(t)dt/\beta(1)) \leq \frac{2e_k}{\beta(1) - e_k} + \frac{1}{q}. \quad (\text{SQ7})$$

The final $1/q$ accounts for the chosen step interpolation. It follows by integrating the CDF difference. This probability describes a scalar distribution of determinant loss; no physical source measure or norm has been normalized away.

In particular the first half of the window carries the exact asymptotic fraction in (PR13), approximately 69.7084 percent, not one half. The full cumulative loss also has the new evaluated integrated value

$$\frac{1}{mq^2} \sum_{j=0}^{q-1} \log \frac{\det H_{K, N_0}}{\det H_{K, N_j}} \rightarrow \int_0^1 \beta(t)dt, \quad (\text{SQ8})$$

whose certified interval is (PR12). This is an explicitly defined additional observable, not a reassignment of the original four-sign total.

6. The full original word spectrum through the cutoff interval

Retain the actual word $T_k = D_k(M)$ and its original root split. Its kernel V has dimension $d = q - \Delta = (k - 7)^2$, its image W has dimension $\Delta = 16k - 48$, and $K \cap V = 0$. The full boundary-value map π_∂ has kernel V . Its attained metric is

$$G_N^\partial = (\pi_\partial G_N^{-1} \pi_\partial^*)^{-1}.$$

All positive word eigenvalues are those of the inherited image form against this attained boundary metric; the d exact zero directions remain present.

6.1 Uniform localization of the whole image

A polynomial in W is $D_k h$, $\deg h < \Delta$, with every root of D_k of modulus at most $R_k = k\sqrt{\delta^2 + \gamma^2}$. Unlike the full conductor inverse, this factorization has no finite-symbol-pole correction. If b satisfies (LC6), define

$$e_W = (1 - \rho)^{-q} (1 - \rho)^{-d} 2^d \rho^{b/4} + \frac{4^q \rho^{b/2}}{1 - \rho}, \quad \zeta_W = e^{12q} (1 + \rho)^q e_W. \quad (\text{WE1})$$

The complete leading-coefficient inversion of $D_k h$, followed by the same triangular map (LC8), proves (LC9) for every vector of W , with ζ_W and with no exceptional quotient. Consequently

$$e^{-q\beta(t_j) - a_{W,k}} G_{W, N_0} \preceq G_{W, N_j} \preceq e^{-q\beta(t_j) + a_{W,k}} G_{W, N_0}, \quad (\text{WE2})$$

where

$$a_{W,k} = \mathcal{J}_{q,b} + 2C_Q + 2 \log \frac{1 + \zeta_W}{1 - \zeta_W} + \log \kappa_k.$$

Use the finite guards $\zeta_W \leq 1/4$ and (LC6). The choice

$$b = 8 \left\lceil \frac{128q}{8 \log(1/\rho_k)} \right\rceil \quad (\text{WE3})$$

satisfies them eventually, and gives

$$a_{W,k} = O_h(q \log \log k / \log k + \log \kappa_k + 1) = o(q).$$

This is a form comparison on all Δ image directions, not an inference from their determinant.

6.2 A direct finite low-end word band

For completeness the low-end band can be obtained without a large-degree zero law. Put $\epsilon_* = 2^{-32}$, $T_* = \epsilon_* q$, and require $R_k \leq T_*/2$, $\Delta \leq q/10$, $d \geq 36$. Define

$$\begin{aligned} r_* &= R_k/T_*, & K_D &= 2\sqrt{2}\sqrt{2\pi} \Delta^2 q^{-1/2} 10^{2\Delta} e^{\pi q} (\epsilon_* + R_k/q)^{2d}, \\ E_D &= \frac{2dr_*}{1-r_*} + \log(1 + K_D). \end{aligned} \quad (\text{WE4})$$

Then, for every $\deg h < \Delta$,

$$\boxed{e^{-E_D} \|y^d h\|_\sigma^2 \leq \|D_k h\|_\sigma^2 \leq e^{E_D} \|y^d h\|_\sigma^2.} \quad (\text{WE5})$$

Outside $|y| < T_*$, sum the logarithmic error from all d root factors. Inside, the full Legendre evaluation bound on $[q, 2q]$ gives the factor $\Delta^2 10^{2\Delta}/q$; the reference norm on that interval is bounded below by $q^{2d} (2\sqrt{2q})^{-1} e^{-\pi q} \int_q^{2q} |h|^2$. Both inner integrals are bounded by K_D times the reference outer integral. This proves (WE5), including the original region near zero and every nonreal root. Here $E_D = O_h(k)$ on the eventual domain.

Let $\mathcal{L}_{q,\Delta}$ be (JB2), with $n = d$. In boundary-value coordinates the interpolation matrix is $V_{ij} = (\omega_i/q)^j$, $0 \leq j < \Delta$. Retaining every root difference gives

$$\|V\| \leq \Delta, \quad \|V^{-1}\| \leq V_- := \Delta \left(\frac{q + R_k}{2\delta} \right)^{\Delta-1}.$$

The actual word sends a boundary value b to $D_k h$, with $h(\omega_i) = b_i$. Thus (JB2) and (WE5) give its image form in this exact coordinate map.

The original Gamma boundary quotient has the finite bounds

$$\begin{aligned} (A_\partial)^{-1} I &\preceq G_N^{\partial,\sigma} \preceq B_\partial I, \\ A_\partial &= \frac{\Delta e^2}{\sqrt{2\pi}} (2q+1)^{3/2} (4q+1)^{R_k+1}, \quad B_\partial = \Delta \sqrt{2\pi} \left(\frac{2\Delta + R_k}{2\delta} \right)^{2(\Delta-1)}. \end{aligned} \quad (\text{WE6})$$

These follow from the full evaluation kernel and full lower-degree Lagrange sections. They are the retained original finite boundary constants, with mass $\sqrt{2\pi}$.

Let $c_0 = 2q \log q + \eta_0 q$. A common finite error is

$$\begin{aligned} e_{0,k} &= E_D + \log \mathcal{L}_{q,\Delta} + \log \kappa_k + 2 \log \Delta + 2 \log \max(1, V_-) \\ &\quad + \log \max(1, A_\partial) + \log \max(1, B_\partial) + |\log d_d - c_0|. \end{aligned} \quad (\text{WE7})$$

Then all Δ positive eigenvalues at $N_0 = q - 1$ satisfy

$$e^{c_0 - e_{0,k}} \leq \lambda_j^+(T_k^{\dagger G_{N_0}} T_k) \leq e^{c_0 + e_{0,k}}. \quad (\text{WE8})$$

The numerator is the full word image form; the denominator is the attained boundary quotient. Applying the same native comparison to both costs $\log \kappa_k$, not either source mass independently. By (HK3),

$$|\log d_d - c_0| \leq 2\Delta(\log q + 1) + |\eta_0|\Delta + 5.$$

Hence $e_{0,k} = O_h(k \log q)$. This finite derivation can replace the inherited low-end word asymptotic in the following statements.

6.3 Uniform positive energy band

Put

$$\omega_k = \log \kappa_k + \log \max(1, A_\partial) + \log \max(1, B_\partial), \quad e_k^{\text{word}} = e_{0,k} + a_{W,k} + \omega_k.$$

The domain boundary metric changes by at most this fixed logarithmic comparison at every cutoff. Therefore

$$e^{c_0 - q\beta(t_j) - e_k^{\text{word}}} \leq \lambda_l^+(T_k^{\dagger G_{N_j}} T_k) \leq e^{c_0 - q\beta(t_j) + e_k^{\text{word}}}, \quad (\text{WE9})$$

for every $1 \leq l \leq \Delta$, simultaneously in j . Here $e_k^{\text{word}} = O_h(q \log \log k / \log k + k \log q) = o(q)$ on the finite guards above.

No eigenvalue is assigned to one of the d zero directions. The physical word remains $D_k(M_k)$; its positive square is formed with the actual cutoff metric.

7. The entire original cutoff-frequency surface

7.1 The kernel-energy logarithms cancel their common cutoff scale

Let

$$A_{K,N} = H_{K,N}^{-1/2} I_K^* T_k^* G_N T_k I_K H_{K,N}^{-1/2} > 0.$$

The numerator is the restriction of the inherited image metric under the fixed injective map $T_k I_K$. Equation (WE2) gives its full relative Loewner band; equation (LC13) gives the full mean-absolute logarithmic change of its denominator.

The required metric perturbation inequality has a direct proof. For a fixed positive form B , move a positive metric by $H(s) = H_0^{1/2} e^{sK} H_0^{1/2}$. The representative of $(B, H(s))$ is $e^{-sK/2} B_0 e^{-sK/2}$. At every differentiability point, each eigenvalue logarithm has derivative $-u_i^* K u_i$; at a repeated eigenvalue, diagonalize the compressed derivative on its eigenspace. Thus the sum of absolute derivatives is at most $\text{Tr} |K|$. Integration gives

$$\sum_i |\log \lambda_i(B, H_1) - \log \lambda_i(B, H_0)| \leq \text{Tr} |\log(H_0^{-1/2} H_1 H_0^{-1/2})|. \quad (\text{FR1})$$

All eigenvalues are ordered consistently. This is a metric comparison, not an assumed common eigenvector frame.

Let $R_{K,k}$ denote the right side of (LC13), uniform in cutoff, and put

$$R_{A,k} = R_{K,k} + m a_{W,k}.$$

Apply (FR1) after multiplying both numerator and denominator at cutoff N_j by the same scalar $e^{q\beta(t_j)}$. That simultaneous multiplication leaves the generalized operator unchanged. One obtains

$$\sup_j \sum_{i=1}^m |\log \alpha_{i,N_j} - \log \alpha_{i,N_0}| \leq R_{A,k} = o(kq), \quad (\text{FR2})$$

where $\alpha_{i,N}$ are the actual positive kernel energies.

7.2 All positive regularizers, without a hidden endpoint restriction

Use the original response

$$\mathcal{Y}_N(z) = \Lambda z (zI + T_k^{\dagger G_N} T_k)^{-1} L_N, \quad z > 0.$$

The full Schur determinant identity, retaining the entire kernel and all word zeros, is

$$\det \mathcal{Y}_N(z) = \frac{\det(I_m + A_{K,N}/z)}{\det(I_q + T_k^{\dagger G_N} T_k/z)}. \quad (\text{FR3})$$

The function $x \mapsto \log(1 + e^x/z)$ is 1-Lipschitz for every $z > 0$. Thus (FR2) and (WE9) imply the finite uniform bound

$$\sup_{j,z>0} \left| \log \frac{\det \mathcal{Y}_{N_j}(z)}{\det \mathcal{Y}_{N_0}(z)} - \Delta \log \frac{z + e^{c_0}}{z + e^{c_0 - q\beta(t_j)}} \right| \leq R_{A,k} + 2\Delta e_k^{\text{word}}. \quad (\text{FR4})$$

Its error is $o(kq)$. In particular it is valid when the regularizer tends to zero faster than any prescribed exponential; no unresolved original-kernel factor has been held fixed to obtain it.

For $\xi_k(z) = (\log z - 2q \log q)/q$, the scalar softplus estimate adds at most $\log 2/q$, and yields

$$\sup_{j,z>0} \left| \frac{1}{\Delta q} \log \frac{\det \mathcal{Y}_{N_j}(z)}{\det \mathcal{Y}_{N_0}(z)} - [(\eta_0 - \xi_k(z))_+ - (\eta_0 - \beta(t_j) - \xi_k(z))_+] \right| \rightarrow 0. \quad (\text{FR5})$$

This is the cutoff-frequency surface for the unchanged Λ . At $t = 1$ it agrees with the integrated endpoint theorem, including its signs. It is not the earlier enlarged observation Λ_8 .

7.3 Every inverse exterior rank

Assume additionally $d \geq m$, which holds throughout the eventual sequence (and already for $k \geq 21$). Flattening the positive energy spectrum to its common centre retains the exact primary angles Γ_N . The inverse-response spectrum then consists of

$$1 + x_N \quad (\Delta - m \text{ entries}), \quad \frac{1 + x_N}{1 + x_N \gamma_{i,N}} \quad (m \text{ entries}),$$

and $d-m$ unit entries, where $x_N = e^{c_0 - q\beta(t_j)}/z$. The actual word-energy error changes the logarithm of each nonunit inverse eigenvalue by at most e_k^{word} .

The boundary numerator of Γ_N changes by at most ω_k per logarithmic direction. Applying (FR1) to that numerator and $H_{K,N}$ shows

$$\sum_i |\log \gamma_{i,N_j} - \log \gamma_{i,N_0} - q\beta(t_j)| \leq R_{K,k} + m\omega_k.$$

Hence $x_N \gamma_{i,N}$ has cutoff change controlled by the same right side. For $p_0 = \min(p, \Delta)$, if $p_0 \leq \Delta - m$, no angle factor occurs in the largest exterior product. Otherwise $p_0 \geq \Delta - m + 1 = 8k - 31$, and the entire angle error divided by $p_0 q$ tends to zero. Therefore

$$\sup_{j,z>0} \sup_{1 \leq p \leq q-m} \left| \frac{\log \|\wedge^p \mathcal{Y}_{N_0}(z)^{-1}\| - \log \|\wedge^p \mathcal{Y}_{N_j}(z)^{-1}\|}{p_0 q} - [(\eta_0 - \xi_k(z))_+ - (\eta_0 - \beta(t_j) - \xi_k(z))_+] \right| \rightarrow 0. \quad (\text{FR6})$$

Each norm is in its original observed metric. The rank-one operator norm and full determinant are included, but no identification with the fixed conductor's inverse is made.

7.4 The measured phase becomes an explicit moving interval

For $z = ie^s$, define the continuous determinant phase by the factorized positive spectrum:

$$\Phi_N(s) = (\Delta - m)\pi/2 + \sum_i \arctan(e^s/\alpha_{i,N}) - \sum_l \arctan(e^s/\lambda_{l,N}^+). \quad (\text{FR7})$$

This is the original scalar measured-determinant phase with its full multiplicities, not the phase of a single response entry. The fixed zero-power term cancels between cutoffs.

For $K(s) = \arctan e^s$, monotonicity and translation give the exact identity

$$\int_{\mathbb{R}} |K(s-a) - K(s-b)| ds = \frac{\pi}{2} |a-b|.$$

Equations (FR2) and (WE9) therefore imply

$$\sup_j \left\| -\frac{2}{\pi \Delta} [\Phi_{N_j}(c_0 + q \cdot) - \Phi_{N_0}(c_0 + q \cdot)] - \mathbf{1}_{(-\beta(t_j), 0)} \right\|_{L^1(\mathbb{R})} \leq \frac{R_{A,k}}{\Delta q} + \frac{2e_k^{\text{word}}}{q} + \frac{8}{\pi q}. \quad (\text{FR8})$$

For the last term, $\arctan u \leq u$ bounds the two soft transition tails. This retains both frequency ends. It does not imply pointwise convergence at the interval boundaries or assign the separate complex arithmetic-current sign.

7.5 Averaging the original heat over the cutoff interval

Use the common physical heat time

$$\tau_k(b) = e^{-c_0 + qb}.$$

The original measured heat is $\text{Tr}_B[\Lambda e^{-\tau H_N} L_N]$, not the heat of the arithmetic operator compressed first. The exact primary-angle identity gives its zero-energy part $d - m + \text{Tr} \Gamma_N$. Moreover the difference between its positive-energy contribution and the full positive trace, after adding $\text{Tr} \Gamma_N$, lies in $[0, \text{Tr} \Gamma_N]$.

The original finite angle-tail bound and whole-source comparison give a common $C_{\text{ang}} = O_{h,A}(1)$ with $\text{Tr} \Gamma_N \leq C_{\text{ang}}$ over the entire window. All the native pole and period factors remain in that finite provider.

Define

$$\mathcal{H}(b) = \begin{cases} 1, & b \leq 0, \\ 1 - \beta^{-1}(b), & 0 < b < \beta(1), \\ 0, & b \geq \beta(1). \end{cases}$$

Then

$$\sup_{b \in \mathbb{R}} \left| \frac{1}{q\Delta} \sum_{j=0}^{q-1} [\text{Tr}_B \Lambda e^{-\tau_k(b) H_{N_j}} L_{N_j} - (d - m)] - \mathcal{H}(b) \right| \leq \frac{25}{4} \frac{e_k^{\text{word}} + \log q}{q} + \frac{4}{q} + \frac{C_{\text{ang}}}{\Delta}. \quad (\text{FR9})$$

To prove it, outside $|\beta(t_j) - b| \leq \epsilon$, (WE9) bounds the heat by its limiting indicator with error at most $1/q$, for $\epsilon = (e_k^{\text{word}} + \log q)/q$. Since $\beta' \geq \beta'(1) > 8/25$ on $(0, 1]$, the critical set occupies length at most $(25/4)\epsilon$ and at most two additional grid cells. The measured/full difference is at most C_{ang}/Δ . This proves (FR9), including both threshold endpoints, without assigning a pointwise critical heat value.

8. Integration with edition 025 and new shrinking-window laws

The repository changed from the baseline 024 edition while this work was in progress. Its current complete proof files are INTEGRATED_PROFILE_AND_RECEIVERS.md (IR1–24), POWER_CUTOFF_PROFILE_PROOF.md (PP1–45), ELLIPTIC_PROFILE_PROOF.md (EP1–19), and PHASE_HEAT_PROOFS.md (PH1–41), at 76473e0564b32f7566e191829bf67477c8950f0b. The inspected ranges are recorded in the ledger. The structural full-window, all-pair, fixed-subquotient, and phase/heat results are now upstream results. Their profile is $J(t) = \beta(t)$. Their proof still retains the monic zero-statistic remainder $r_U(n, M) = o(n)$ in PP23–28. The new harmonic-measure estimate (HK10) instead treats the evaluation minimum itself, which is why (HK11) does not need a rate for that separate monic statistic.

The native error (LC13) can consequently be calculated from finite constants, and (LC14)–(LC15) supplies a single-sequence rate instead of first fixing a band, then taking $k \rightarrow \infty$, then shrinking the band. This does not imply a useful small numerical threshold; every very conservative eventual guard is retained. The original native period and unit remain fixed.

8.1 Quantitative comparison between arbitrary cutoffs

Set

$$e_k = \frac{R_{K,k}}{mq}, \quad F_k(t_j) = \frac{1}{mq} \log \frac{\det H_{K,N_0}}{\det H_{K,N_j}}.$$

Then $F_k(0) = 0$ exactly, and $|F_k(t_j) - \beta(t_j)| \leq e_k$ on every original cutoff. The finite all-direction version is

$$\sum_{a=1}^m \left| \log \lambda_a(H_{K,N_j}^{-1/2} H_{K,N_i} H_{K,N_j}^{-1/2}) - q[\beta(t_j) - \beta(t_i)] \right| \leq 2R_{K,k}. \quad (\text{ME1})$$

For a direct proof, multiply each metric by its specified scalar $e^{q\beta(t)}$. The sum of absolute relative logarithms from each rescaled metric to H_{K,N_0} is at most $R_{K,k}$. Apply the derivative argument of (FR1) along the two concatenated metric geodesics. Its instantaneous trace-norm length is invariant under the explicitly changing isometries; integration bounds the endpoint relative logarithms by the sum of the two lengths. This is not an assumption that different cutoff Grams commute. The same proof applies to the complete fixed subquotients in (SQ1).

8.2 A resolved local mean of the actual rank-one source responses

Let u_N be the remainder class of the next original unit-norm source polynomial. Appending that source gives exactly

$$G_{N+1}^{-1} = G_N^{-1} + u_N u_N^*.$$

Define the actual total and observed leverages

$$\rho_N = u_N^* G_N u_N, \quad b_N = (\Lambda u_N)^* Q_{B,N} (\Lambda u_N).$$

The original orthogonal splitting into K and its attained observation complement gives the exact increment (IR10-13)

$$A_{j,k} = \log \frac{\det H_{K,N_j}}{\det H_{K,N_{j+1}}} = \log \frac{1 + \rho_{N_j}}{1 + b_{N_j}} \geq 0. \quad (\text{ME2})$$

For a positive integer window length L , put $h = L/q$, and retain the geometric mean in its literal exponent:

$$\mathcal{G}_{j,L} = \left[\prod_{l=j}^{j+L-1} \frac{1 + b_{N_l}}{1 + \rho_{N_l}} \right]^{1/(mL)}.$$

No individual observed leverage is reconstructed from the product. Telescoping gives

$$\left| -\log \mathcal{G}_{j,L} - \frac{\beta(t_j + h) - \beta(t_j)}{h} \right| \leq \frac{2e_k}{h}. \quad (\text{ME3})$$

At $j = 0$, the right side improves to e_k/h , because the low endpoint is exact.

For $a > 0$, and $a \leq t_j < t_j + h \leq 1$, let

$$M_a = \frac{10(1+a)}{9a}.$$

On this interval, $-\beta'' \leq M_a$. Indeed the variance representation in (PR6) gives $\text{Var}(n) \geq p_0 p_1 = z/(4F(z)^2)$, and $F(z)^2 \leq (1-z)^{-1}$. Using $z \geq a/(1+a)$, $z \leq 11/20$, proves the bound. Thus

$$\left| \log \frac{\mathcal{G}_{j,L}}{\sqrt{z_{t_j}}} \right| \leq \frac{2e_k}{h} + \frac{1}{2} M_a h. \quad (\text{ME4})$$

The target $\sqrt{z_t}$ follows from the exact identity $e^{-\beta'(t)} = \sqrt{z_t}$. It is a scalar value of this precisely defined geometric mean, not an identification of the arithmetic operator with the elliptic coordinate.

Choose the actual integer $L = \lceil 2q\sqrt{e_k/M_a} \rceil$ once it fits in the specified interior interval. Then the right side is at most

$$2\sqrt{M_a e_k} + \frac{M_a}{2q}. \quad (\text{ME5})$$

Accordingly this is a shrinking-window calculation whenever the finite bound $e_k \rightarrow 0$; each such window still contains a growing number of original source steps. No single-step limit is asserted.

At the initial endpoint there is an equally explicit nonstationary law. For $h = L/q \leq 1$, (PR8) and the sharper $j = 0$ error give

$$e^{-h/4 - e_k/h} \leq \frac{\mathcal{G}_{0,L}}{\sqrt{h/e}} \leq e^{e_k/h}. \quad (\text{ME6})$$

For any $h_k \downarrow 0$ with $e_k/h_k \rightarrow 0$ and integer qh_k , $\mathcal{G}_{0,qh_k}/\sqrt{h_k/e} \rightarrow 1$. The source mass and all original phases already enter the leverages in (ME2); none is adjusted to force this limit.

8.3 Negative mesoscopic curvature of the original determinant loss

The explicit error also permits a second local difference, on windows larger than its square-root scale. Fix $0 < a < 1/2$, and use

$$L_a = 1280 \left(\frac{1+a}{a} \right)^3.$$

On $[a, 1]$,

$$\boxed{|\beta'''(t)| \leq L_a, \quad -\beta''(t) > 1/80.} \quad (\text{ME7})$$

For verification, if κ_3 is the third centred moment of the coefficient distribution, then

$$\beta''' = \frac{\kappa_3}{32 \text{Var}(n)^3}.$$

Since $a_n \leq 1$ and $F \geq 1$, at $z \leq 11/20$

$$\begin{aligned} \mathbb{E}n^2 &\leq \frac{z(1+z)}{(1-z)^3} \leq \frac{6820}{729} < 10, \\ \mathbb{E}n^3 &\leq \frac{z(1+4z+z^2)}{(1-z)^4} \leq \frac{102740}{2187}. \end{aligned}$$

Also $\mathbb{E}n = t/4 \leq 11/40$. Hence $|\kappa_3| \leq \mathbb{E}n^3 + 3(11/40)\mathbb{E}n^2 + 2(11/40)^3 < 55$. The variance is at least $9a/[80(1+a)]$ by the proof of (ME4). These rational bounds prove the first part of (ME7); $\text{Var}(n) < 10$ proves the second.

For $h = L/q$, $t_j - h \geq a$, and $t_j + h \leq 1$, the full finite result is

$$\boxed{\left| \frac{F_k(t_j + h) - 2F_k(t_j) + F_k(t_j - h)}{h^2} - \beta''(t_j) \right| \leq \frac{4e_k}{h^2} + \frac{L_a h}{3}.} \quad (\text{ME8})$$

The three native errors contribute $4e_k/h^2$. For the profile itself use

$$\frac{\beta(t+h) - 2\beta(t) + \beta(t-h)}{h^2} = \int_{-1}^1 (1-|u|)\beta''(t+hu)du;$$

the third-derivative bound costs $L_a h \int |u|(1-|u|)du = L_a h/3$. Whenever the right side of (ME8) is below $1/80$, the native second difference is strictly negative. Thus the mean kernel-volume loss in the later of two adjacent original windows is smaller than in the earlier one. This is a sign theorem for the stated *mesoscopic scalar determinant difference*, not for the separate complex-current product.

Choosing, for example, $h \asymp e_k^{1/3}$ proves convergence in (ME8). The resulting effective rate follows from (CU6). In contrast, differentiating an unquantified $o(kq)$ remainder would not justify any shrinking-window conclusion.

8.4 An effective native quantile location

For $0 < \theta < 1$, let the actual grid quantile be

$$\hat{t}_{\theta,k} = \min\{t_j : F_k(t_j) \geq \theta F_k(1)\}.$$

Its target is $t_\theta = \beta^{-1}(\theta\beta(1))$. The cutoff profile is strictly increasing, and $\beta' \geq 8/25$ on $(0, 1]$. The same finite error e_k therefore gives

$$\boxed{|\hat{t}_{\theta,k} - t_\theta| \leq \frac{25}{8}(1+\theta)e_k + \frac{1}{q}.} \quad (\text{ME9})$$

If the right side exceeds one, the estimate is still valid but uninformative. Otherwise, below the target by more than $25(1+\theta)e_k/8$, the upper bound $F_k \leq \beta + e_k$ lies strictly below $\theta F_k(1)$; above it by the same amount the lower bound lies above the target. Passing to the next grid point costs at most $1/q$. This proves the statement without bounding an individual jump by its asymptotic derivative.

The new repository already certifies the half-volume location

$$0.28581630680480845015 < t_{1/2} < 0.28581630680480845016.$$

The scalar code included here independently replays that interval by bisection in the exact Landen coordinate and provides 30-decimal outward endpoints. The new native conclusion is the finite error

$$\boxed{|\hat{t}_{1/2,k} - t_{1/2}| \leq \frac{75}{16}e_k + \frac{1}{q}.} \quad (\text{ME10})$$

It depends on the actual finite constants in (LC12)–(LC13), not a supplied numerical native packet.

9. Verification and scope

The scalar certificate uses integer directed rounding and exact rational coefficients, 260 elliptic terms, and 130 certified bisections on a 90-digit integer grid. Its elliptic tail is bounded by the first omitted positive coefficient divided by $1 - z$; the derivative tail retains its additional linear index factor. The root is bracketed on $(0, 16/25)$, and bisection stops with a rational interval, not a floating midpoint. The logarithm uses the convergent atanh series after exact power-of-two range reduction. The independent integrated-profile bound is (PR11).

The finite exact checker tests the hypergeometric coefficient identities, the Riccati equation, series reversion, interval constants, Gamma moments, complete quotient and layer minima, non-commuting metric similarities, and measured determinant and exterior identities. Negative controls include dropping the original mass, replacing a quotient by a raw block, suppressing the word kernel, treating a linear cutoff profile as correct, and replacing the actual measured energy by a compressed arithmetic square.

REFERENCE_KERNEL_DIAGNOSTICS.json contains exact rational values of $d_q K_{q,M}(0,0)$ for several finite Gamma reference examples, with separately identified non-interval logarithmic diagnostics. Small examples are not asserted to satisfy an asymptotic native guard. The actual native xi moments, admitted period, and invariant coefficient frames are not numerically supplied by those fixtures.

The new quantitative conclusions are (HK10)/(CU7a), the single-sequence native bound (LC13)–(LC15), the evaluated integrated loss (PR12), and the shrinking-window and native-quantile laws (ME3)–(ME10). The full-cutoff profile, derivative representation, subquotient structure, and qualitative phase/heat limits are also independently present in the new repository edition; the finite errors here can be inserted into those receivers. The separate native complex-current values/signs remain unfinished. No theorem here establishes an offcritical zeta zero or a proof of RH. No remote repository file was modified.

References and exact source use

The source ledger gives canonical links and read ranges. Standard analytic identities are distinguished from the programme-specific derivations.

National Institute of Standards and Technology. (n.d.). *NIST Digital Library of Mathematical Functions*, §§5.11, 18.23, 19.4, 19.5, and 19.8. <https://dlmf.nist.gov/5.11> ; <https://dlmf.nist.gov/18.23> ; <https://dlmf.nist.gov/19.8> . The log-Gamma remainder, Meixner-Pollaczek convention, elliptic series, and Landen identities are the classical formulas used explicitly above.

Split-Zero research programme. (2026a, September 22). *The original kernel value and its receiving calculations* [CK1–19]. *Zeta-function research reader*, commit 2b1445b21cc2fcd7e8b7ff34c6aea7139fb6afb5. The fixed-symbol quotient maps, complete original source comparison, and full affine-fibre map are retained in (LC4), (LC8), and (LC10).

Split-Zero research programme. (2026b, September 22). *The complete Gamma-power jet minimum* [PJ1–70]. Same repository and commit. PJ1–23 supplies the finite Laguerre/near-zero comparison and full parity kernels; the finite right-inverse mechanism is independently made explicit in (JB2)–(JB8). The new scalar minimum proof replaces the use of the polynomial zero-statistic limit in this particular step.

Split-Zero research programme. (2026c, September 22). *Exact logarithmic moment and Euler constant* [ELLIPTIC_LOG_MOMENT_PROOF.md, EL1–14]. Same repository and commit. Its evaluated moment and Euler constant are retained in (PR3)–(PR4).

Exact equilibrium for the square-root logarithmic potential. (2026). [EIQ1–33, original uploaded cumulative source]. The full positive density, normalization, Euler inequality on both exterior regions, and uniqueness were read in the contiguous original section. The extracted bytes and their SHA-256 are included under `source_extracts/`.

Split-Zero research programme. (2026d, September 22). *The original kernel across the complete cutoff window* [SZ-20260922-025; IR, PP, EP and PH proof files]. Same repository, commit 76473e0564b32f7566e191829bf67477c8950f0b. The source ledger distinguishes the read portions and mathematical overlaps. In particular (HK11) sharpens PP28 without asserting a rate for its different monic remainder PP23.